

Momentum in the Cryptocurrency Market: A Comprehensive Analysis under Realistic Assumptions

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Abstract

Prior studies on cryptocurrency momentum overlook important real-world factors and inadequately assess its profitability. We evaluate time-series and cross-sectional momentum addressing these issues. When daily price fluctuations are considered, many momentum portfolios are liquidated, and portfolios with statistically significant mean returns often earn negative profits. The mean return is insufficient for testing profitability in the cryptocurrency market, where returns are heavily skewed and fat-tailed. Evidence of time-series momentum is strong, whereas evidence of cross-sectional momentum is almost non-existent. Evidence suggests that investor overconfidence and the disposition effect are important drivers of cryptocurrencies' price dynamics.

Keywords: Cryptocurrency, Time-series momentum, Cross-sectional momentum, Liquidation risk, Overreaction.

JEL Classification: G10, G11, G12.

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1 Introduction

Momentum in cryptocurrency markets has been extensively examined in the literature. While some scholars document the existence of momentum (Yang, 2019; Liu et al., 2020; Liu and Tsyvinski, 2021; Liu et al., 2022), others find no evidence to support it (Grobys and Sapkota, 2019; Dong et al., 2022). These conflicting findings may be attributable to differences in sample periods, sample selection criteria, look-back and holding periods, or test methods. Given the relatively short history of the cryptocurrency market, small changes in empirical design can significantly impact empirical results.

However, a more critical issue is that these studies overlook important real-world frictions and therefore fail to evaluate momentum strategies rigorously. Most importantly, they underestimate liquidation risk by ignoring interim price fluctuations during portfolio holding. Cryptocurrencies frequently soar and plunge, and a strategy that appears profitable based on weekly or monthly returns could have triggered a margin call or liquidation event mid-period. Liquidation risk is particularly severe when the holding period is long.

To illustrate this point, consider a momentum portfolio of \$1 million. Without loss of generality, assume that it takes a \$1 million long position in one cryptocurrency and a \$1 million short position in another and holds them for one week. If the price of the long asset remains unchanged while the short asset appreciates by 50%, the portfolio incurs a loss of \$500,000 and its value reduces to \$500,000. Consequently, the effective leverage of the short leg increases to 3 ($=1.5/0.5$). If the short asset appreciates by another 34% or more before the next rebalancing day—assuming the long asset remains flat—the total accumulated loss exceeds 100%, resulting in portfolio liquidation.

Even without incurring losses, the portfolio can become subject to higher liquidation risk during the holding period. Assume that both the long and short assets appreciate by 100% on the first day. While the net profit and loss remain zero, the leverage of both legs doubles. Therefore, if the short asset outperforms the long asset by more than 50% before the next rebalancing day, the portfolio is liquidated. If the loss is recovered and falls below 100% by the next rebalancing day,

these liquidation events would not be captured without daily mark-to-market (evaluation).

The above examples demonstrate that fluctuations in asset values can expose the portfolio to excessive leverage during the holding period, thereby creating significant liquidation risk, especially when the holding period is long. This phenomenon is called leverage drift. While leverage drift is unlikely to lead to liquidation in traditional markets, such risks should not be ruled out in cryptocurrency markets, where three-digit returns are not rare and portfolios are more concentrated due to the scarcity of liquid assets. Therefore, it is critical to monitor the portfolio value at least on a daily basis to account for potential liquidation.

Even when the holding period return exceeds -100%, the significance test of the mean return, typically used in the literature, may miss it: If there are 1,000 observations, a single occurrence of a -100% return reduces the mean return by only 0.1%, and it may remain statistically positive, falsely suggesting that the strategy is profitable. Setting aside liquidation events, the mean return is still inadequate for testing the profitability of a cryptocurrency strategy, since its return distribution is significantly fat-tailed and skewed; thus it can earn a negative profit even when the mean return is statistically positive. Other real-world considerations, such as transaction costs, margin mode, and leverage, are also ignored in most studies.

Hence, we aim to comprehensively analyze momentum in the cryptocurrency market accounting for real-world frictions and using up-to-date data, and to provide a more conclusive answer regarding the existence of momentum and its underlying mechanism. We differentiate our study by marking portfolios to market and checking their solvency daily. We find that many momentum portfolios are liquidated when interim price fluctuations are taken into account. We also demonstrate the inadequacy of the mean return as a measure of profitability and propose to use the mean log return instead. In addition, we account for transaction costs, margin modes, and leverage constraints to reflect actual trading environments. To ensure implementability, we include only large and liquid coins in the sample and perform additional tests using the coins listed on the Binance futures market, as coins can be short-sold only in a futures market. We robustify our analysis by evenly

distributing the investment over the holding period.

We find strong evidence of time-series momentum. A strategy that buys the market when its look-back period return falls within the top third of its historical returns outperforms the market across a wide range of look-back and holding periods. The strategy performs best when the look-back period is 28 days and the holding period is five days, yielding a Sharpe ratio of 1.51, compared to 0.85 of the market portfolio. The strategy holds a long position only when the market is bullish and defends well against market downturns. In contrast, a strategy that sells the market when the market falls yields negative profits in most cases, implying that time-series momentum is concentrated in a bullish market. Time-series momentum performs comparably across different coin types, perhaps because coins tend to move in tandem following Bitcoin. The three-factor model of Liu et al. (2022) cannot explain time-series momentum.

Contrary to time-series momentum, evidence of cross-sectional momentum is weak. Among 21 cross-sectional momentum portfolios of selected look-back and holding periods, six are liquidated, and only three outperform the market. The portfolio with a look-back period of 14 days and a holding period of seven days performs best, yielding a Sharpe ratio of 1.28, compared to 1.01 of the market. However, even the best portfolio suffers from liquidations of three of its seven accounts. In fact, 98 of the 138 accounts in our test (71%) are liquidated. A decomposition analysis confirms that the weak performance of cross-sectional momentum, which contradicts earlier studies such as Liu et al. (2022), is mainly attributable to the consideration of liquidation events.

Our empirical analysis also reveals that the mean return is inadequate for testing profitability in the cryptocurrency market. Nine portfolios yield statistically significant mean returns ($t > 2.0$), but only one of them yields a significant mean log return, and seven portfolios with positive mean returns earn negative profits.

In contrast to the equity market, where momentum profit originates mainly from the short leg and small stocks (Hong et al., 2000; Grinblatt and Moskowitz, 2004), it originates mainly from the long leg and large coins in the cryptocurrency market. Except for a few large ones, coins

usually exhibit reversal rather than momentum, and the short leg is exposed to high jump risks and frequently incurs losses. Winners exhibit more predictable patterns—momentum among large coins and reversal among the rest, but the patterns among losers are far less discernible and often opposite to those among winners; when winners show momentum, losers show reversal, and vice versa.

Empirical evidence suggests that investors’ behavioral biases are important drivers of cryptocurrency price dynamics. The cryptocurrency market is characterized by the lack of fundamental value and the dominance of retail investors. Retail investors are more susceptible to behavioral biases, and uncertainty of the fundamental value amplifies them (Odean, 1998b; Barber and Odean, 2000; Kumar, 2009; Baker and Wurgler, 2006). Using the model of Daniel et al. (1998), we show that overreaction induced by overconfidence can cause momentum, but it can also cause short-term reversal when investors are overly overconfident or valuation uncertainty is high.

Time-series momentum appears to be driven by investor overreaction. A factor related to overreaction explains a significant portion of the time-series momentum premium, and the momentum effect is stronger among high-attention/overreaction group coins. Our findings contradict the argument of Liu and Tsyvinski (2021), who find that low-attention coins exhibit stronger time-series momentum and conclude that underreaction is the main driver of time-series momentum. The contradicting results may stem from different samples: They compare only ten well-known coins, whereas we compare all coins.

The reversal effect observed among past winners (excluding large coins) can be explained by investors’ overreaction, followed by profit realization and correction. Past winners becoming losers earn significantly higher look-back period returns, are smaller, and have higher turnover than those remaining winners, implying investors’ overreaction to these coins during the look-back period. Investors use higher leverage when trading smaller coins, exposing themselves to higher margin call risk. Fear of forced liquidation can intensify trading activities and short-term reversal effects among small coins. The fact that many investors are overconfident retail traders (Chhatwani and

Parija, 2023; Kim et al., 2022) also supports the overreaction explanation. The weak momentum effect of past losers can be explained by the disposition effect: Capital gains overhang predicts future returns when it is negative, *i.e.*, when investors are at a loss.

We conjecture that large past winners exhibit momentum because they have lower valuation uncertainty and higher growth potential, thus attracting long-term and institutional investors. Major coins are bigger, older, and less volatile, which are signs of lower valuation uncertainty. These coins have lower turnover and are less subject to margin trading, suggesting that a large portion of their investors are long-term and institutional investors, who are known to be less overconfident. Lower levels of valuation uncertainty and overconfidence can prolong overreaction and cause momentum. Kogan et al. (2024) also find that major cryptocurrency investors follow a momentum-like strategy.

Liu et al. (2022) attribute cross-sectional momentum to attention-based overreaction, based on the stronger momentum effect observed among high-attention coins. However, our findings do not support the attention-based explanation: A momentum portfolio formed of high-attention coins performs comparably to that of low-attention coins; winners, where the momentum effect is concentrated, perform worse when attention is high; and the momentum strategy does not perform better during periods of high attention. A moderate level of attention may induce continued overreaction and momentum, but a significant increase in attention appears to cause an abnormal overreaction that is quickly reversed. The difference between our findings and those of Liu et al. (2022) are due to our exclusion of illiquid coins.

While there is some evidence of momentum, especially in the time series, our results should be interpreted with caution. We test various pairs of look-back and holding periods and choose their optimal combinations, thus introducing a data snooping bias. This empirical design is intentional, as we aim to test the existence of momentum: If momentum remains weak even under these optimized, look-ahead biased conditions, the anomaly is unlikely to exist. Also, due to the high tail risk of the momentum strategy, investors are likely to impose certain stop-loss rules, which can

significantly change its risk-return profile.

Short positions' jumps inflict a significant loss on momentum strategies, while long-only strategies are exposed to the high risk of the cryptocurrency market. A momentum-based long-short strategy that can generate steady, market-neutral profits appears unachievable. The maximum Sharpe ratio achieved by a momentum strategy is about 1.5. Meanwhile, several studies document a Sharpe ratio greater than 2.0 in the equity market: *e.g.*, Gu et al. (2020); Han (2022); Jiang et al. (2023). Considering the high tail risk, the small number of liquid coins, and the high dominance of a few major coins, it is difficult to argue that a cryptocurrency momentum strategy can be an attractive alternative investment vehicle for institutional investors.

We contribute to the cryptocurrency literature by conducting a comprehensive analysis of momentum and unveiling its true nature. We reveal the limitations of prior studies, propose alternative testing methods, and uncover the true risk of a momentum-based strategy. We also present evidence related to the underlying mechanism of momentum, which contradicts previous findings in the literature. By accounting for real-world considerations, we bridge the gap between theory and practice.

The rest of the paper is organized as follows. Section 2 describes the sample data and the methodology employed for the empirical analysis. Sections 3 and 4 respectively conduct various tests on time-series momentum and cross-sectional momentum. Section 5 concludes.

2 Data and methodology

2.1 Data

Our sample includes all available cryptocurrencies in CoinMarketCap. CoinMarketCap collects data from over 200 exchanges and provides daily data on opening, closing, high, and low prices, trading volume, and market capitalization, all in USD. The dataset includes both active and inactive cryptocurrencies and therefore is free of survivorship bias. The trading volume has been available

since December 28, 2013, and our sample starts on this date and ends on August 28, 2023.

The market capitalization data have some negative values without any description, which are treated as missing values. Split events not reflected in the CoinMarketCap data are identified, and affected prices are manually adjusted. Also, 97 stablecoins are excluded from the sample. The list of stablecoins and the manually adjusted events can be found in the Internet Appendix (Table IA1).

While most prior studies employ market capitalization as the sole data filtering criterion, *e.g.*, Liu and Tsyvinski (2021); Liu et al. (2022), some large cryptocurrencies suffer from low liquidity and may not be tradable without significantly impacting the market. Figure IA1 draws a scatter plot of market capitalization against trading volume. The scatter plot of all coins (left graph) suggests a linear relationship between market capitalization and trading volume, with a correlation coefficient of 0.813. However, once we zoom in on the area where most coins lie (right graph), it becomes evident that there are many small coins with a large trading volume and vice versa.

Since some coins have extremely low liquidity relative to their size, we consider three data filtering criteria: 1) market capitalization of 1 million USD; 2) trading volumes of 1 million USD; and 3) a combination of both.¹ The filters are applied to daily data and 30-day moving averages to mitigate the time-series variation of coins in the sample.

Table IA1 (c) reports the number of coins in the sample after applying the filters. Without any filter, the number of coins increases rapidly, especially in recent years, and reaches almost 9,000 in 2023. However, the number of coins with a market capitalization of at least 1 million USD remains stable at around 1,300 over the past three years, implying that most newly minted coins during this period do not meet the minimum market capitalization requirement. When coins are filtered using both market capitalization and trading volume, the number is further reduced by more than half. The sample from the volume filter is similar to the sample from both filters, indicating that trading volume is the binding condition. Consequently, we apply both market capitalization and volume filters to ensure that portfolio strategies are implementable.

¹The trading volume provided by CoinMarketCap is an aggregate value from all exchanges. Therefore, the trading volume of a coin that passes the volume filter can be significantly lower than 1 million USD on each exchange.

Table 1 presents the performance of the market portfolio, the equal-weighted market portfolio, Bitcoin, Ethereum, and NASDAQ 100. The market portfolio is defined as the value-weighted portfolio of all coins in the sample. Cryptocurrencies demonstrate substantial returns, with the market, Bitcoin, and Ethereum yielding annualized mean returns of 62.37%, 63.13%, and 146.05%, respectively. Yet, they also record maximum drawdowns (MDDs) of 89.1%, 83.4%, and 94.0%. While the cryptocurrency market offers potentially high returns, they are inherently paired with substantial risks. As a result, the Sharpe ratios of the market portfolio and Bitcoin, 0.85 and 0.88, respectively, are even lower than that of NASDAQ 100, which is 0.96.

Table 1: Performance of cryptocurrencies

This table reports the performance of the market portfolio (MKT), equal-weighted portfolio (MKT-EW), Bitcoin, Ethereum, and NASDAQ 100. ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). All values are calculated using daily returns during the sample period from December 28, 2013 to August 28, 2023, except for Ethereum, whose sample starts on August 8, 2015.

	Mean	Std	Sharpe	Cum	MDD
MKT	62.37	73.63	0.85	2695.9	89.1
MKT-EW	75.60	88.30	0.86	2970.3	97.0
Bitcoin	63.13	72.05	0.88	3451.5	83.4
Ethereum	146.05	113.22	1.29	59509.8	94.0
NASDAQ 100	25.43	26.43	0.96	325.8	35.6

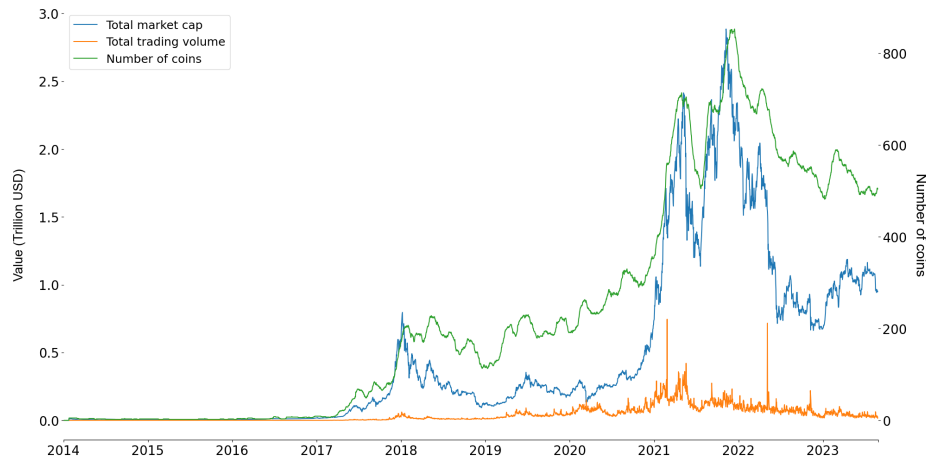


Figure 1: Cryptocurrency market overview

Figure 1 plots the total market capitalization, the total trading volume, and the number of coins in our sample. The number of coins is 5 on the first day of the sample, peaks at 784 in December 2021, and is 433 on the last day. The total market capitalization peaks in early 2022 reaching almost 3 trillion USD, then drops significantly and remains around 0.95 trillion USD. The cryptocurrency market is a highly concentrated market dominated by a couple of coins, particularly Bitcoin and Ethereum. The market dominance of Bitcoin and Ethereum reaches its highest value of 96.2% in 2016 and the lowest value of 48.9% in 2018, and has an average of 79.0% during the sample period.

We verify our data and code by reproducing some tables in Liu and Tsyvinski (2021) and Liu et al. (2022) (Section B of the Internet Appendix). When the same sample period and filtering rules are used, the reproduced tables are comparable to those in these references.

2.2 Methodology

2.2.1 Procedure

We analyze both time-series and cross-sectional momentum in the cryptocurrency market. We first employ regression analyses to examine various pairs of look-back and holding periods ranging from one day to 56 days and identify their optimal combinations. The chosen look-back and holding period pairs are used to construct momentum strategies, which are then evaluated thoroughly. For time-series momentum, strategies trading the market portfolio are tested, and for cross-sectional momentum, coin-level long-short strategies are tested. Finally, various tests are conducted to identify the driver of cryptocurrency price dynamics and to assess the robustness of our findings.²

2.2.2 Portfolio formation

We use value-weighted portfolios for the main analysis and explore alternative portfolio construction methods; volume-weight, equal-weight, capped-value-weight, and capped-volume-weight, as a

²Robustness checks are reported in the Internet Appendix.

robustness check. Due to the significant dominance of major coins, a value-weighted portfolio can become heavily concentrated in a few major coins, potentially leading to findings that do not represent the entire market. Capped-value-weighted portfolios mitigate major coins' dominant effect by capping the weights of the largest 5% coins at the 5% value. Large coins with low trading volume may not be viable investments, especially when the weight is proportional to the size. Therefore, volume-weighted and capped-volume-weighted portfolios are also considered. The capped-volume-weighted portfolio is similarly defined to the capped-value-weighted portfolio.

2.2.3 Marking to market

Portfolios are marked to market (evaluated) daily regardless of their holding periods. Cryptocurrencies frequently soar and plunge within a short interval, and ignoring interim price fluctuations during the holding period underestimates short-term volatility and liquidation risk. As an example, consider the dramatic rebound of Terra Luna. After plummeting for nine consecutive days culminating in a total loss of 99.99%, it surged by 349.75% on May 14, 2022. Such an extreme turnaround can inflict considerable losses on short positions, potentially leading to portfolio liquidation. Nevertheless, the weekly return of Luna for that week is still a drop of 99.99%. The magnitude of the decline overshadows the rebound, and the risk of liquidation would not be captured if daily fluctuations were ignored. Similarly, on June 7, 2022, Unifi Protocol DAO (UNFI) surged over 1,000% and closed the day at 432.78%, only to sharply decline by 55.44% the next day. A short position of UNFI on July 7 could have liquidated the entire portfolio. Yet, its return of the week is a rather modest gain of 210.85%, which masks the true level of risk.

In our sample, there are three instances of five consecutive days of price increase, followed by a drop of 50% or more on the subsequent day, and 136 instances of five consecutive days of price decline, followed by a hike of 50% or more on the next day. These observations highlight the extreme volatility of the cryptocurrency market and the importance of daily mark-to-market to adequately assess the profitability of momentum strategies.

Portfolios are marked to market using daily close prices, which can be significantly different from daily highs and lows. Thus, despite being more accurate than using holding period returns, our approach still underestimates liquidation risk. An important implication of such extreme price movements is that most investors would not be able to tolerate huge short-term losses and are likely to impose certain types of stop-loss rules, which can change the characteristics of a momentum strategy in a nontrivial fashion. Therefore, even if a naïve test of a momentum strategy indicates a significant profit, the actual profitability cannot be guaranteed, and the results should be interpreted with caution.

Portfolios are marked to market daily as follows. Let V_τ denote the dollar value of the portfolio on the last rebalancing date τ . For any date t ($> \tau$) before next rebalancing, the portfolio value V_t is defined as the sum of its initial value and the accumulated unrealized profit and loss from both long and short positions Π_t^L and Π_t^S :

$$V_t = V_\tau + \Pi_t^L + \Pi_t^S, \quad (1)$$

$$\Pi_t^L = \sum_{i \in \mathcal{L}} n_{i,\tau} \cdot (P_{i,t} - P_{i,\tau}), \quad (2)$$

$$\Pi_t^S = \sum_{j \in \mathcal{S}} n_{j,\tau} \cdot (P_{j,t} - P_{j,\tau}), \quad (3)$$

where $n_{i,\tau}$ is the number of coins held for coin i at time τ , $P_{i,t}$ is its price at time t , and $\mathcal{L}$ and $\mathcal{S}$ denote long and short legs, respectively. If $V_t \leq 0$, the portfolio is deemed liquidated. The portfolio return on date t is defined as $V_t/V_{t-1} - 1$. Note that a simple return-based formula—the weighted sum of individual coin returns—cannot be used because the weights change during the holding period.

2.2.4 Transaction costs

An important aspect of real-world investment is transaction costs. We assume a transaction cost of 15 basis points (bps) for every trade for the following reasons. At the time of writing this article,

Binance, the leading cryptocurrency exchange, charges a transaction fee of 4.5 bps to regular users in the futures market, and the average tick size relative to the price is 3.26 bps.³ The minimum, maximum, and average slippages per coin calculated from 15,661,698 actual market orders in the Binance futures market from June 24, 2023 to August 20, 2023 are 0.01, 11.81, and 1.53 bps, respectively.⁴ The average order size is 56 USD, only 0.02% of the daily trading volume. Larger orders would have a greater impact on the market, resulting in larger slippage. Using the average slippage, the average transaction cost for a futures contract is estimated to be 9.29 bps. Binance offers one of the lowest fees in the market, and the fees have decreased over time. In light of these circumstances, we consider a transaction cost of 15 bps a reasonable estimate (perhaps closer to the lower limit) of the actual transaction costs and apply it when evaluating portfolios.

2.2.5 Distribution of investment

Given the relatively short sample period, the start date of a backtest can have a nontrivial impact on the empirical results. As shown later, the performance of a strategy varies significantly depending on the rebalancing day, *e.g.*, every Monday or every Sunday. To mitigate the impact of the rebalancing day, we conduct our empirical analysis in the following manner. For a time-series momentum strategy, we run independent tests starting on each day of the first holding period and report the average performance. For instance, if a strategy’s holding period is three days, we test the strategy three times starting on the first three days of the sample period. For a cross-sectional momentum strategy, we evenly allocate the initial wealth over the days in the first holding period and treat the investments as if they are in separate accounts, *i.e.*, a profit from one account is not transferred to another, and if an account’s loss exceeds 100%, it is liquidated and the loss is not propagated to the other accounts. These methods neutralize the rebalancing-day effect and render more robust findings.

³Binance runs a VIP program and an investor with a large trading volume can get discounts on the fees. The fees applied to a regular user are the upper limits.

⁴The slippages of all coins are reported in Table IA2.

2.2.6 Leverage and margin mode

In the cryptocurrency market, short positions are typically taken via perpetual futures (swaps). Perpetual futures are futures contracts with indefinite maturity, and their prices are synced with the spot prices through a funding fee mechanism.⁵ Cryptocurrency exchanges demand a margin for both long and short positions, and opening a long position for 100 USD and a short position for 100 USD requires a margin of 200 USD without leverage. Hence, we assume a leverage of 2 so that a long-short strategy can invest 100% of the wealth in each leg. A consequence of such an assumption is that the maximum amount a strategy can invest is limited to a rather small value. This is because exchanges limit the maximum amount a leveraged position can hold, and the limit is stricter for smaller, less liquid coins. Thus, we also assess strategies assuming no leverage and investment of half of the wealth in each leg. Under this assumption, a strategy sacrifices potential profits but also has a lower chance of liquidation.

Another aspect to consider is when a portfolio is deemed liquidated. Cryptocurrency exchanges offer two types of margin mode in the futures market. Under the cross-margin mode, the entire account balance is utilized to margin all open positions, and gains from profitable positions counter-balance losses from the others. The margin level is determined by the combined value of the assets and obligations within the cross-margin account, and in the event of liquidation, every position in the account is impacted. Under the isolated-margin mode, a specific collateral amount backs each trade and every position functions independently. Thus, if the margin level of a position drops below the maintenance margin, the position is liquidated without affecting the other positions.

Liquidation occurs more frequently under the isolated-margin mode, but the loss is limited to the value of the liquidated position. Liquidation is less likely under the cross-margin mode, but if it happens, it can wipe out the entire portfolio value. Investors prefer the cross-margin mode for its mutual support. For the primary analysis, we adopt the cross-margin mode with 0 maintenance margin. In reality, the maintenance margin is greater than 0 and there is a higher chance of

⁵If the futures price is higher than the spot price, long positions pay a fee to short positions, and vice versa.

liquidation. Later, we also explore the isolated-margin mode as a robustness check.

3 Time-series momentum

3.1 Regression analysis

To develop a time-series momentum strategy, we first conduct regression analyses using different pairs of look-back and holding period returns of the market. The regression equation has the form

$$p_{t,t+k} = \alpha + \beta p_{t-j,t} + \epsilon_t, \quad (4)$$

where $p_{t,t+k}$ denotes the time-series percentile rank of the return over t to $t+k$, $r_{t,t+k}$, and ϵ_t is an error term. The time-series percentile rank $p_{t,t+k}$ is determined by comparing $r_{t,t+k}$ to all previous k -day returns. We use percentile rank instead of raw return because it mitigates the impact of extreme returns on the regression. The percentile rank also provides information on the market status, with a high rank indicating a bullish market and a low rank a bearish market, thereby allowing us to analyze the dependency of time-series momentum on the market condition.

Table 2 reports the regression results, where the rows and columns respectively represent the look-back and the holding periods. The values in the table are the estimates of β and their t -statistics. Notably, all the coefficients are positive except for the (1, 1) pair, providing strong evidence of time-series momentum.⁶ Nonetheless, not all coefficients are statistically significant. For a given look-back period, the magnitude of the coefficient and its statistical significance tend to increase and then decrease as the holding period increases. When the look-back period is longer, the most significant coefficient is observed at a shorter holding period. Similarly, for a given holding period, the coefficient tends to increase and then decrease, and the most significant value is found at a shorter look-back period when the holding period is longer. These results suggest that the price moves in the same direction for a certain period before it reverses. From the table, the trend

⁶We henceforth use the notation (j, k) to denote a pair of j -day look-back period and k -day holding period.

appears to last for about 30 to 40 days. The coefficient has the highest t -statistic of 4.08 when $j = 28$ and $k = 1$. The coefficient loses statistical significance when both the look-back and the holding periods extend beyond 28 days.

To develop time-series momentum strategies, we choose look-back and holding period pairs whose coefficients are large and statistically significant. Based on this criterion, look-back periods from 7 to 28 days and holding periods from 1 to 14 days are selected.

Table 2: Regression of time-series momentum

This table reports the results of the time-series momentum regression defined in Equation (4). The dependent variable is the time-series percentile rank of the holding period return, and the independent variable is the time-series percentile rank of the look-back period return. The figures are the estimates of β and Newey-West adjusted t -statistics.

Look-back (j)	Holding period (k)										
	1	3	5	7	14	21	28	35	42	49	56
1	-0.032 (-1.79)	0.026 (1.43)	0.020 (1.06)	0.036 (1.93)	0.045 (2.29)	0.041 (1.93)	0.054 (2.63)	0.052 (2.46)	0.041 (1.93)	0.048 (2.23)	0.047 (2.25)
3	0.026 (1.43)	0.055 (2.37)	0.067 (2.60)	0.063 (2.38)	0.077 (2.54)	0.068 (2.13)	0.083 (2.70)	0.084 (2.67)	0.066 (1.97)	0.070 (2.13)	0.065 (2.00)
5	0.023 (1.26)	0.068 (2.67)	0.070 (2.51)	0.069 (2.31)	0.092 (2.54)	0.075 (1.95)	0.096 (2.57)	0.090 (2.36)	0.075 (1.84)	0.079 (1.96)	0.070 (1.72)
7	0.049 (2.73)	0.075 (2.81)	0.078 (2.57)	0.074 (2.31)	0.108 (2.67)	0.091 (2.11)	0.110 (2.61)	0.099 (2.23)	0.086 (1.82)	0.087 (1.84)	0.078 (1.66)
14	0.061 (3.21)	0.099 (3.33)	0.107 (3.01)	0.117 (3.02)	0.127 (2.76)	0.132 (2.71)	0.134 (2.57)	0.119 (2.08)	0.103 (1.68)	0.100 (1.6)	0.091 (1.45)
21	0.065 (3.45)	0.089 (3.04)	0.092 (2.63)	0.102 (2.71)	0.126 (2.69)	0.123 (2.40)	0.124 (2.15)	0.109 (1.67)	0.095 (1.36)	0.090 (1.27)	0.090 (1.27)
28	0.075 (4.08)	0.101 (3.57)	0.105 (3.20)	0.120 (3.28)	0.134 (2.69)	0.132 (2.30)	0.127 (2.00)	0.111 (1.54)	0.099 (1.32)	0.100 (1.31)	0.108 (1.42)
35	0.074 (3.83)	0.101 (3.33)	0.102 (2.82)	0.112 (2.71)	0.125 (2.23)	0.119 (1.88)	0.117 (1.65)	0.103 (1.38)	0.098 (1.25)	0.106 (1.35)	0.116 (1.47)
42	0.069 (3.38)	0.091 (2.80)	0.084 (2.13)	0.091 (2.03)	0.103 (1.73)	0.097 (1.43)	0.095 (1.26)	0.097 (1.23)	0.100 (1.27)	0.107 (1.35)	0.116 (1.45)
49	0.065 (2.95)	0.083 (2.43)	0.077 (1.87)	0.086 (1.83)	0.092 (1.49)	0.082 (1.14)	0.092 (1.19)	0.106 (1.32)	0.108 (1.34)	0.111 (1.39)	0.124 (1.53)
56	0.067 (3.06)	0.084 (2.37)	0.073 (1.68)	0.079 (1.63)	0.085 (1.36)	0.091 (1.28)	0.111 (1.43)	0.125 (1.55)	0.130 (1.60)	0.140 (1.72)	0.150 (1.83)

3.2 Portfolio analysis

3.2.1 Time-series momentum portfolios

To construct a time-series momentum strategy, we categorize the past returns of the market portfolio into terciles. When the look-back period return falls within the top third of the historical

returns, the strategy takes a long position on the market portfolio. Conversely, if the return falls within the bottom third, it takes a short position on the market portfolio. Otherwise, it clears all positions and holds cash. Note that this strategy holds either a long position or a short position on a given day and is different from the cross-sectional momentum strategy that simultaneously holds positions on both sides. Table 3 reports the performance of the long-only, short-only, and long-short portfolios with different look-back and holding periods.⁷

Remarkably, most long-only portfolios exhibit superior performance compared to the market portfolio in terms of the Sharpe ratio and the cumulative return. Although these portfolios often earn lower mean returns, their Sharpe ratios and cumulative returns are higher owing to their lower standard deviations. The (28, 5) portfolio yields the highest annualized mean return of 72.85%, Sharpe ratio of 1.51, and cumulative return of 36,686%, which are significantly higher than those of the market, 62.37%, 0.85, and 2,696%. It holds a position for 48% of the sample period. Transaction costs impact performance more when the holding period is shorter due to more frequent rebalancing. Still, the impact is not severe since the strategy takes positions only when the momentum condition is met.

Contrary to the long-only portfolios, the short-only portfolios yield unfavorable outcomes, with all portfolios incurring losses at the end of the sample period. It appears that time-series momentum is almost non-existent when the market is bearish. Consequently, the long-short portfolios underperform their long-only counterparts.

Since short positions do not improve the strategies, we focus on the best-performing long-only portfolio, (28, 5), for the detailed analysis below. Figure 2 displays the log-scale cumulative returns of the (28, 5) portfolio and the market. The strategy defends well against market downturns. With its nature of buying the market only when the market is in an upward trend, it underperforms the market when the market is bullish, but it successfully times a bearish market and avoids large drawdowns. The year-by-year performance reported in Table IA9 confirms that the superior

⁷Performance before transaction costs can be found in Table IA8.

performance of the (28, 5) strategy is not due to a few fortunate years. It has a higher Sharpe ratio than the market in eight of ten years and a lower standard deviation and MDD in all years.

Overall, the results suggest that there is a serial correlation in the cryptocurrency market, and a time-series momentum strategy with carefully chosen look-back and holding periods can earn significant profits. Nevertheless, as a naked strategy, it is exposed to the high risk of the cryptocurrency market and can generate profits in the future only if the market continues to grow.

Table 3: Performance of time-series momentum portfolios

This table reports the performance of time-series momentum portfolios of various look-back and holding periods, (j, k) . ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 26, 2014 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(7, 1)	50.39	43.73	1.15	5096.1	56.2	-20.66	47.89	-0.43	-95.5	96.7	31.88	64.80	0.49	190.2	80.9
(7, 3)	52.51	48.27	1.09	5108.0	63.2	-12.01	51.01	-0.24	-91.0	95.3	48.89	68.46	0.71	1092.7	73.3
(7, 5)	50.22	52.81	0.95	3144.8	77.6	-19.07	54.54	-0.35	-96.2	97.6	26.08	70.29	0.37	9.9	94.5
(7, 7)	45.12	54.81	0.82	1675.0	81.7	-14.85	53.78	-0.28	-94.0	96.7	21.61	70.28	0.31	-28.5	91.6
(7, 14)	66.56	60.86	1.09	9989.0	79.9	-31.31	61.19	-0.51	-99.2	99.6	53.65	72.86	0.74	1241.0	94.3
(14, 1)	51.98	45.48	1.14	5463.8	54.1	-5.25	48.77	-0.11	-80.7	86.9	47.73	66.43	0.72	1100.3	60.4
(14, 3)	57.12	47.98	1.19	8065.2	54.9	-6.09	49.36	-0.12	-82.7	88.3	53.12	67.97	0.78	1738.3	71.5
(14, 5)	60.98	49.36	1.24	11009.8	58.2	-4.49	50.09	-0.09	-80.8	86.9	59.66	68.34	0.87	3252.3	61.0
(14, 7)	64.11	50.62	1.27	14025.6	56.6	-10.60	52.34	-0.20	-90.4	95.0	67.35	69.10	0.97	6654.9	72.6
(14, 14)	50.61	56.25	0.90	2742.5	73.3	-1.96	30.90	-0.06	-47.5	74.3	27.50	71.44	0.38	17.8	96.2
(21, 1)	54.90	45.28	1.21	7343.6	60.4	-2.61	48.11	-0.05	-74.5	85.3	52.49	65.46	0.80	1911.2	63.7
(21, 3)	56.23	47.78	1.18	7470.8	60.1	1.78	48.74	0.04	-62.1	81.6	59.83	67.31	0.89	3546.5	65.4
(21, 5)	46.72	49.54	0.94	2648.3	69.0	7.10	49.53	0.14	-38.8	73.6	45.97	68.36	0.67	779.4	80.4
(21, 7)	48.13	51.57	0.93	2756.5	71.8	10.37	49.10	0.21	-14.8	73.8	47.29	69.18	0.68	840.7	81.8
(21, 14)	56.84	54.75	1.04	5504.3	77.7	-6.82	53.62	-0.13	-87.2	95.2	60.24	71.10	0.85	2790.4	88.9
(28, 1)	65.04	45.00	1.45	19991.6	63.5	-8.78	47.19	-0.19	-85.4	90.9	56.63	64.72	0.88	3031.6	64.5
(28, 3)	65.39	47.17	1.39	18782.8	61.9	-3.66	47.68	-0.08	-76.6	87.7	61.40	66.19	0.93	4438.9	65.0
(28, 5)	72.85	48.34	1.51	36685.8	61.8	-2.44	48.49	-0.05	-74.8	88.9	72.17	67.27	1.07	11780.6	60.2
(28, 7)	71.71	48.28	1.49	32921.4	62.4	-0.02	48.95	0.00	-68.6	82.1	73.50	67.55	1.09	13227.4	62.4
(28, 14)	64.60	54.12	1.19	12206.5	71.9	2.36	52.66	0.04	-66.7	87.6	64.88	70.16	0.92	4790.0	82.6
Market											62.37	73.63	0.85	2695.9	89.1

3.2.2 Factor regression

Liu et al. (2022) propose a cryptocurrency three-factor model, which consists of the market, size, and momentum factors. This section examines whether these factors can explain the time-series momentum premium. We construct the factors following Liu et al. (2022), as summarized below.

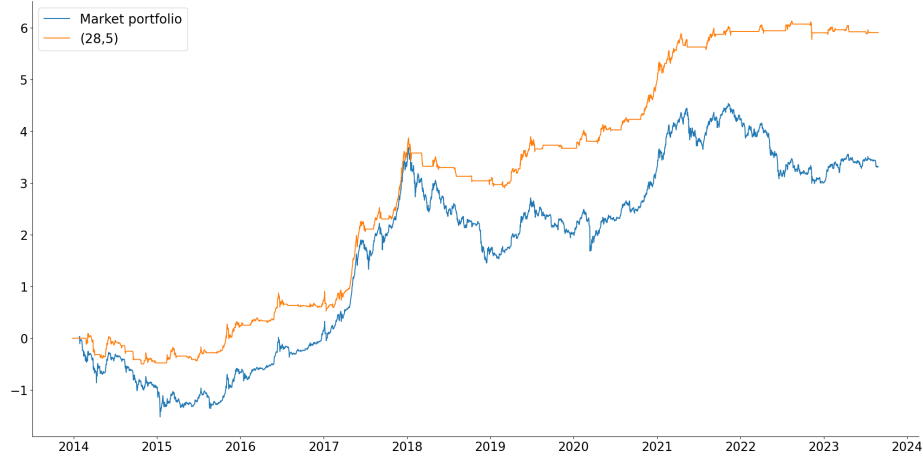


Figure 2: Time-series momentum portfolio log-scale cumulative returns

The market factor is defined as the excess return of the value-weighted market portfolio. The three-month U.S. treasury bill rate is used as a proxy for the risk-free rate. For the size factor, coins are split into three groups on market capitalization; Small (bottom 30%), Middle (middle 40%), and Big (top 30%), and a value-weighted portfolio is formed within each size group. The size factor is the return of the Small-minus-Big portfolio. For the momentum factor, coins are divided into two size groups and further into three momentum groups within each size group: Low (bottom 30%), Middle (middle 40%), and High (top 30%), based on their past three-week returns. The momentum factor portfolio is constructed as

$$MOM = 1/2(Small\ High + Big\ High) - 1/2(Small\ Low + Big\ Low). \quad (5)$$

In addition to the three factors, we also consider two factors related to overreaction. Daniel et al. (1998) propose that continuing overreaction induced by investors' overconfidence and biased self-attribution can cause momentum. Motivated by their theoretical work, Byun et al. (2016) develop a measure of continuing overreaction and empirically show that it can predict future returns better

than past returns. The continuing overreaction measure (CO) is defined as follows:

$$CO_{i,t} = \frac{\sum_{j=1}^J (J - j + 1) \cdot \text{sign}(r_{i,t-j}) \cdot V_{i,t-j}}{\sum_{j=1}^J V_{i,t-j} / J}, \quad (6)$$

where $\text{sign}(r_{i,t})$ is the sign of the return of coin i on day t , and $V_{i,t}$ is its trading volume on the same day. Using the trading volume as a proxy for the degree of investor overconfidence and overreaction (Odean, 1998b; Gervais and Odean, 2001), the authors measure the direction and magnitude of investor overconfidence with the signed volume, $\text{sign}(r_{i,t}) \cdot V_{i,t}$. By assigning more weights to more recent observations, CO aims to capture the trend of overconfidence. A high (low) value of CO implies increasing investor overconfidence in their positive (negative) signal.

We derive two overreaction factors from the continuing overreaction measure, a cross-sectional overreaction factor (CS-CO) and a time-series overreaction factor (TS-CO). The cross-sectional overreaction factor is constructed following the method for the momentum factor, *i.e.*, coins are divided into two size groups and further into three groups based on the continuing overreaction measure, and a long-short portfolio that goes long on high CO coins and short on low CO coins is formed. The time-series overreaction factor is the return of a portfolio that goes long on the market when its CO measure is positive. The TS-CO factor portfolio is designed to take only long positions, as the time-series momentum portfolio takes only long positions. The period J is set to three weeks to be consistent with the momentum factor.

Table 4 reports the factor regression results of the (28, 5) long-only portfolio. When the portfolio return is regressed on individual factors, the time-series overreaction factor is the most significant with a t -statistic of 18.81. The TS-CO factor also has the most explanatory power (adjusted $R^2 = 0.670$) and is the only factor that results in a nonsignificant alpha. The other significant factor is the market factor (t -statistic = 9.33). The momentum factor (MOM) is derived from cross-sectional momentum, and the fact that its coefficient is not significant implies that time-series momentum is different from cross-sectional momentum. The Liu et al. (2022) three-factor model cannot explain the return of the time-series momentum portfolio, as shown in regression (6). When the return

is regressed on all four factors (regression (7)), only the TS-CO factor remains significant, and the alpha becomes nonsignificant. The explanatory power of the TS-CO factor aligns with the model of Daniel et al. (1998) and suggests that overreaction is likely the main driver of time-series momentum.

Table 4: Time-series momentum portfolio factor regression

This table reports the factor regression results of the (28, 5) time-series momentum long-only portfolio. The factors are market (MKT), size (SIZE), momentum (MOM), and time-series continuing overreaction (TS-CO). The definitions of the factors can be found in Section 3.2.2. The sample period is January 26, 2014 to August 28, 2023. The t -statistics are Newey-West adjusted t -statistics.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Const	0.0055 (3.58)	0.0092 (3.72)	0.0081 (3.47)	0.0019 (1.50)	0.0055 (3.59)	0.0049 (3.32)	0.0017 (1.429)
MKT	0.4539 (9.33)				0.4547 (9.44)	0.4445 (9.25)	0.0634 (1.86)
SIZE		0.0394 (0.89)			-0.0079 (-0.29)	-0.288 (-1.06)	-0.0208 (-1.26)
MOM			0.0971 (1.76)			0.0535 (1.64)	0.0350 (1.65)
TS-CO				0.7803 (18.81)			0.6955 (12.39)
Adj R^2	0.459	0.003	0.044	0.670	0.459	0.471	0.678

3.2.3 Market dependency of time-series momentum

Prior studies document that time-series momentum depends on the market status. Moskowitz et al. (2012) find that a time-series momentum portfolio is more profitable when the market is volatile. Lim et al. (2018) observe that time-series momentum performs exceptionally well during an extreme bear market, but exhibits weak performance during an extreme bull market. On the contrary, Liu and Tsyvinski (2021) find that cryptocurrency time-series momentum is stronger when the market is bullish. The superior performance of the long-only portfolios in Table 3 also supports this view.

To investigate time-series momentum’s market dependency more in detail, we divide look-back and holding period returns into quintiles (Q1 to Q5 in ascending order) and calculate transition probabilities between them.⁸ The heatmaps in Figure 3 present the results for different look-back

⁸The breakpoints are determined using all historical returns on each day.

and holding periods. In the figure, $P(Q5 \rightarrow Q4, Q5)$ denotes the transition probability from Q5 to Q4 or Q5. $P(Q1 \rightarrow Q1, Q2)$, $P(Q5 \rightarrow Q1, Q2)$, and $P(Q1 \rightarrow Q4, Q5)$ are similarly defined. If momentum prevails, $P(Q5 \rightarrow Q4, Q5)$ and $P(Q1 \rightarrow Q1, Q2)$ would be high, whereas if reversal prevails, $P(Q5 \rightarrow Q1, Q2)$ and $P(Q1 \rightarrow Q4, Q5)$ would be high. If the transition occurs randomly, $P(Q_i \rightarrow Q_j, Q_k)$ would be 40% for all i , j , and k .

The probability $P(Q5 \rightarrow Q4, Q5)$ is higher than 40%, whereas $P(Q5 \rightarrow Q1, Q2)$ is lower than 40% for most periods, suggesting a strong time-series momentum effect in a bullish market. The momentum effect is particularly strong when the look-back period is between 7 and 28 days and the holding period is between 14 and 28 days. In contrast, when the look-back period return is in the lowest quintile (Q1), momentum is not observable. Rather, the figure indicates weak reversal in some periods. Specifically, for the (28, 5) strategy, $P(Q5 \rightarrow Q4, Q5)$ is 52.32% and $P(Q5 \rightarrow Q1, Q2)$ is 32.52%, whereas $P(Q1 \rightarrow Q1, Q2)$ is 40.03% and $P(Q1 \rightarrow Q4, Q5)$ is 46.49%, implying momentum in a bullish market and reversal in a bearish market.

As a robustness check, we perform a regression analysis in different market states. The market state is defined as good, normal, or bad if the look-back period return is in the top, middle, or bottom third, and the percentile rank regression in Equation (4) is conducted in each market state. Table IA10 reports the results in good and bad states. The results show that when the market is in a good state, the coefficient is significant for the majority of the look-back and holding periods, with t -statistics higher than those reported in Table 2. On the contrary, when the market is in a bad state, the coefficient becomes nonsignificant for all the periods. This result reaffirms that time-series momentum is more pronounced in a bullish market.

3.2.4 Time-series momentum in different coin groups

This section examines whether time-series momentum is concentrated in a specific coin type. Coins are divided into five groups based on size, Google trend, trading volume, relative trading volume (volume/size), trading volume change, and continuing overreaction, and a value-weighted portfolio

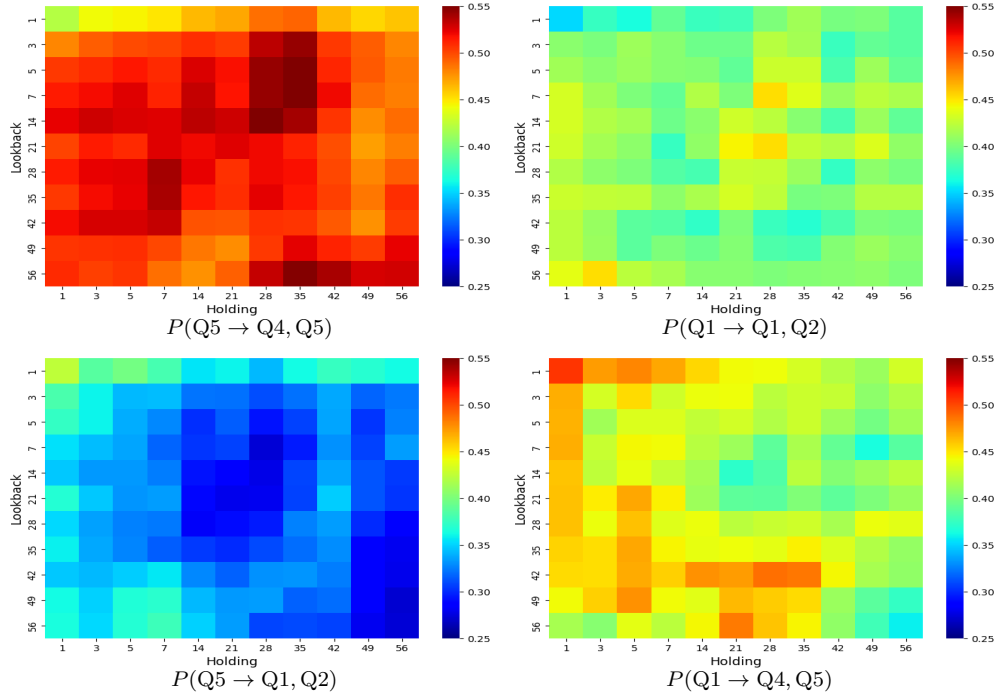


Figure 3: Transition probabilities between look-back and holding period returns of the market portfolio

is constructed within each group.⁹ Trading volume change intends to measure the temporal change in investors' attention. Following Liu and Tsyvinski (2021), we regress the portfolio's holding period (5 days) return on its look-back period (28 days) return in each group. Since there are only a handful of coins before 2017, the sample from the beginning of 2017 is used. Table 5 reports the regression results.

Overall, the coefficients do not exhibit distinct patterns across coin groups. Due to the lack of fundamental value, coins tend to move in tandem following the movement of Bitcoin and are highly correlated. As the performance of a time-series momentum strategy is determined by the aggregate performance of the coins in the portfolio, not by their relative performance, it does not vary significantly across coin types. That being said, the coefficient tends to be more significant

⁹The trading volume is defined as the average daily trading volume over the look-back period, and the trading volume change is defined as $(V_e - V_b)/V_b$, where V_b (V_e) is the daily trading volume on the first (last) day of the look-back period. Details of the Google trend measure can be found in the Internet Appendix.

in high-attention groups than in low-attention groups, supporting the overreaction explanation of time-series momentum.

Table 5: Regression of time-series momentum in different coin groups

This table reports the results of the regression of the holding period return on the look-back period return in different coin groups. The look-back and holding periods are 28 days and 5 days, respectively. Columns M, G, V, VM, ΔV , and CO respectively denote size, Google trend, trading volume, relative trading volume, trading volume change, and continuing overreaction groups. The figures are estimates of the coefficient on the look-back period return and their Newey-West adjusted t -statistics. The sample period is January 1, 2017 to August 28, 2023.

	M	G	V	VM	ΔV	CO
Low	0.006 (0.62)	0.042 (1.84)	0.003 (0.43)	0.050 (1.83)	0.003 (0.55)	-0.001 (-0.38)
2	0.076 (3.84)	0.056 (2.68)	0.001 (0.23)	0.055 (3.47)	0.023 (1.57)	0.041 (2.87)
3	0.074 (3.37)	0.078 (3.59)	0.061 (3.24)	0.056 (3.01)	0.043 (2.65)	0.048 (2.30)
4	0.058 (3.92)	-0.002 (-1.68)	0.061 (2.79)	0.062 (3.33)	0.058 (3.30)	0.041 (4.06)
High	0.046 (2.57)	0.049 (3.10)	0.045 (2.55)	0.024 (2.03)	0.068 (2.69)	0.007 (0.61)

3.3 Summary of findings and discussion

Empirical evidence suggests that time-series momentum is widespread across diverse look-back and holding periods when the market is trending upward. The trend typically persists for about 30 to 40 days. In contrast, the momentum effect is not discernible in a bearish market, and a long-short portfolio performs worse than a long-only portfolio with identical look-back and holding periods. Time-series momentum performs comparably across different size, Google trend, volume, and overreaction groups.

Liu and Tsyvinski (2021) attribute the time-series momentum effect to underreaction, based on the observation that low-attention (measured by Google trend) coins exhibit stronger time-series momentum. Our findings, however, do not support their argument. Time-series momentum does not show any distinct pattern across the Google trend and volume groups, and the coefficient is more significant in the highest groups. The results from other attention-related groups are similar and more in line with an overreaction mechanism. The result from continuing overreaction groups

also supports overreaction rather than underreaction. The TS-CO factor’s explanatory power documented in Section 3.2.2 is also more in line with an overreaction mechanism. Our findings differ from those of Liu and Tsyvinski (2021) perhaps because they compare only ten major coins, whereas we compare all the coins in our sample.

4 Cross-sectional momentum

This section examines cross-sectional momentum in the cryptocurrency market. As before, we begin with a regression analysis to identify look-back and holding period pairs that are likely to render profitable momentum strategies and conduct a comprehensive analysis of those strategies. As there are only a handful of coins (fewer than 10) before 2017, we test cross-sectional momentum using the sample from the beginning of 2017.

4.1 Regression analysis

For each pair of look-back and holding periods, we compute the returns of all coins in the sample and assign cross-sectional percentile ranks. We then run a pooled regression of the form

$$k_{t,t+k}^i = \alpha + \beta k_{t-j,t}^i + e_t^i, \quad (7)$$

where $k_{t,t+k}^i$ denotes the cross-sectional percentile rank of coin i ’s return over the period from t to $t + k$. Panel (a) of Table 6 reports the regression results.

Contrary to the results from the time-series regression in Section 3.1, the majority of the coefficients are negative, suggesting reversal rather than momentum. The reversal pattern is particularly evident when both the look-back and holding periods are short; *e.g.*, the coefficient and its t -statistic have their minimums of -0.055 and -41.25 when $j = 3$ and $k = 1$. The reversal effect diminishes as the holding period extends, but most coefficients remain negative.

While the result largely suggests reversal, there are a few pairs with positive coefficients. Specif-

ically, the one-day look-back period paired with a holding period of 28 days or longer presents a positive coefficient that is statistically significant at the 1% level. However, the magnitude is not economically significant, being only 0.007 or smaller. Except for the above case, none of the positive coefficients are statistically significant.

Liu et al. (2022) observe that cross-sectional momentum works better among bigger coins. Following their observation, we repeat the regression analysis using the top 5% of coins ranked by market capitalization and report the results in Panel (b) of Table 6. Contrary to the previous result from all coins, many coefficients turn positive. The momentum effect is generally stronger when the sum of the look-back and holding periods is between 20 and 40 days. As both periods extend, the momentum effect diminishes and a negative coefficient appears more frequently. Negative coefficients prevail when the look-back and holding periods are longer than a month, suggesting long-term reversal among large coins.

Given the results above, we examine both momentum and reversal strategies. Since portfolios are value-weighted, look-back and holding period pairs signaling momentum among the top 5% are selected. Specifically, for each look-back period up to 28 days, three holding periods with the highest t -statistics in Table 6 (b) are chosen. These are (1, 14), (1, 21), (1, 28), (3, 14), (3, 21), (3, 28), (5, 14), (5, 21), (5, 28), (7, 14), (7, 21), (7, 28), (14, 7), (14, 14), (14, 21), (21, 5), (21, 7), (21, 14), (28, 3), (28, 5), and (28, 7). For the reversal strategy, three zones—short-term reversal, long-term look-back and short-term holding, and long-term reversal—are identified, and 12 pairs are chosen in these zones: (1, 1), (1, 3), (3, 1), (3, 3), (42, 1), (42, 3), (49, 1), (49, 3), (49, 42), (49, 49), (56, 42), and (56, 49).

4.2 Portfolio analysis

4.2.1 Cross-sectional momentum portfolios

This section constructs cross-sectional momentum portfolios using the look-back and holding periods selected in the previous section and evaluates them. Coins are sorted into quintiles based

Table 6: Regression of cross-sectional momentum

This table reports the results of the cross-sectional momentum regression defined in Equation (7). The dependent variable is the cross-sectional percentile rank of the holding period return, and the independent variable is the cross-sectional percentile rank of the look-back period return. The figures are the estimates of β and Newey-West adjusted t -statistics.

Look-back (j)	Holding period (k)										
	1	3	5	7	14	21	28	35	42	49	56
(a) All coins											
1	-0.053 (-40.11)	-0.033 (-25.30)	-0.027 (-20.57)	-0.015 (-11.11)	-0.004 (-3.10)	0.001 (0.53)	0.005 (3.13)	0.007 (4.38)	0.005 (3.08)	0.007 (4.16)	0.006 (3.95)
3	-0.055 (-41.25)	-0.042 (-25.50)	-0.029 (-16.03)	-0.023 (-11.84)	-0.007 (-3.46)	-0.004 (-1.75)	0.003 (1.37)	0.003 (1.52)	0.000 (0.19)	0.002 (0.69)	0.001 (0.28)
5	-0.054 (-39.92)	-0.036 (-19.75)	-0.029 (-14.45)	-0.025 (-11.18)	-0.010 (-4.01)	-0.006 (-2.34)	0.002 (0.65)	0.000 (0.06)	-0.003 (-1.01)	-0.003 (-0.88)	-0.004 (-1.26)
7	-0.044 (-32.43)	-0.034 (-17.62)	-0.029 (-13.01)	-0.025 (-10.45)	-0.012 (-4.22)	-0.008 (-2.70)	-0.008 (-0.10)	-0.004 (-1.19)	-0.006 (-1.79)	-0.006 (-1.82)	-0.008 (-2.39)
14	-0.038 (-27.05)	-0.028 (-13.32)	-0.025 (-9.73)	-0.022 (-7.78)	-0.015 (-4.38)	-0.008 (-2.11)	-0.008 (-1.94)	-0.012 (-2.85)	-0.015 (-3.45)	-0.018 (-3.99)	-0.021 (-4.59)
21	-0.035 (-23.86)	-0.027 (-12.19)	-0.024 (-8.90)	-0.022 (-7.38)	-0.013 (-3.33)	-0.013 (-3.11)	-0.015 (-3.33)	-0.020 (-4.22)	-0.025 (-4.90)	-0.028 (-5.38)	-0.028 (-5.23)
28	-0.032 (-21.39)	-0.023 (-10.45)	-0.019 (-6.87)	-0.018 (-5.65)	-0.016 (-4.07)	-0.019 (-4.10)	-0.024 (-4.86)	-0.030 (-5.76)	-0.035 (-6.40)	-0.036 (-6.22)	-0.035 (-5.88)
35	-0.031 (-19.62)	-0.023 (-10.03)	-0.022 (-7.72)	-0.023 (-7.11)	-0.024 (-5.64)	-0.027 (-5.58)	-0.033 (-6.36)	-0.040 (-7.25)	-0.042 (-7.25)	-0.042 (-6.78)	-0.042 (-6.53)
42	-0.031 (-19.22)	-0.027 (-11.37)	-0.027 (-9.22)	-0.027 (-8.24)	-0.029 (-6.71)	-0.033 (-6.53)	-0.039 (-7.14)	-0.043 (-7.40)	-0.045 (-7.39)	-0.045 (-7.02)	-0.044 (-6.57)
49	-0.030 (-18.29)	-0.027 (-11.27)	-0.028 (-9.37)	-0.029 (-8.55)	-0.033 (-7.43)	-0.037 (-7.13)	-0.042 (-7.30)	-0.045 (-7.39)	-0.047 (-7.38)	-0.046 (-6.96)	-0.043 (-6.16)
56	-0.030 (-17.38)	-0.028 (-11.48)	-0.030 (-9.84)	-0.032 (-9.20)	-0.037 (-7.91)	-0.039 (-7.29)	-0.043 (-7.19)	-0.047 (-7.39)	-0.047 (-7.02)	-0.044 (-6.37)	-0.043 (-6.00)
(b) Top 5%											
1	0.004 (0.78)	0.020 (3.56)	0.019 (3.31)	0.033 (5.52)	0.039 (6.00)	0.043 (6.25)	0.042 (5.82)	0.042 (5.52)	0.039 (5.04)	0.038 (4.80)	0.039 (4.75)
3	0.004 (0.76)	0.004 (0.75)	0.015 (2.52)	0.024 (3.86)	0.042 (6.20)	0.043 (5.67)	0.047 (5.81)	0.040 (4.61)	0.039 (4.22)	0.037 (3.82)	0.036 (3.54)
5	-0.003 (-0.55)	0.007 (1.12)	0.010 (1.70)	0.015 (2.38)	0.040 (5.79)	0.040 (5.10)	0.044 (5.22)	0.029 (3.20)	0.028 (2.92)	0.028 (2.69)	0.025 (2.30)
7	0.009 (1.48)	0.009 (1.37)	0.011 (1.72)	0.015 (2.36)	0.044 (6.38)	0.040 (5.10)	0.041 (4.79)	0.025 (2.69)	0.024 (2.41)	0.024 (2.27)	0.020 (1.84)
14	0.009 (1.35)	0.023 (3.24)	0.029 (4.00)	0.037 (5.18)	0.048 (6.81)	0.042 (5.21)	0.029 (3.30)	0.018 (1.81)	0.013 (1.27)	0.007 (0.63)	0.005 (0.42)
21	0.011 (1.47)	0.021 (2.61)	0.028 (3.49)	0.030 (3.79)	0.039 (4.81)	0.024 (2.99)	0.009 (0.98)	-0.007 (-0.68)	-0.015 (-1.37)	-0.019 (-1.64)	-0.016 (-1.36)
28	0.008 (1.09)	0.022 (2.50)	0.029 (3.24)	0.029 (3.26)	0.022 (2.41)	0.003 (0.35)	-0.011 (-1.19)	-0.029 (-2.91)	-0.036 (-3.28)	-0.035 (-3.07)	-0.031 (-2.58)
35	0.001 (0.19)	0.008 (0.81)	0.008 (0.77)	0.007 (0.74)	-0.001 (-0.05)	-0.016 (-1.61)	-0.033 (-3.28)	-0.049 (-4.84)	-0.049 (-4.45)	-0.044 (-3.75)	-0.037 (-2.97)
42	-0.006 (-0.79)	-0.000 (-0.01)	-0.003 (-0.24)	-0.003 (-0.29)	-0.014 (-1.27)	-0.030 (-2.79)	-0.046 (-4.15)	-0.055 (-4.90)	-0.048 (-4.31)	-0.042 (-3.46)	-0.031 (-2.44)
49	-0.009 (-1.06)	-0.004 (-0.42)	-0.007 (-0.62)	-0.010 (-0.89)	-0.019 (-1.72)	-0.033 (-2.89)	-0.046 (-3.90)	-0.049 (-4.11)	-0.041 (-3.40)	-0.032 (-2.63)	-0.023 (-1.82)
56	-0.011 (-1.33)	-0.010 (-0.97)	-0.013 (-1.18)	-0.016 (-1.37)	-0.023 (-1.99)	-0.033 (-2.77)	-0.041 (-3.34)	-0.042 (-3.35)	-0.029 (-2.28)	-0.024 (-1.85)	-0.017 (-1.38)

on their returns over the look-back period, and a long-short portfolio is constructed by taking long positions in the highest-return quintile (Q5) and short positions in the lowest-return quintile (Q1). Panel (a) of Table 7 reports the performance of these portfolios, where each row represents a look-back and holding period pair as indicated by the first column.¹⁰ If a portfolio is liquidated during the sample period, as indicated by a cumulative return (Cum) of -100%, the performance until liquidation is reported.

Despite being the best 21 pairs from the regression analysis, only three of them outperform the market in terms of the Sharpe ratio, and two have lower MDDs than the market. Moreover, six portfolios, (1, 28), (3, 21), (3, 28), (5, 21), (5, 28), and (7, 28), are liquidated during the sample period.¹¹ The (14, 7) portfolio achieves the highest Sharpe ratio of 1.28, while the (1, 14) portfolio has the lowest MDD of 62.8%—the market portfolio’s Sharpe ratio and MDD are 1.01 and 89.1%, respectively. However, it should be noted that three of the seven accounts of the (14, 7) portfolio are liquidated during the sample period, implying that even the best-performing portfolio could have been liquidated had its investments not been distributed over the holding period.¹²

Surprisingly, long-only portfolios generally outperform their long-short counterparts in terms of the Sharpe ratio. Their standard deviations are comparable to those of long-short portfolios, suggesting that short positions offer a limited benefit in offsetting market fluctuations in the cryptocurrency market. Indeed, most short-only portfolios are liquidated during the sample period. The Sharpe ratios of the long-only portfolios range between 0.90 and 1.42 and are usually higher than the market Sharpe ratio of 1.01.

Most long-short portfolios are more volatile than the market and frequently experience large losses. Extreme surges and plunges are rather common in the cryptocurrency market, and a single such event can have a drastic impact on portfolio performance. As shown in Table 7, liquidated

¹⁰Performance before transaction costs can be found in Table IA14.

¹¹In terms of accounts, 98 of the 139 accounts (70.5%) are liquidated, and liquidation occurs more frequently when the rebalancing frequency is longer due to leverage drift (Table IA16). The 139 accounts are the total number of accounts of 21 portfolios: *e.g.*, a portfolio with a holding period of three days consists of three accounts.

¹²The number of liquidated accounts and the liquidation dates are reported in Table IA15.

portfolios generally have a positive mean return before liquidation, indicating that liquidations are often driven by a few large shocks rather than by gradual accumulation of losses. Extreme returns should not be ruled out as outliers. Our filtering rule requires large market capitalizations and trading volumes, and is stricter than those adopted by most previous studies. Inclusion of smaller or less liquid coins would increase the frequency of extreme events.

Table IA16 compares the performance of cross-sectional momentum portfolios i) without mark-to-market, ii) with mark-to-market, and iii) with mark-to-market and transaction costs. It reveals that most short-only portfolios are liquidated once marked to market daily: Since returns cannot fall below -100%, only short portfolios are liquidated. Even when a holding period return is -100%, it can be hidden in the mean return unless explicitly examined: If there are 1,000 observations, a return of -100% reduces the mean return by only 0.1%. Consequently, without mark-to-market, none of the long-short portfolios is liquidated, and many yield Sharpe ratios exceeding 1.0. This result confirms that the weak performance of cross-sectional momentum, which contradicts earlier studies such as Liu et al. (2022), is primarily due to daily mark-to-market. Transaction costs also reduce performance, especially when the holding period is short, but mark-to-market is what reveals the true risk of the momentum strategy.

Figure 4 (a) plots the log-scale cumulative returns of the (1, 14) and (14, 7) long-short portfolios. The (14, 7) portfolio experiences significant volatility in 2017, recording an MDD of 86.5%, but performs well thereafter. The (1, 14) portfolio exhibits relatively stable performance compared to the (14, 7) portfolio, but it persistently declines from mid-2021 onward.

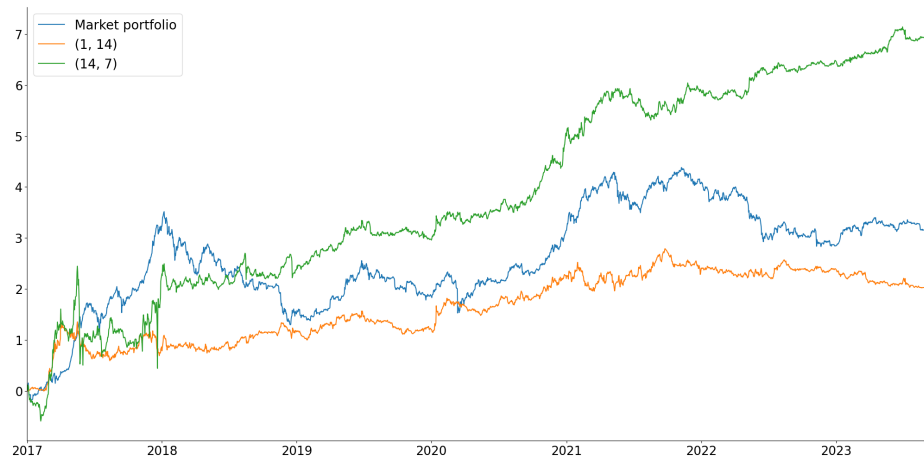
4.2.2 Cross-sectional momentum portfolios using the top 5%

This section evaluates cross-sectional momentum portfolios comprising the top 5% of the coins. Since the top 5% contains only a small number of coins, as few as one in 2017 and at most 40 in 2021, we include at least the four largest coins in the sample and divide them into two groups instead of five. The long-short portfolio buys coins above the median and sells those below the

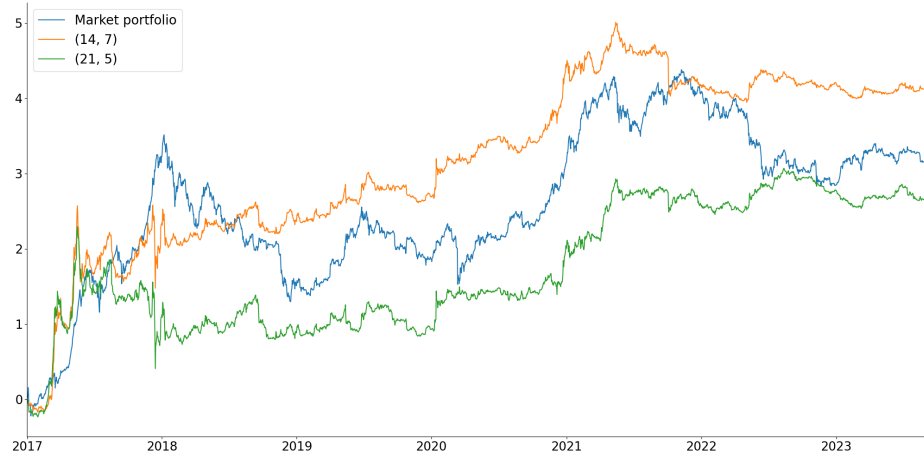
Table 7: Performance of cross-sectional momentum portfolios

This table reports the performance of cross-sectional momentum portfolios of various look-back and holding periods, (j, k) . ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2017 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(a) All coins															
(1, 14)	107.21	95.85	1.12	5751.9	95.5	-78.02	108.92	-0.72	-100.0	100.0	49.18	61.55	0.80	687.6	62.8
(1, 21)	101.21	94.41	1.07	4091.4	95.0	-1357.04	263.45	-5.15	-100.0	100.0	24.11	104.39	0.23	-87.2	99.1
(1, 28)	105.58	102.04	1.03	3810.3	95.2	-569.07	226.64	-2.51	-100.0	100.0	47.12	257.66	0.18	-100.0	100.0
(3, 14)	106.11	95.02	1.12	5609.2	95.6	-734.00	198.59	-3.70	-100.0	100.0	39.36	76.10	0.52	90.9	97.5
(3, 21)	84.12	93.32	0.90	1309.0	95.7	-577.53	361.17	-1.60	-100.0	100.0	-351.57	290.98	-1.21	-100.0	100.0
(3, 28)	95.79	93.93	1.02	2913.1	94.4	-658.42	235.38	-2.80	-100.0	100.0	3.52	328.68	0.01	-100.0	100.0
(5, 14)	119.89	103.52	1.16	9696.5	95.7	-62.30	121.70	-0.51	-100.0	100.0	108.66	93.52	1.16	6359.1	91.5
(5, 21)	106.98	101.44	1.05	4371.4	94.1	38.12	899.16	0.04	-100.0	100.0	194.01	366.65	0.53	-100.0	100.0
(5, 28)	104.48	102.79	1.02	3452.8	94.4	-698.21	218.38	-3.20	-100.0	100.0	94.18	220.08	0.43	-100.0	100.0
(7, 14)	135.35	113.83	1.19	17485.2	95.9	-9.50	117.64	-0.08	-99.7	100.0	110.19	101.45	1.09	7758.8	89.9
(7, 21)	117.74	113.45	1.04	5288.1	94.6	-997.99	222.40	-4.49	-100.0	100.0	72.74	118.07	0.62	68.4	97.7
(7, 28)	117.47	113.91	1.03	5073.3	95.1	-636.56	252.14	-2.52	-100.0	100.0	126.56	292.82	0.43	-100.0	100.0
(14, 7)	149.95	105.74	1.42	55166.6	93.9	77.73	231.28	0.34	-99.2	99.9	174.73	136.93	1.28	101218.3	86.5
(14, 14)	125.43	103.35	1.21	12736.2	94.6	-534.08	237.92	-2.24	-100.0	100.0	146.57	162.83	0.90	4553.6	96.6
(14, 21)	110.85	106.87	1.04	3777.9	93.4	-644.20	249.42	-2.58	-100.0	100.0	153.15	215.97	0.71	915.3	98.8
(21, 5)	129.52	106.14	1.22	13904.9	95.5	-48.24	114.62	-0.42	-100.0	100.0	103.39	110.37	0.94	1870.3	95.7
(21, 7)	116.23	104.41	1.11	6302.3	95.3	-581.68	257.71	-2.26	-100.0	100.0	79.07	112.00	0.71	147.3	98.5
(21, 14)	112.98	103.24	1.09	5355.7	94.9	-33.84	137.60	-0.25	-100.0	100.0	94.36	116.29	0.81	380.6	98.2
(28, 3)	121.62	108.25	1.12	6987.8	94.3	-63.04	107.82	-0.58	-100.0	100.0	57.98	98.06	0.59	87.4	95.9
(28, 5)	113.67	105.60	1.08	4806.5	94.3	-51.96	115.73	-0.45	-100.0	100.0	62.38	97.77	0.64	163.5	96.0
(28, 7)	97.38	104.08	0.94	1704.0	94.4	-41.58	122.00	-0.34	-100.0	100.0	41.32	97.29	0.42	-37.7	97.9
(b) Top 5%															
(1, 14)	100.45	85.69	1.17	6704.3	89.1	-104.08	94.68	-1.10	-100.0	100.0	13.82	59.77	0.23	-27.4	95.2
(1, 21)	94.61	84.98	1.11	4695.3	90.8	-651.88	204.82	-3.18	-100.0	100.0	-33.39	181.69	-0.18	-100.0	100.0
(1, 28)	90.16	83.23	1.08	3794.8	89.9	-591.36	197.04	-3.00	-100.0	100.0	-93.31	198.39	-0.47	-100.0	100.0
(3, 14)	103.82	88.25	1.18	7165.9	90.0	-67.41	98.29	-0.69	-99.9	99.9	34.21	68.76	0.50	92.7	92.2
(3, 21)	99.15	83.97	1.18	6682.5	90.4	-30.09	191.74	-0.16	-100.0	100.0	67.04	109.50	0.61	256.4	80.7
(3, 28)	98.03	87.27	1.12	5139.9	91.5	-624.10	218.13	-2.86	-100.0	100.0	-67.51	235.22	-0.29	-100.0	100.0
(5, 14)	118.53	90.02	1.32	17794.5	90.7	-105.20	106.28	-0.99	-100.0	100.0	42.12	71.61	0.59	208.0	75.2
(5, 21)	97.49	85.14	1.14	5574.1	90.4	-809.24	196.73	-4.11	-100.0	100.0	1.24	94.68	0.01	-95.0	98.7
(5, 28)	102.68	88.75	1.16	6572.5	93.7	-547.70	206.54	-2.65	-100.0	100.0	108.56	216.13	0.50	-100.0	100.0
(7, 14)	109.83	89.20	1.23	10411.3	89.9	-707.01	225.40	-3.14	-100.0	100.0	-75.10	205.34	-0.37	-100.0	100.0
(7, 21)	94.15	87.09	1.08	4030.1	89.3	-752.77	190.31	-3.96	-100.0	100.0	13.31	87.30	0.15	-81.0	96.5
(7, 28)	84.32	87.12	0.97	2053.8	93.5	-590.52	210.59	-2.80	-100.0	100.0	-37.66	183.88	-0.20	-100.0	100.0
(14, 7)	131.99	94.59	1.40	34419.3	89.6	-44.87	86.56	-0.52	-99.6	99.8	89.38	74.85	1.19	6056.5	66.8
(14, 14)	113.18	92.91	1.22	10727.5	91.6	-70.86	94.05	-0.75	-99.9	99.9	63.12	74.76	0.84	961.4	77.7
(14, 21)	87.46	89.97	0.97	2148.1	92.3	-720.10	190.21	-3.79	-100.0	100.0	-6.46	93.80	-0.07	-96.6	99.3
(21, 5)	115.88	95.00	1.22	11294.9	90.3	-51.11	86.15	-0.59	-99.7	99.8	65.87	72.90	0.90	1315.0	84.8
(21, 7)	114.74	93.89	1.22	11178.8	90.3	-60.93	85.94	-0.71	-99.8	99.9	60.28	72.66	0.83	888.2	80.7
(21, 14)	92.09	90.52	1.02	2869.8	91.6	-91.76	97.83	-0.94	-99.9	100.0	22.14	69.91	0.32	-13.1	93.2
(28, 3)	111.86	98.50	1.14	6778.6	90.6	-49.18	86.20	-0.57	-99.6	99.8	58.36	76.91	0.76	591.5	90.8
(28, 5)	106.62	94.78	1.13	5954.1	91.2	-54.32	86.44	-0.63	-99.7	99.9	52.68	74.60	0.71	416.7	92.7
(28, 7)	104.66	93.63	1.12	5619.9	91.9	-66.22	87.51	-0.76	-99.9	99.9	39.57	75.50	0.52	105.9	94.9
Market											78.86	77.80	1.01	2333.3	89.1



(a) All coins



(b) Top 5%

Figure 4: Cross-sectional momentum portfolio log-scale cumulative returns

median. Panel (b) of Table 7 reports the performance of these portfolios.

When the investment pool is restricted to the top 5% of the coins, long-short portfolios have lower mean returns and standard deviations compared to those using all coins. Consequently, the best portfolios earn lower cumulative returns, but also have lower MDDs. Although the regression analysis finds a stronger momentum effect among large coins, only five of the 21 portfolios of the top 5% yield higher Sharpe ratios than those of all coins, and only the (14, 7) portfolio marginally outperforms the market, with a Sharpe ratio of 1.19. Despite being composed of large coins, five long-short portfolios are liquidated during the sample period.

Most long-only portfolios consisting of the top 5% outperform their all-coin counterparts. In contrast, short-only portfolios underperform those of all coins, indicating that losses from the short leg are not solely attributable to extreme jumps of small coins. Most short-only portfolios plunge by 99% during the 2017 bull market and another 99% during the 2020 bull market.

Figure 4 (b) presents the cumulative log returns of the two highest Sharpe ratio portfolios, (14, 7) and (21, 5). Both portfolios perform steadily until early 2021, after which they stagnate or decline, indicating a waning of the momentum effect among large coins.

4.2.3 Long-term profitability of momentum strategies

The t -statistic of the mean return is commonly used to assess the profitability of a strategy. However, when the return distribution is skewed and fat-tailed, a strategy with a statistically positive mean return can earn a negative profit, and the usual t -test can fail. Appendix A discusses this matter in detail and suggests using the mean log return to assess the long-term profitability of a cryptocurrency strategy. Table 8 reports both the mean returns and the mean log returns of the cross-sectional momentum portfolios, along with their t -statistics.

Among the portfolios of all coins, nine yield mean returns with t -statistics above 2.0, of which only one, (14, 7), passes the cutoff value of 3.0 proposed by Harvey et al. (2016). In contrast, only the (14, 7) portfolio has a mean log return with a t -statistic above 2.0. Moreover, seven portfolios have

negative mean log returns despite positive mean returns. Similarly, among the portfolios of the top 5%, two have mean returns with t -statistics above 2.0, but none exhibit significant mean log returns, and nine have negative mean log returns despite positive mean returns. These findings highlight the limitations of the mean return and the Sharpe ratio as a profitability measure and provide a more accurate assessment of the profitability of the momentum strategy in the cryptocurrency market.

Table 8: Cross-sectional momentum portfolio t -test

This table reports the mean daily returns (%) and mean daily log returns (%) of the cross-sectional momentum long-short portfolios and their Newey-West adjusted t -statistics. When a portfolio is liquidated, the returns before liquidation are reported. The sample period is January 1, 2017 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	All coins				Top 5%			
	Return		Log return		Return		Log return	
	Mean	t -statistic	Mean	t -statistic	Mean	t -statistic	Mean	t -statistic
(1, 14)	0.135	2.044	0.085	1.287	0.038	0.472	-0.013	-0.154
(1, 21)	0.066	0.721	-0.085	-0.871	0.193	0.868	-0.140	-0.401
(1, 28)	0.416	0.542	-1.115	-0.687	0.021	0.029	-0.460	-0.459
(3, 14)	0.108	1.188	0.027	0.275	0.094	1.003	0.027	0.268
(3, 21)	-0.682	-0.998	-1.825	-1.847	0.184	2.040	0.052	0.650
(3, 28)	0.291	0.712	-0.994	-0.838	0.092	0.130	-0.932	-0.653
(5, 14)	0.298	2.809	0.172	1.538	0.115	1.325	0.046	0.542
(5, 21)	0.816	1.363	-0.409	-0.702	0.003	0.036	-0.124	-1.070
(5, 28)	0.546	1.207	-0.103	-0.142	0.575	1.299	0.077	0.119
(7, 14)	0.302	2.564	0.180	1.675	0.078	0.142	-0.474	-0.619
(7, 21)	0.199	1.620	0.021	0.163	0.036	0.437	-0.068	-0.681
(7, 28)	0.634	0.963	-0.412	-0.396	0.181	0.300	-0.188	-0.251
(14, 7)	0.479	3.649	0.285	2.604	0.245	2.628	0.170	1.882
(14, 14)	0.402	2.834	0.158	1.428	0.173	1.966	0.097	1.092
(14, 21)	0.420	2.046	0.095	0.933	-0.018	-0.177	-0.140	-1.140
(21, 5)	0.283	2.657	0.123	1.121	0.180	1.990	0.109	1.251
(21, 7)	0.217	2.006	0.037	0.302	0.165	1.961	0.094	1.160
(21, 14)	0.259	2.434	0.065	0.561	0.061	0.769	-0.006	-0.071
(28, 3)	0.159	1.505	0.026	0.249	0.160	1.592	0.080	0.801
(28, 5)	0.171	1.735	0.040	0.414	0.144	1.554	0.068	0.716
(28, 7)	0.113	1.153	-0.019	-0.197	0.108	1.172	0.030	0.310

4.2.4 Factor regression

This section examines whether the three factors proposed by Liu et al. (2022) and the continuing overreaction factor (CS-CO) can account for the cross-sectional momentum. Table 9 reports the regression results of the (14, 7) long-short portfolio. When the return is regressed on individual

factors, only the overreaction factor is significant with a t -statistic of 8.07 and an adjusted R^2 of 0.20, while the three factors of Liu et al. (2022) are nonsignificant. When the return is regressed on all four factors (regression (7)), the CS-CO factor remains significant with a t -statistic of 4.62. However, none of these factors or their combinations can explain the cross-sectional momentum premium: The t -statistic of alpha exceeds 3.0 in all cases.

The unexpectedly low explanatory power of MOM is because it is constructed using past three-week returns, whereas the (14, 7) portfolio’s look-back period is two weeks. The CS-CO factor also uses past three-week returns, but it is significant because it puts more emphasis on recent returns. When MOM and CS-CO are constructed using past two-week returns (Table IA17), the coefficients on both factors become statistically significant with t -statistics of 4.93 and 8.34, respectively. When the portfolio return is regressed on both factors, only MOM remains significant with a t -statistic of 2.94.

Table 9: Cross-sectional momentum portfolio factor regression

This table reports the factor regression results of the (14, 7) cross-sectional momentum long-short portfolio. The factors are market (MKT), size (SIZE), momentum (MOM), and cross-sectional continuing overreaction (CS-CO). The definitions of the factors can be found in Section 3.2.2. The sample period is January 1, 2017 to August 28, 2023. The t -statistics are Newey-West adjusted t -statistics.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Const	0.0294 (4.28)	0.0302 (4.36)	0.0234 (3.90)	0.0197 (3.04)	0.0291 (4.44)	0.0224 (3.99)	0.0166 (3.10)
MKT	0.0926 (1.32)				0.0779 (1.18)	0.0804 (1.44)	0.0324 (0.65)
SIZE		0.1039 (0.77)			0.0887 (0.66)	-0.0597 (-0.51)	-0.1061 (-0.91)
MOM			0.3432 (1.54)			0.3519 (1.56)	0.2217 (1.58)
CS-CO				0.8466 (8.07)			0.7073 (4.62)
Adj R^2	0.004	0.004	0.120	0.199	0.007	0.123	0.239

4.2.5 Transition probabilities

This section examines cross-sectional transition probabilities. Coins are divided into quintiles based on their look-back and holding period returns (Q1 to Q5 in ascending order), and the transition

probabilities from the look-back period quintiles to holding period quintiles are calculated. For the top 5%, coins are divided into two groups. Figure 5 presents heatmaps of transition probabilities for various pairs of look-back and holding periods.

When all coins are considered, the probability $P(Q5 \rightarrow Q4, Q5)$ is below 40% across all look-back and holding periods, while $P(Q5 \rightarrow Q1, Q2)$ consistently exceeds 40%, implying reversal among winners. The reversal effect is stronger when the look-back and holding periods are shorter than two weeks or longer than five weeks, and weaker when the look-back period is long and the holding period is short. In contrast, the transition from Q1 indicates weak momentum, especially when the look-back period is less than two weeks and the holding period is longer than three weeks. These results suggest that past winners tend to reverse and become losers, while past losers continue to underperform across a wide range of look-back and holding periods.

The transition probabilities of the top 5% (Panel (b)) draw a markedly different picture, with past winners (Q2) exhibiting momentum while past losers (Q1) showing reversal.¹³ The momentum effect among winners is stronger when the look-back period is shorter than two weeks and the holding period is longer than two weeks, whereas the reversal effect among losers is particularly pronounced when both the look-back and holding periods are longer than three weeks, indicating long-term reversal among losers.

A closer examination of the transition using the (14, 7) strategy reveals several interesting observations (Table 10). First, when all coins are used, past winners becoming losers earn significantly higher returns during the look-back period than those remaining winners (70.64% vs 52.06%), implying that coins appreciating significantly during the look-back period tend to fall sharply during the holding period. These coins are also significantly smaller and have higher turnover than those remaining winners. Second, while past losers are slightly more likely to remain losers, those becoming winners have an extremely high standard deviation (310.24%) and skewness (82.05) of the holding period returns, indicating that many losers jump aggressively during the holding period.

¹³Note the difference in the range of the probability between Panels (a) and (b). The same color represents different probabilities.

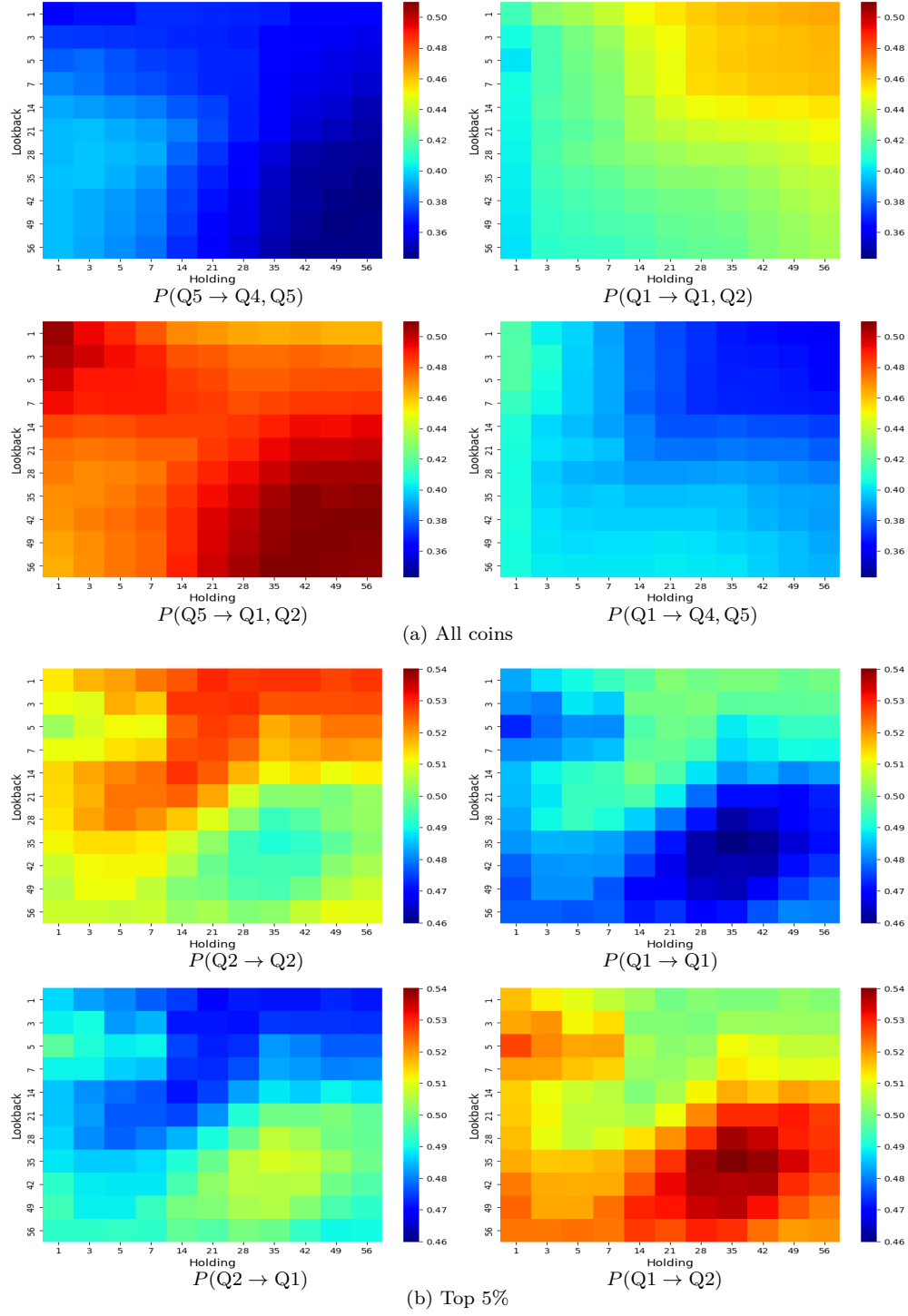


Figure 5: Transition probabilities between look-back and holding period returns

As mentioned in Section 2.2, 136 coins in the sample fall for five consecutive days before surging more than 50% the following day. Such jumps incur significant losses on short positions.

Among the top 5%, past winners are more likely to remain winners than to become losers (52.33% vs 47.67%), suggesting weak momentum among large winners. As in the case of all coins, past winners becoming losers have significantly higher look-back period returns than those remaining winners (36.27% vs 14.99%), again implying the reversal of the most appreciated coins. The probabilities of past losers becoming winners (50.61%) and remaining losers (49.39%) are almost identical, indicating neither momentum nor reversal. Unlike small losers, large losers transitioning into winners do not exhibit extremely volatile or skewed holding period returns.

Overall, the transition probabilities reveal that winners exhibit more evident patterns—reversal among all coins and momentum among large coins—whereas losers exhibit less distinct and often opposing transition patterns. These inconsistent transition patterns make it difficult to develop a profitable momentum strategy. The contrasting results between all coins and large coins imply an interaction between size and momentum in the cryptocurrency market.

4.2.6 Cross-sectional momentum in different coin groups

This section examines cross-sectional momentum across different coin groups. Coins are divided into quintiles based on size, Google trend, trading volume, relative trading volume (volume/size), trading volume change, and continuing overreaction, and a (14, 7) cross-sectional momentum portfolio is constructed within each group. Since coins need to be divided into 25 groups, we require a minimum of 100 coins in the cross-section and start backtesting from December 2017. In addition, we ignore daily price fluctuations and calculate returns using the prices on rebalancing days, since the objective of this analysis is to examine performance variation across the groups, without the confounding effects of liquidation events. Table 11 reports the mean holding period (one week) returns of the portfolios.

Panel (a) presents the relationship between size and cross-sectional momentum. The perfor-

Table 10: Transition probabilities of the (14, 7) strategy

This table reports the transition probabilities and return statistics of the (14, 7) cross-sectional momentum portfolio. Rows (columns) indicate the return groups in the look-back (holding) period. ‘Prob’, ‘Mean0’, ‘Mcap0’, ‘Vol0’, ‘Mean’, ‘Std’, and ‘Skew’ denote the transition probability (%), mean look-back period return (%), mean market capitalization (\$10 million), mean trading volume (\$10 million), and the mean (%), standard deviation (%), and skewness of the holding-period return, respectively.

		Q1	Q2	Q3	Q4	Q5
Q1	Prob	23.74	18.99	18.23	18.63	20.41
	Mean0	-23.14	-18.87	-18.56	-19.50	-23.03
	Mcap0	35.90	85.06	115.36	77.63	42.29
	Vol0	3.74	8.40	10.91	7.11	3.74
	Mean	-17.30	-6.78	-1.98	3.42	27.29
	Std	14.01	11.35	11.89	13.23	310.24
	Skew	-1.25	-0.78	-0.11	0.60	82.05
Q2	Prob	14.14	20.88	23.71	23.03	18.24
	Mean0	-9.07	-7.83	-7.51	-7.76	-7.92
	Mcap0	65.37	196.12	315.23	329.50	127.08
	Vol0	7.45	17.06	26.13	23.07	10.47
	Mean	-15.57	-6.79	-2.07	3.58	20.99
	Std	12.77	11.15	11.65	12.99	36.65
	Skew	-1.00	-0.76	-0.16	0.73	10.48
Q3	Prob	13.44	21.49	23.86	23.21	18.00
	Mean0	-1.36	-1.19	-1.39	-1.05	-0.17
	Mcap0	96.04	233.14	367.73	400.36	238.29
	Vol0	10.65	19.73	27.41	27.53	18.28
	Mean	-14.61	-6.60	-2.09	3.24	21.79
	Std	12.56	10.92	11.56	12.93	40.26
	Skew	-1.09	-0.74	-0.19	0.63	19.47
Q4	Prob	17.54	21.68	20.72	20.59	19.47
	Mean0	9.43	7.36	6.47	7.26	9.55
	Mcap0	94.79	258.22	411.75	538.09	341.56
	Vol0	11.09	23.51	34.58	38.99	24.28
	Mean	-14.66	-6.46	-2.10	2.98	23.23
	Std	12.27	10.85	11.47	13.36	41.10
	Skew	-1.10	-0.71	-0.31	0.61	16.22
Q5	Prob	31.65	16.68	13.22	14.27	24.18
	Mean0	70.64	39.32	39.03	39.41	52.06
	Mcap0	67.88	219.80	373.94	408.68	253.70
	Vol0	13.06	23.80	35.73	38.85	26.53
	Mean	-17.19	-6.57	-2.52	2.84	29.38
	Std	13.23	11.10	11.86	13.80	54.85
	Skew	-1.09	-0.69	-0.22	0.76	18.80

(a) All coins

		Q1	Q2
Q1	Prob	49.39	50.61
	Mean0	-7.24	-6.22
	Mcap0	2862.98	3819.12
	Vol0	227.39	289.22
	Mean	-5.84	5.53
	Std	11.88	17.72
	Skew	-1.15	6.07
Q2	Prob	47.67	52.33
	Mean0	36.27	14.99
	Mcap0	3569.82	5440.24
	Vol0	312.41	421.75
	Mean	-6.41	7.94
	Std	11.74	20.22
	Skew	-1.00	3.84

(b) Top 5%

mance of the momentum strategy improves with size, earning a mean weekly return of 0.53% in the smallest size group (M1) and 2.36% (t -statistic = 4.25) in the largest size group (M5). This result reaffirms the stronger momentum effect among larger coins.

Panels (b) to (f) present results when coins are sorted on a variable related to investor attention and overreaction. When coins are sorted on Google trend (Panel (b)) or trading volume (Panel (c)), the momentum portfolio earns the highest mean return in the middle group, while returns in the extreme groups are smaller and of similar magnitude. The decreasing mean return with relative volume appears to support an underreaction explanation of momentum (Panel (d)). However, this result is largely driven by the size effect, as relative volume is inversely related to size. The mean return decreases with volume change and is significant only in the first two groups, implying that a significant increase in attention does not induce momentum (Panel (e)). The mean return also decreases with continuing overreaction (Panel (f)). Overall, no systemic patterns can be observed across attention-related groups, except that the momentum strategy tends to earn lower returns in the highest attention groups, which contradicts the attention-based overreaction mechanism of momentum.

4.2.7 Binance futures

In the cryptocurrency market, investors can take short positions only in the futures market. Futures markets are more liquid than spot markets, and the coins traded in them are relatively large and liquid. Hence, we test the cross-sectional momentum strategy using the coins listed on the Binance futures market. We obtain futures price and trading volume data from Binance’s website.¹⁴ The dataset contains 239 unique coins and spans the period from September 8, 2019 to November 17, 2023. We require a minimum of 20 coins in the cross-section, and the backtesting commences on February 12, 2020. Table IA18 reports the performance of the cross-sectional momentum portfolios formed of the coins listed on the Binance futures market.

¹⁴<https://data.binance.vision/?prefix=data/futures/cm/daily/klines/>

Table 11: Performance of cross-sectional momentum in different coin groups

This table reports the performance of the (14, 7) cross-sectional momentum portfolio in different coin groups. Coins are grouped on size, Google trend, trading volume, relative trading volume (volume/size), trading volume change, and continuing overreaction, and a value-weighted cross-sectional momentum portfolio is formed within each group. The reported values are mean holding period returns and their Newey-West adjusted t -statistics. The sample period is December 1, 2017 to August 28, 2023, except for Google trend groups, which use coins listed on the Binance futures market from September 2019 to August 2023 for data availability. See Section A.1 of the Internet Appendix for more details.

	Low	2	3	4	High	H-L
M1	-0.051 (-0.058)	1.394 (1.191)	0.906 (1.054)	0.695 (0.774)	0.484 (0.481)	0.534 (0.863)
M2	2.590 (0.980)	0.772 (0.941)	0.926 (1.074)	1.139 (1.237)	1.271 (1.222)	-1.319 (-0.510)
M3	-0.256 (-0.342)	0.790 (0.945)	0.683 (0.815)	0.763 (0.899)	0.346 (0.381)	0.602 (1.158)
M4	-0.297 (-0.402)	0.685 (0.831)	0.744 (0.899)	0.455 (0.567)	1.271 (1.244)	1.568 (2.778)
M5	-0.473 (-0.635)	0.585 (0.813)	0.539 (0.798)	1.434 (1.980)	1.890 (2.430)	2.363 (4.247)
M5-M1	-0.422 (-0.764)	-0.810 (-0.886)	-0.367 (-0.683)	0.740 (1.428)	1.406 (2.073)	

(a) Size groups

	Low	2	3	4	High	H-L
G1	0.648 (0.805)	1.108 (1.085)	0.835 (0.920)	0.913 (0.879)	2.226 (1.362)	1.577 (1.179)
G2	1.220 (1.267)	0.840 (0.947)	1.828 (1.320)	1.583 (1.394)	1.296 (1.173)	0.077 (0.131)
G3	1.274 (1.247)	4.661 (1.709)	0.853 (0.796)	2.092 (1.608)	4.173 (2.000)	2.899 (2.049)
G4	1.272 (1.054)	1.544 (1.689)	1.756 (1.670)	2.611 (1.632)	1.600 (0.958)	0.328 (0.297)
G5	0.176 (0.196)	1.089 (0.810)	0.982 (0.774)	2.664 (1.902)	1.475 (1.474)	1.300 (1.378)
G5-G1	-0.473 (-1.248)	-0.020 (-0.028)	0.146 (0.137)	1.750 (2.334)	-0.751 (-0.534)	

(b) Google trend groups

	Low	2	3	4	High	H-L
V1	1.135 (0.933)	0.044 (0.061)	0.557 (0.749)	0.968 (1.282)	2.506 (2.491)	1.371 (1.181)
V2	-0.245 (-0.304)	0.029 (0.039)	0.376 (0.535)	0.826 (0.959)	1.604 (1.815)	1.849 (2.772)
V3	-0.186 (-0.246)	1.191 (1.421)	1.287 (1.472)	1.374 (1.597)	2.229 (2.117)	2.415 (3.260)
V4	-0.827 (-1.083)	0.680 (0.802)	0.893 (1.082)	0.385 (0.497)	1.329 (1.315)	2.156 (3.326)
V5	-0.163 (-0.211)	0.692 (0.981)	0.545 (0.830)	1.656 (2.186)	1.315 (1.697)	1.479 (2.601)
V5-V1	-1.298 (-1.214)	0.648 (1.570)	-0.012 (-0.027)	0.688 (1.626)	-1.190 (-1.723)	

(c) Volume groups

	Low	2	3	4	High	H-L
VM1	-1.098 (-1.478)	0.495 (0.603)	0.850 (1.052)	0.995 (1.340)	2.221 (2.455)	3.319 (4.739)
VM2	-0.038 (-0.051)	0.650 (0.840)	0.910 (1.146)	1.173 (1.541)	1.297 (1.613)	1.336 (2.282)
VM3	-0.444 (-0.540)	0.311 (0.405)	0.604 (0.787)	0.944 (1.285)	1.854 (1.923)	2.297 (3.836)
VM4	-0.335 (-0.418)	1.025 (1.183)	1.480 (1.487)	0.852 (0.996)	0.874 (1.010)	1.209 (1.793)
VM5	-0.575 (-0.690)	0.179 (0.202)	0.128 (0.148)	0.388 (0.485)	-0.248 (-0.255)	0.327 (0.423)
VM5-VM1	0.523 (0.953)	-0.316 (-0.775)	-0.722 (-1.468)	-0.607 (-1.250)	-2.469 (-2.985)	

(d) Relative volume groups

	Low	2	3	4	High	H-L
$\Delta V1$	-0.670 (-0.872)	-0.031 (-0.040)	0.612 (0.732)	0.378 (0.507)	1.016 (1.255)	1.686 (3.016)
$\Delta V2$	-0.485 (-0.663)	0.183 (0.258)	1.190 (1.399)	1.205 (1.464)	1.237 (1.472)	1.722 (3.760)
$\Delta V3$	-0.257 (-0.352)	0.509 (0.714)	0.833 (1.136)	0.643 (0.903)	1.216 (1.655)	1.473 (3.337)
$\Delta V4$	0.308 (0.383)	0.440 (0.568)	0.619 (0.843)	1.501 (1.867)	1.715 (2.145)	1.407 (2.714)
$\Delta V5$	-0.769 (-0.839)	0.692 (0.824)	0.458 (0.570)	1.614 (1.787)	0.411 (0.405)	1.181 (1.546)
$\Delta V5-\Delta V1$	-0.099 (-0.147)	0.724 (1.571)	-0.155 (-0.335)	1.236 (2.608)	-0.605 (-0.794)	

(e) Volume change groups

	Low	2	3	4	High	H-L
CO1	-1.757 (-2.303)	-0.577 (-0.781)	0.099 (0.135)	0.389 (0.538)	0.475 (0.606)	2.232 (3.558)
CO2	-0.637 (-0.828)	0.305 (0.383)	0.973 (1.255)	0.719 (0.996)	1.125 (1.411)	1.762 (3.932)
CO3	-0.091 (-0.114)	0.454 (0.583)	1.202 (1.417)	1.132 (1.397)	1.664 (1.980)	1.755 (3.470)
CO4	0.940 (1.105)	0.862 (1.081)	0.765 (0.982)	0.920 (1.164)	1.662 (1.877)	0.722 (1.465)
CO5	0.270 (0.364)	0.676 (0.892)	1.227 (1.553)	1.472 (1.793)	0.875 (0.903)	0.605 (0.833)
CO5-CO1	2.027 (3.516)	1.253 (3.060)	1.128 (2.864)	1.083 (2.160)	0.401 (0.556)	

(f) Continuing overreaction groups

Most long-short portfolios earn profits and five of them outperform the market. They also have smaller MDDs than the market. The (5, 21) and (7, 28) strategies perform best with Sharpe ratios of 1.32 and 1.21, respectively. Nevertheless, the cumulative returns in Figure IA3 reveal that most profits are accrued prior to 2022, and the performance thereafter is mediocre. This result aligns with the previous observation that a cross-sectional momentum portfolio formed of the top 5% does not perform well since 2021.

The previous best strategy, (14, 7), performs comparably to the market with a Sharpe ratio of 0.83, whereas the (1, 14) strategy with the lowest MDD now has the highest MDD of 72.4% and the lowest Sharpe ratio of -0.35. These inconsistent results cast doubt on the robustness of momentum in the cryptocurrency market.

4.2.8 Reversal portfolios

Since the regression analysis using all coins suggests reversal, we test reversal strategies. Most reversal portfolios are liquidated during the bull market in early 2017. Therefore, we test them from the beginning of 2018. Table IA19 reports the performance of reversal strategies. All long-short portfolios with a holding period shorter than a week yield negative mean returns. Portfolios with look-back and holding periods longer than 40 days yield positive mean returns, but only two of them earn positive profits. These portfolios, (49, 42) and (56, 42), outperform the market yielding Sharpe ratios of 0.64 and 0.57, respectively (the market Sharpe ratio is 0.37). However, they earn negative returns for a prolonged period until early 2021. Although reversal is evident in the cryptocurrency market, developing a profitable strategy exploiting it remains challenging.

4.3 Summary of findings and discussion

The empirical evidence in this study suggests that past winners, especially large ones, exhibit momentum, whereas past losers are subject to high jump risks, and short-selling them incurs significant losses. Although several pairs of look-back and holding periods lead to profitable momentum

strategies, few yield statistically significant mean log returns. Given that these pairs are identified through regression analysis using the entire sample, it is questionable whether their profitability will persist in the future.

Since evidence of cross-sectional momentum is weak, we investigate what drives cryptocurrency prices in general, rather than probing the drivers of momentum. The cryptocurrency market is characterized by the lack of fundamental value and the dominance of retail investors. When valuation uncertainty is high, investors’ behavioral biases are amplified (Kumar, 2009), and prices become more susceptible to investor sentiment (Baker and Wurgler, 2006). Furthermore, without knowing the fundamental value, the “smart money” (rational arbitrageurs) cannot promptly correct mispricing, leading to excessive volatility and recurrent boom-burst cycles. Consequently, it is likely that cryptocurrency prices are governed primarily by investors’ behavioral biases and sentiment.

One of the most well-known behavioral biases is overconfidence. The literature suggests that retail investors are overconfident and that overconfident behavior leads to frequent trading and stronger reactions to market signals (Odean, 1998b; Barber and Odean, 2000). Odean (1998b) suggests that overconfident traders increase volatility, underreact to abstract, statistical, and highly relevant information, while overreact to salient, anecdotal, and extreme information. Regarding the cryptocurrency market, Chhatwani and Parija (2023) and Kim et al. (2022), using the National Financial Capability Study, find that overconfident retail investors are more likely to invest in cryptocurrencies.

Daniel et al. (1998) theoretically show that continuing overreaction, induced by overconfidence and self-attribution bias, can cause short-term autocorrelation (momentum) and long-term reversal. However, overconfidence and self-attribution bias can also cause short-term reversal when valuation uncertainty is high and investors are overly overconfident. To show this, we repeat the simulation in Daniel et al. (1998) using different values of valuation uncertainty and overconfidence.¹⁵ We use the same parameter values used in Daniel et al. (1998) to construct the baseline case and increase

¹⁵The model and the simulation process are described in Appendix B.

the uncertainty of the fundamental value (σ_θ) and the level of overconfidence ($1/\sigma_c$), respectively.

Figure 6 (a) shows the impact of valuation uncertainty on price dynamics. When the uncertainty increases, the price responds to the private signal more strongly, and when it is sufficiently large, the price decreases immediately after the arrival of the private signal without exhibiting any momentum effects. When the level of overconfidence increases (Figure 6 (b)), the price again overreacts more to the private signal and reverses more quickly. These results suggest that overconfidence can cause momentum, but it can also cause short-term reversal when valuation uncertainty is high or investors are overly overconfident.

Another prominent behavioral bias of retail investors is the disposition effect, which suggests that they sell assets quickly at gains and hold on to them when facing losses (Shefrin and Statman, 1985; Odean, 1998a). “HODL” (Hold On for Dear Life), an Internet meme referring to holding assets until losses are recouped, aptly represents cryptocurrency investors’ reluctance to realize losses. While the disposition effect is generally used to explain momentum through an underreaction mechanism (Grinblatt and Han, 2005; Frazzini, 2006), it can also cause short-term reversal of appreciated coins if selling pressure is high and investors are skeptical about their fundamental value. If the fundamental value is believed to be close to zero, investors are likely to view price appreciation as a short-term phenomenon rather than an increase of the fundamental value. Therefore, selling pressure can change the sentiment surrounding the coin and cause a decline of the price, rather than to cause future price increase through underreaction. On the other hand, holding loser coins can delay their price decline, resulting in downward momentum.

Another characteristic of the cryptocurrency market is its low-margin, high-leverage environment, which allows even a leverage of 100 or higher. Low margin requirements exacerbate overpricing and bubble-and-crash patterns (Bhojraj et al., 2009) and allow investors to employ high leverage to boost profits. High leverage increases margin call risks, and investors’ fear of forced liquidation can intensify trading activities and exert opposing pressure on prices, thus inhibiting the persistence of momentum. As short positions are subject to higher liquidation risk, such pressure

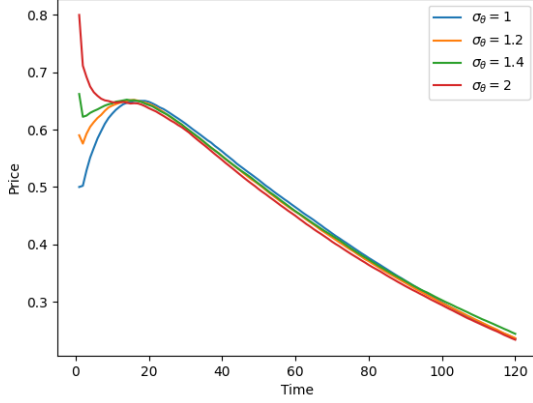
is higher and can cause a jump of depreciated coins.

Finally, cryptocurrencies are uniquely influenced by their close ties to online communities and real-time information flow via social media. Retail investors often exchange and collect information on social media platforms, which allows them to collectively behave like a singular, large trader (herding). It is also widely known that price manipulations such as wash trading, rug pull, and pump-and-dump schemes exist in the cryptocurrency market (Cong et al., 2023). The interplay between online sentiments and price dynamics, combined with the potential price manipulation, can create bubble-burst patterns of cryptocurrencies.

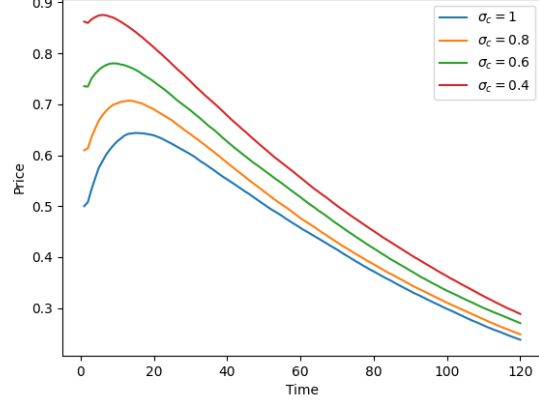
Put together, the excessive volatility and bubble-burst cycles of the cryptocurrency market can be explained by 1) the overconfidence of retail investors, 2) high valuation uncertainty, 3) margin trading, and 4) investor herding and price manipulation. As explained above, these factors are also associated with short-term reversal, which explains why the reversal effect is more pronounced in the cryptocurrency market. The weak momentum effect of past losers may be attributable to the disposition effect. Regarding the momentum effect observed among large past winners, we conjecture that it is because major coins have lower valuation uncertainty and higher growth potential, thus attracting more institutional and long-term investors. Major coins possess features associated with lower valuation uncertainty, *i.e.*, they are bigger, older, and less volatile, and institutional and long-term investors are considered less overconfident, which can together contribute to the continuation of overreaction and momentum. The remainder of this section examines whether our conjecture can be supported by the data.

4.3.1 Momentum and reversal of all coins

The reversal effect observed from winners among all coins can be explained by overreaction, followed by profit realization and correction. Table 10 shows that past winners transitioning into losers earn significantly higher look-back period returns, are smaller, and have higher turnover than those remaining winners, implying investors' overreaction to these coins during the look-back period.



(a) Valuation uncertainty and price dynamics



(b) Overconfidence and price dynamics

Figure 6: Price dynamics for different levels of valuation uncertainty and overconfidence

The low returns of the winners in the highest-attention and overreaction groups (G5, V5, VM5, $\Delta V5$, and CO5 in Table 11) also imply that coins subject to significant overreaction during the look-back period tend to reverse during the holding period. Table 11 further shows that the momentum strategy earns lower returns in smaller coin groups, suggesting that the reversal effect is stronger when valuation uncertainty is higher.

Cryptocurrency news is usually salient and anecdotal, and investors are likely to overreact to it (Odean, 1998b). Cai and Zhao (2024) also argue that investors tend to overvalue coins with salient upside returns and undervalue those with salient downside returns, leading to subsequent reversals through a correction mechanism. These observations strengthen the overreaction-correction explanation.

Figure 7 (a) plots futures/spot trading volume ratio against size. It reveals that smaller coins have higher futures trading volume, indicating that they are traded by more overconfident, speculative investors employing high leverage. Chhatwani and Parija (2023) and Kim et al. (2022) also find that cryptocurrency investors are more overconfident.

Overall, evidence supports our conjecture that the reversal effect is attributable to investor overconfidence, high valuation uncertainty, and margin trading.

The weak momentum effect of losers can be explained by the disposition effect, as investors’ reluctance to realize losses can make the price drift downward slowly. To test the disposition effect, we run a Fama-MacBeth regression of the future return on the capital gains overhang, following Grinblatt and Han (2005). The capital gains overhang is calculated using the daily returns of the past twelve weeks, and the future return is defined as the one-week ahead return.

As shown in Table IA20, the coefficient on the capital gains overhang is statistically significant with a t -statistic of 3.63, supporting the disposition effect explanation. However, when the sample is split into two groups—positive and negative capital gains overhang groups, the coefficient is statistically significant only when capital gains overhang is negative. Furthermore, when the return is negative, trading volume is lower—85.3% of the trading volume when the return is positive, which also reflects investors’ reluctance to realize losses (unreported). These findings support our conjecture that the disposition effect drives the momentum of past losers.

4.3.2 Momentum and reversal of top 5%

Retail investors and speculators are likely to prefer small, alternative coins for their high volatility and potential jackpot returns, whereas long-term and institutional investors prefer major coins for their liquidity, lower volatility, and higher growth potential. Since long-term and institutional investors are less overconfident and trade less frequently, overreaction can last longer, driving the momentum effect of large winners.

Although there is no concrete evidence supporting this conjecture, speculative investors’ preference for small coins can be inferred indirectly. For instance, Korean investors, known for their speculative investment style, prefer meme coins, and these coins often rank among the highest volume coins on Upbit, the largest Korea-based crypto exchange. The average volatility of trading volume change and the average relative volume of the smallest 95% are significantly higher at 130.5% and 36.3% than those of the largest 5%, 74.9% and 15.1% (unreported). Figure 7 (a) also shows that smaller coins are more subject to margin trading. These findings suggest more

speculative trading targeting small coins.

Kogan et al. (2024) find that while retail investors are contrarian in general, they follow a momentum-like strategy in cryptocurrencies: they hold on to their cryptocurrency investments even after large price movements. The authors conjecture that retail investors interpret an increase in price as a signal of a higher likelihood of future widespread adoption and update their expectations accordingly. They draw their conclusion based on trading data of three major cryptocurrencies—Bitcoin, Ethereum, and Ripple. Therefore, their findings likely represent the behaviors of investors trading major cryptocurrencies rather than those of general cryptocurrency investors, and are consistent with our conjecture that major cryptocurrency investors are mainly long-term investors. As shown in Figure 7 (b), most coins, except the largest ones, have declined more than 80% from their all-time-high prices, reflecting investors’ skeptical view on them. Thus, investors trading them are unlikely to expect long-term capital appreciation or widespread adoption, but to follow trends seeking short-term gains.

Overall, evidence supports our conjecture that overreaction induced by overconfidence, valuation uncertainty, and margin trading is an important driver of cryptocurrencies’ price dynamics. Overreaction appears stronger but short-lived among small coins due to stronger overconfidence and higher valuation uncertainty, while it lasts longer among larger coins, causing momentum.

4.3.3 Comparison to Liu et al. (2022)

Liu et al. (2022) attribute cross-sectional momentum to attention-based overreaction. They employ trading volume and Google trend data as indicators of investor attention and find that the momentum effect is concentrated among high-attention coins. Since the cryptocurrency market comprises a small number of well-known large coins and other lesser-known small coins, they argue that large coins attract more investor attention and exhibit stronger momentum.

However, our findings do not support the attention-based explanation. The momentum portfolio formed of higher Google trend or volume coins (high-attention coins) performs comparably to

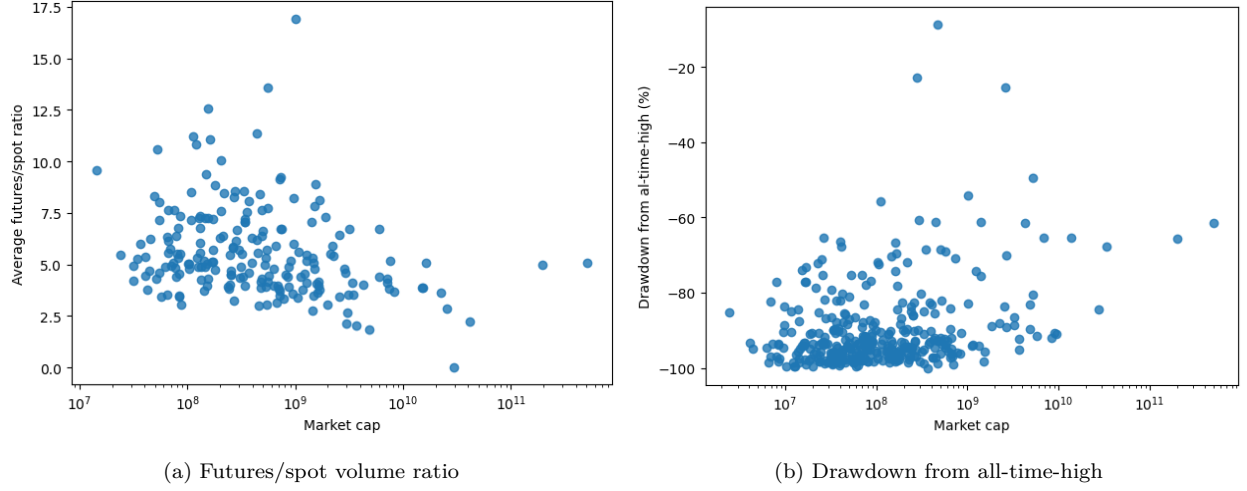


Figure 7: Size effects on margin trading and drawdowns

that of lower Google trend or volume coins (low-attention coins). In addition, winners, where the momentum effect is concentrated, perform worse when attention is high. When attention is measured using relative volume or volume change, the momentum portfolio performs better among low-attention coins. These results contradict the attention-based explanation. A moderate level of attention may induce continued overreaction and momentum, but a significant increase in attention causes reversal following overreaction.

We also do not find evidence that the momentum strategy performs better during periods of high attention. When the return of the (14, 7) strategy is regressed on Google trend and its interaction with look-back period return, neither is significant (Table IA21). The return is also not significantly correlated with market trading volume.

The differences between our findings and those of Liu et al. (2022) can be attributed to different filtering methods, rather than different sample periods. Table IA7 replicates Table AI XI in Liu et al. (2022), which examines the momentum effect in different volume groups, using their filtering method (market capitalization > 1 million USD) and sample period (January 2014 to July 2020). The table also reports the results when our filtering method (both market capitalization and trading volume > 1 million USD) is used and when the sample period is extended to our sample period

(January 2014 to August 2023).

When the same filtering method and sample period are used, our results are consistent with those of Liu et al. (2022); the momentum effect is more evident in the high-volume group. When our sample period is used, the momentum effect remains stronger in the high-volume group. However, when coins with trading volume below 1 million USD are excluded, the momentum effect disappears in the high-volume group, and the momentum strategy earns a lower return in that group than in the low-volume group. Applying the volume filter eliminates more than half of the coins from the sample. Thus, the low-volume group contains coins with relatively large trading volume that were included in the high-volume group in Liu et al. (2022), and only coins with significantly higher trading volume remain in the high-volume group. As discussed earlier, these coins tend to be overvalued during the look-back period and experience a reversal over the holding period.

5 Conclusion

We conduct a comprehensive analysis of both time-series and cross-sectional momentum in the cryptocurrency market, accounting for real-world considerations and providing a more accurate assessment of their performance. We find strong evidence of time-series momentum, but evidence of cross-sectional momentum is very limited. Momentum profits predominantly arise from the long leg, while the short leg incurs significant losses, posing a threat of liquidation to the strategy. A momentum-based market-neutral strategy that can generate stable profits appears unattainable. The maximum Sharpe ratio we obtain from a momentum strategy is about 1.5. Considering the substantial tail risk, the limited number of liquid coins, and the high dominance of a few major coins, it is questionable whether a cryptocurrency momentum strategy can be qualified as an alternative investment vehicle for institutional investors. Furthermore, it is important to recognize that we introduce a data-snooping bias by examining various pairs of look-back and holding periods; thus, the findings in this paper should be considered a best-case scenario.

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A Long-term profitability of a portfolio

This section demonstrates the limitations of the significance test of the mean return in the cryptocurrency market and proposes alternative testing methods. Let P_t denote the value (price) of a portfolio at time t . When the rate of return, $r_t = P_t/P_{t-1} - 1$, is sufficiently small, the log return, $l_t = \log(P_t/P_{t-1}) = \log(1 + r_t)$, can be approximated by r_t , and if the sample mean of r_t , $t = 1, \dots, T$, $\bar{r} = \sum_{t=1}^T r_t/T$, is statistically significantly positive, one may conclude that the sample mean of l_t , $\bar{l} = \sum_{t=1}^T l_t/T = \log(P_T/P_0)$, will also be positive and the portfolio will be profitable. However, if the return is highly volatile and often involves jumps, which is the case in the cryptocurrency market, this approximation is no longer valid and the cumulative return can be negative even when the mean return is significantly positive, due to Jensen's inequality. This is a well-known fact but often disregarded because returns are usually small enough in the securities market. In the cryptocurrency market, on the other hand, three-digit returns are not rare and examining only the mean return can lead to a misleading conclusion. We investigate this point in detail under the assumption that the price follows a geometric Brownian motion or a jump-diffusion process.

A.1 Diffusion process

Suppose P_t follows a geometric Brownian motion of the form:

$$dP_t/P_t = \mu dt + \sigma dW_t, \quad (8)$$

where μ denotes the drift, σ volatility, and W_t a standard Wiener process. The log return between time 0 and t , $X_t = \log(P_t/P_0)$, is given by

$$X_t = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t, \quad (9)$$

and its expectation has the form

$$E[X_t] = \left(\mu - \frac{\sigma^2}{2} \right) t. \quad (10)$$

Equation (10) shows that if $\mu - \sigma^2/2 < 0$, the expected value of the portfolio converges to 0 even when μ is positive.

When returns are measured at a short interval, the sample mean $\bar{r}$ roughly estimates μ , and one would reject the null hypothesis $\mu = 0$ if its t -statistic is greater than a critical value. If both μ and σ increase by a factor of $a > 1$, $\mu - \sigma^2/2$ decreases and eventually becomes negative as a increases, but the test statistic will remain at the same level. That is, if the mean and the standard deviation are sufficiently high, there is a considerable chance of the long-term value converging to zero even when $\bar{r}$ is statistically significantly positive.

A.2 Jump-diffusion process

Cryptocurrency returns are skewed and fat-tailed and might be better described by a jump-diffusion process. Following Merton (1976), we assume X_t has the distribution

$$X_t = \left(\mu - \frac{\sigma^2}{2} - \lambda k \right) t + \sigma W_t + \sum_{i=0}^{N_t} Y_i, \quad (11)$$

where N_t is a Poisson variable with the intensity parameter λ , $Y_i \sim N(\nu, \delta^2)$ is a random jump size, and $k = \exp(\nu + \delta^2/2) - 1$. The first four moments of l_t are given by (Matsuda, 2004):

$$E[l_t] = \mu - \frac{\sigma^2}{2} - \lambda k + \lambda \nu, \quad (12)$$

$$Var[l_t] = \sigma^2 + \lambda \delta^2 + \lambda \nu^2, \quad (13)$$

$$Skew[l_t] = \frac{\lambda(3\delta^2\nu + \nu^3)}{(\sigma^2 + \lambda\delta^2 + \lambda\nu^2)^{3/2}}, \quad (14)$$

$$Kurt[l_t] = \frac{\lambda(3\delta^4 + 6\nu^2\delta^2 + \nu^4)}{(\sigma^2 + \lambda\delta^2 + \lambda\nu^2)^2}. \quad (15)$$

The second-order approximation of k yields

$$E[l_t] \approx \mu - \frac{s^2}{2} - \frac{\lambda\delta^2}{2} \left(\frac{\delta^2}{4} + v \right), \quad (16)$$

where $s^2 = \sigma^2 + \lambda\delta^2 + \lambda\nu^2$ is the variance. For the same μ and variance, the expected log return will be smaller under a jump-diffusion process than under a diffusion process if $\delta^2/4 + \nu > 0$. As shown in the paper, cryptocurrency returns are positively skewed, which implies a positive ν . Thus, if the value of a portfolio follows a jump-diffusion process, *ceteris paribus*, its expected log return will be smaller and there will be a higher chance of the long-term value converging to zero.

A.3 Simulation

Under the diffusion process, the sample mean log return $\bar{l}$ has the interval

$$\bar{l} \in \left(\mu - \frac{\sigma^2}{2} - c \frac{\sigma}{\sqrt{T}}, \mu - \frac{\sigma^2}{2} + c \frac{\sigma}{\sqrt{T}} \right), \quad (17)$$

where c is a critical value. Assuming that the time interval is sufficiently small such that the variance of r_t is approximately equal to σ^2 , the interval of the sample mean return $\bar{r}$ becomes

$$\bar{r} \in \left(\mu - c \frac{\sigma}{\sqrt{T}}, \mu + c \frac{\sigma}{\sqrt{T}} \right). \quad (18)$$

Similarly, under the jump-diffusion process, these intervals are given by

$$\bar{l} \in \left(\mu - \frac{\sigma^2}{2} - \lambda k + \lambda\nu - c \frac{s}{\sqrt{T}}, \mu - \frac{\sigma^2}{2} - \lambda k + \lambda\nu + c \frac{s}{\sqrt{T}} \right), \quad (19)$$

and

$$\bar{r} \in \left(\mu - c \frac{s}{\sqrt{T}}, \mu + c \frac{s}{\sqrt{T}} \right). \quad (20)$$

When there are jumps, the variance of r_t cannot be approximated by the variance of l_t , and the interval in Equation (20) can be significantly different from the true interval. Thus, we also estimate the intervals via simulation.

We draw the intervals of $\bar{l}$ and $\bar{r}$ using the parameters estimated from the daily returns of the best cross-sectional momentum strategy, (14, 7), identified in Section 4.2. The mean, standard deviation, skewness, and kurtosis of the daily returns are respectively 0.005, 0.072, 14.442, and 466.270. The corresponding values of the daily log returns are 0.003, 0.060, 1.805, and 94.329.

The parameters of the diffusion process are estimated using the maximum likelihood estimator, which gives $\hat{\mu} = 0.005$ and $\hat{\sigma} = 0.060$. For the jump-diffusion process, a constrained maximum likelihood estimator described in Section A.4 is used, which gives $\hat{\mu} = 0.005$, $\hat{\sigma} = 0.032$, $\hat{\nu} = 0.051$, $\hat{\delta} = 0.394$, and $\hat{\lambda} = 0.016$. The Kolmogorov–Smirnov (KS) statistic is 0.1551 for the diffusion process and 0.0404 for the jump-diffusion process. Although both processes are rejected, the jump-diffusion process has a smaller statistic.

Based on these estimates, we draw the intervals for the following range of the parameters: $\mu = 0.0015a$, $\sigma = 0.0200a$ for the diffusion process, and $\mu = 0.0015a$, $\sigma = 0.0100a$, $\nu = 0.0150a$, $\delta = 0.1095a$, $\lambda = 0.0150$ for the jump-diffusion process, where $a \in [1, 10]$. These parameter values are chosen so that they generate the same variance under both processes and pass (roughly) through the estimated parameters. The skewness and the kurtosis of the jump-diffusion process are independent of a and remain the same at 1.615 and 105.729. The sample period T and the confidence level are set to 1,000 days and 95%, respectively.

Figure A.1 presents the simulation results, where Panels (a) and (b) display the intervals under the diffusion process and the jump-diffusion process, respectively. The shaded regions represent the intervals: the regions with a smooth boundary are obtained from Equations (17) to (20), and the regions with a wiggly boundary are obtained from simulations with 10,000 iterations. The dark blue region is where $\bar{r}$ is statistically significantly positive.

The graphs reveal several important points. First, as μ increases, the expected log return, and

therefore the expected profit, becomes negative, whereas the chance of rejecting the null hypothesis $E[r] = 0$ remains the same. The expected log return turns negative when $\mu > 0.011$ under the diffusion process and when $\mu > 0.009$ under the jump-diffusion process. For the same μ and variance, the expected log return is lower under the jump-diffusion assumption, and the gap widens as μ increases. If the value of a portfolio follows a jump-diffusion process, it is more likely to converge to 0 even when the mean return is statistically significantly positive. Second, under the jump-diffusion process, the interval of $\bar{r}$ is significantly wider and the t -test of $\bar{r}$ becomes almost meaningless, especially when μ is high. Third, even when $\bar{l}$ is positive, there is a nontrivial chance of its true mean being negative. Put together, in a highly volatile market such as the cryptocurrency market, examining the simple mean return or the Sharpe ratio is not sufficient. To adequately test the long-term profitability, it is necessary to check the significance of log returns.

A.4 Jump-diffusion parameter estimation

The parameters of the jump-diffusion process are estimated via a constrained maximum likelihood estimator (MLE) as described below.

$$\begin{aligned}
& \max_{\theta} \sum_{t=1}^T \log f(l_t; \theta) \\
& \text{subject to} \\
& E[l_t] = \mu - \frac{\sigma^2}{2} - \lambda k + \lambda \nu = \bar{l} \\
& Var[l_t] = \sigma^2 + \lambda \delta^2 + \lambda \nu^2 = \bar{v} \\
& Skew[l_t] = \frac{\lambda(3\delta^2\nu + \nu^3)}{(\sigma^2 + \lambda\delta^2 + \lambda\nu^2)^{3/2}} = \bar{s} \\
& Kurt[l_t] = \frac{\lambda(3\delta^4 + 6\nu^2\delta^2 + \nu^4)}{(\sigma^2 + \lambda\delta^2 + \lambda\nu^2)^2} = \bar{k},
\end{aligned} \tag{21}$$

where $\theta = \{\mu, \sigma, \nu, \delta, \lambda\}$, and $\bar{l}, \bar{v}, \bar{s}$, and $\bar{k}$ are the first four sample moments. The density function $f(l_t; \theta)$ is given by

$$f(l_t; \theta) = \sum_{i=0}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!} N\left(l_t; \mu - \frac{\sigma^2}{2} - \lambda k + i\nu, \sigma^2 + i\delta^2\right), \quad (22)$$

where $N(\cdot)$ denotes a normal density function.

The moment constraints are imposed because the usual MLE does not have a unique maximum and the moments of the estimated process by MLE, skewness and kurtosis in particular, can be significantly different from the sample moments. Below are the estimation results.

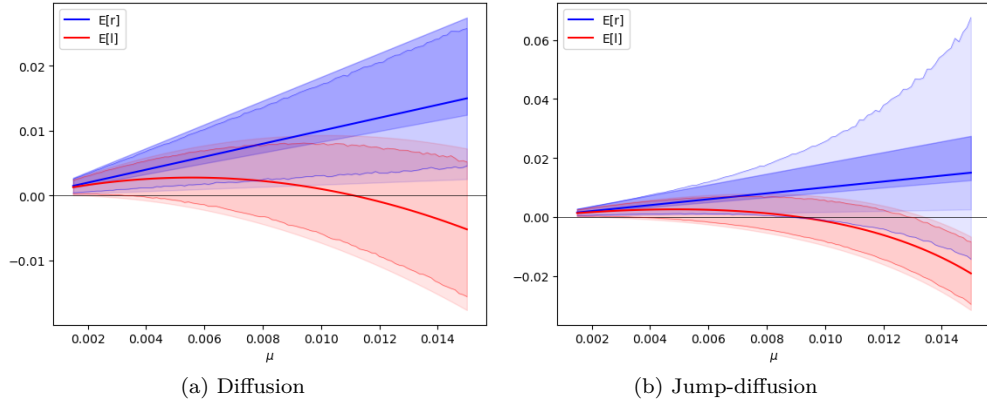


Figure A.1: Intervals of sample mean returns and sample mean log returns.

Sample moments			Estimation			
	r_t	l_t	$\hat{\theta}$		Estimated moments	
mean	0.00479	0.00285	μ	0.00477	mean	0.00285
std	0.07166	0.06004	σ	0.03200	std	0.06003
skew	14.44195	1.80449	ν	0.05096	skew	1.80338
kurt	466.26990	94.32867	δ	0.39418	kurt	94.13231
			λ	0.01633		

B The model of Daniel et al. (1998)

Daniel et al. (1998) assume two types of investors, an informed, overconfident investor I and an

uninformed investor U . At time 1, I receives a private signal about the underlying security:

$$s_1 = \theta + \epsilon, \quad (23)$$

where $\theta \sim N(0, \sigma_\theta^2)$ is the terminal value of the security and $\epsilon \sim N(0, \sigma_\epsilon^2)$ is noise. From time 2 to T , public signals ϕ_t arrive:

$$\phi_t = \theta + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \quad (24)$$

I initially estimates the error variance of the private signal as $\sigma_{c,1}^2$ and updates his precision, $v_{c,t} = 1/\sigma_{c,t}^2$, as follows: If $\text{sign}(s_1 - \Phi_{t-1}) = \text{sign}(\phi_t - \Phi_{t-1})$, $v_{c,t} = (1 + \bar{k})v_{c,t-1}$, otherwise, $v_{c,t} = (1 - \underline{k})v_{c,t-1}$, where $\bar{k} > \underline{k} > 0$, and Φ_t is the average of all public signals up to t :

$$\Phi_t = \frac{1}{t-1} \sum_{\tau=2}^t \phi_\tau = \theta + \frac{1}{t-1} \sum_{\tau=2}^t \eta_\tau. \quad (25)$$

The price of the security at t is then given by

$$P_t = \frac{(t-1)v_\eta \Phi_t + v_{c,t}s_1}{v_\theta + (t-1)v_\eta + v_{c,t}}, \quad (26)$$

where $v_\eta = 1/\sigma_\eta^2$ and $v_\theta = 1/\sigma_\theta^2$.

For simulation, Daniel et al. (1998) assume $\theta = 0$, $s_1 = 1$, $\bar{k} = 0.75$, $\underline{k} = 0.1$, $\sigma_\theta^2 = \sigma_\epsilon^2 = 1$, and $\sigma_\eta^2 = 7.5$. We use the same values to construct the baseline case and examine the effect of valuation uncertainty by increasing σ_θ to 1.2, 1.4, and 2.0, and the effect of overconfidence by decreasing $\sigma_{c,1}$ to 0.8, 0.6, and 0.4. Each simulation consists of 50,000 iterations.

Internet Appendix for

Momentum in the Cryptocurrency Market: A Comprehensive Analysis under Realistic Assumptions

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A Data

Table IA1: CoinMarketCap Data

Coin	Event
Conscious Value Network (CVNT)	200:1 swap on March 8, 2021
Cocos-BCX (COCOS, rebranded as COMBO)	1000:1 swap on January 19, 2021

(a) Manually adjusted events

USDT	USDC	BUSD	DAI	TUSD	FRAX	USDP	USDD
GUSD	FEI	USDJ	LUSD	USTC	EURS	TRIBE	vUSDC
USDx	vBUSD	XSGD	VAI	SUSD	CUSD	SBD	vUSDT
EUROc	OUSD	USDK	RSV	KRT	GYEN	CEUR	HUSD
BIDR	IDRT	vDAI	DGD	BITCNY	XCHF	DGX	EOSDT
BITUSD	ZUSD	ESD	USDS	BAC	ALUSD	YUSD	MIMATIC
MIM	DOLA	USDs	EURT	MUSD	USX	AGEUR	mCUSD
TOR	TRYB	mCEUR	XIDR	1GOLD	MXNT	XSTUSD	USDEX
MTR	BRCP	CUSDT	SEUR	xDAI	DUSD	USDH	USDZ
DJED	JPYC	PAR	DSD	ONC	FUSD	ONEICHI	CADC
XUSD	KBC	USDB	MONEY	USDR	COFFIN	IRON	DPT
WANUSDT							

(b) List of stable coins

Year	Number				Market cap (100 mil)				Volume (100 mil)			
	None	M	V	M&V	None	M	V	M&V	None	M	V	M&V
2014	287	25	3	2	75.21	74.47	64.27	63.97	0.36	0.35	0.29	0.28
2015	483	28	3	2	45.18	44.73	41.35	40.67	0.41	0.39	0.38	0.38
2016	499	52	5	4	107.38	106.47	101.40	100.61	1.23	1.17	1.13	1.12
2017	579	169	54	43	1259.97	1238.87	1227.70	1214.64	53.33	51.69	51.99	51.32
2018	711	448	192	147	2967.85	2946.89	2906.97	2882.72	146.44	143.65	144.24	142.92
2019	1080	553	213	160	2079.26	2073.15	2043.33	2033.46	365.07	361.38	362.55	360.45
2020	1559	709	317	240	3173.84	3151.09	3100.53	3080.29	748.57	739.67	743.33	738.43
2021	3102	1269	832	562	18573.99	18379.11	18278.61	18210.29	1618.12	1564.21	1603.50	1561.90
2022	6570	1382	883	544	12055.43	12024.80	11860.31	11797.77	879.60	804.92	857.45	802.65
2023	8966	1332	694	471	10310.87	10283.47	10125.91	10055.93	446.00	416.43	434.08	414.18

(c) Effects of filtering methods (M: Size > 1 mil USD, V: Volume > 1 mil USD)

Table IA2: Slippage

This table reports the average slippage per coin and its descriptive statistics. The results are from 15,661,698 records of actual market orders in the Binance futures market from June 24, 2023 to August 20, 2023. The slippages are in basis points.

Symbol	Slip	Symbol	Slip	Symbol	Slip	Symbol	Slip	Symbol	Slip	Symbol	Slip
1000LUNC	0.48	BCH	0.21	ENJ	1.16	IOTX	1.77	OCEAN	1.13	STORJ	1.30
1000SHIB	0.51	BEL	0.69	ENS	0.65	JASMY	0.84	OGN	4.02	STX	0.62
1000XEC	1.37	BLUR	1.09	EOS	4.43	JOE	1.01	OMG	0.71	SUI	0.62
11INCH	1.15	BLZ	0.84	ETC	0.25	KAVA	0.49	ONE	2.51	SUSHI	0.56
AAVE	0.64	BNB	0.18	ETH	0.02	KEY	0.96	ONT	1.81	SXP	1.01
ACH	1.54	BTC	0.01	FET	1.52	KLAY	1.94	OP	0.30	THETA	0.58
ADA	1.40	C98	2.57	FIL	0.90	KNC	0.69	PEOPLE	2.76	TLM	2.94
AGIX	1.55	CELO	7.13	FLM	4.94	KSM	1.30	PERP	1.30	TOMO	0.38
ALGO	2.94	CELR	1.91	FLOW	5.88	LDO	0.29	PHB	0.75	TRB	2.76
ALICE	3.48	CFX	2.12	FTM	1.56	LEVER	3.11	QNT	0.54	TRU	1.09
ALPHA	0.45	CHR	2.07	FXS	0.76	LINA	2.79	QTUM	1.47	TRX	0.48
AMB	4.37	CHZ	0.49	GALA	1.61	LINK	0.61	RAD	1.75	T	1.57
ANKR	1.55	CKB	1.60	GAL	0.65	LIT	4.88	RDNT	1.32	UMA	1.87
ANT	0.82	COMP	0.75	GMT	1.77	LPT	0.83	REEF	2.05	UNFI	1.09
APE	2.11	COTI	0.78	GMX	0.62	LQTY	0.63	REN	0.68	UNI	0.65
API3	3.41	CRV	5.78	GRT	0.45	LRC	1.34	RLC	0.42	VET	1.82
APT	0.61	CTK	0.87	GTC	3.38	LTC	0.49	RNDR	0.34	WAVES	0.32
ARB	0.38	CTSI	2.49	HBAR	0.73	MAGIC	0.57	ROSE	0.74	WOO	0.46
ARPA	0.74	CVX	1.20	HFT	1.05	MANA	0.96	RSR	1.52	XEM	11.81
AR	0.89	DAR	3.02	HIGH	2.83	MASK	1.09	RUNE	3.16	XLM	0.38
ASTR	0.89	DASH	1.04	HOOK	2.97	MATIC	0.59	RVN	1.86	XMR	0.25
ATA	3.82	DENT	3.81	HOT	2.40	MAV	1.19	SAND	0.87	XRP	0.80
ATOM	0.45	DGB	4.18	ICP	0.84	MDT	1.06	SFP	0.98	XTZ	3.58
AUDIO	1.83	DODO	3.36	ICX	1.56	MINA	0.90	SKL	1.37	XVG	1.06
AVAX	0.35	DOGE	0.64	IDEX	0.87	MKR	0.46	SNX	1.52	XVS	1.13
AXS	0.63	DOT	0.74	ID	1.39	MTL	0.37	SOL	0.20	YFI	0.54
BAKE	3.53	DUSK	0.79	IMX	0.53	NEAR	2.59	SPELL	1.00	ZEC	1.14
BAL	0.89	DYDX	1.83	INJ	0.52	NEO	0.53	SSV	1.82	ZEN	0.67
BAND	0.46	EDU	0.65	IOST	0.62	NKN	0.87	STG	0.60	ZIL	1.61
BAT	1.67	EGLD	0.95	IOTA	1.78	NMR	2.49	STMX	6.64	ZRX	1.73

(a) Average slippage per coin

Count	Mean	Std	Min	25%	50%	75%	Max
180	1.53	1.49	0.01	0.63	1.05	1.83	11.81

(b) Descriptive statistics

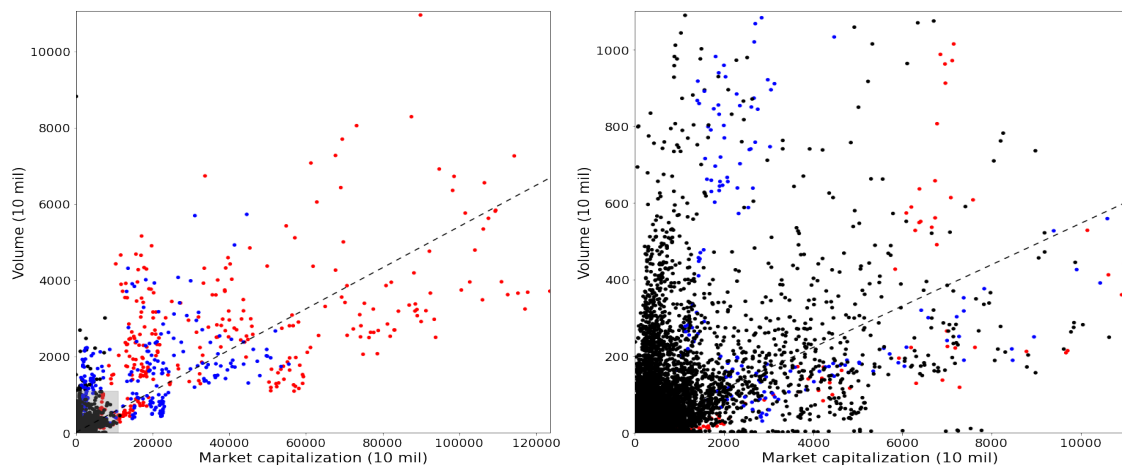


Figure IA1: Market capitalization vs. trading volume

This figure displays scatter plots of market capitalization and trading volume. The market capitalization and trading volume are 30-day averages. Red and blue dots respectively denote Bitcoin and Ethereum. The graph on the right magnifies the shaded area in the graph on the left.

A.1 Google trends

Individual coins' Google Trends data are downloaded from the Google Trends website.¹⁶ Some coins in our sample have names that do not exclusively represent the coins, *e.g.*, 'CHIP', 'Cosmos', 'Dash', and 'Quant'. Therefore, if a coin's name does not include one of the words 'coin', 'network', 'protocol', and 'chain', we add the word 'coin' to the search term, *e.g.*, 'CHIP coin'.

The maximum period for weekly Google Trends data is limited to five years. Therefore, for time-series momentum analysis that requires samples from December 2013, monthly Google Trends data are used. For cross-sectional momentum analysis, weekly Google Trends data from September 2019 to August 2023 are used, and coins are restricted to those listed on the Binance futures market. The reason for this restriction is that many coins are relatively unknown and have 0 Google Trends values in many periods.

Investor attention is proxied by the change in Google Trends. For weekly trend data, the change in Google Trends is defined as the difference between the current week's trend and the average trend over the preceding four weeks. For monthly trend data, it is defined as the difference between the current month's trend and the previous month's trend. The change in Google Trends is standardized to have zero mean and unit variance.

¹⁶<https://trends.google.com/trends/>

B Replication of Liu and Tsyvinski (2021) and Liu et al. (2022)

We replicate some of the results in Liu and Tsyvinski (2021) and Liu et al. (2022) using our sample to ensure that we obtain similar results when following their sample periods and methodologies. In the tables below, HKR denotes our results, and LT and LTW respectively denote the results in Liu and Tsyvinski (2021) and Liu et al. (2022).

Table IA3: Time-series momentum (Table 6 in LT)

LT use Bitcoin returns from CoinDesk for the period January 1, 2011 to April 29, 2013, and from CoinMarketCap thereafter. We use Bitcoin returns from Investing.com for the period when they are not available from CoinMarketCap.

Panel A						
Weekly	$R_{t,t+1}$	$R_{t,t+2}$	$R_{t,t+3}$	$R_{t,t+4}$	$R_{t,t+6}$	$R_{t,t+8}$
R_t	0.20**	0.49**	0.81***	1.07***	1.55**	1.62*
R^2	0.04	0.08	0.09	0.08	0.06	0.02

(a) LT

Weekly	$R_{t,t+1}$	$R_{t,t+2}$	$R_{t,t+3}$	$R_{t,t+4}$	$R_{t,t+6}$	$R_{t,t+8}$
R_t	0.20**	0.50***	0.79***	1.10***	1.59**	1.64*
R^2	0.04	0.08	0.08	0.08	0.06	0.02

(b) HKR

Panel B - with forward looking							
Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-10.70	1.10	(0.92)	3.59	(1.69)	7.21	(1.58)
Middle	1.74	1.21	(1.34)	2.77	(1.92)	8.76	(3.43)
High	19.43	8.01	(4.30)	16.22	(4.94)	39.08	(5.30)
Diff		6.91		12.63		31.87	

(a) LT

Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-10.61	0.69	(0.57)	2.86	(1.36)	5.44	(1.31)
Middle	1.52	1.70	(1.88)	3.79	(2.24)	9.72	(2.82)
High	19.32	7.85	(4.28)	15.53	(4.91)	39.00	(5.51)
Diff		7.16		12.67		33.56	

(b) HKR

Panel B - without forward looking							
Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-10.86	0.80	(0.63)	2.22	(1.18)	2.41	(0.90)
Middle	1.88	1.44	(1.50)	3.05	(1.94)	8.53	(3.08)
High	18.42	6.42	(3.34)	13.25	(3.91)	31.60	(4.27)
Diff		5.62		11.03		29.19	

(a) LT

Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-10.12	1.19	(1.00)	3.32	(1.64)	5.85	(1.48)
Middle	1.52	1.70	(1.88)	3.79	(2.24)	9.72	(2.82)
High	19.52	7.98	(4.29)	15.87	(4.97)	39.23	(5.48)
Diff		6.79		12.55		33.38	

(b) HKR

Table IA4: Google trends (Table 8 in LT)

Panel A						
Weekly	$R_{t,t+1}$	$R_{t,t+2}$	$R_{t,t+3}$	$R_{t,t+4}$	$R_{t,t+6}$	$R_{t,t+8}$
$Google_t$	0.03*** (3.92)	0.05*** (4.33)	0.07*** (4.23)	0.10*** (3.99)	0.09** (1.98)	0.07 (1.30)
R^2	0.03	0.04	0.03	0.02	0.01	0.00

(a) LT

Weekly	$R_{t,t+1}$	$R_{t,t+2}$	$R_{t,t+3}$	$R_{t,t+4}$	$R_{t,t+6}$	$R_{t,t+8}$
$Google_t$	0.02** (2.38)	0.05*** (3.63)	0.06*** (3.70)	0.09*** (4.29)	0.09** (2.45)	0.08 (1.42)
R^2	0.02	0.02	0.02	0.02	0.01	-0.00

(b) HKR

Panel B - with forward looking							
Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-0.45	0.43	(0.42)	0.02	(0.01)	0.10	(0.04)
Middle	-0.02	2.55	(2.03)	6.79	(2.73)	19.77	(3.11)
High	0.48	6.53	(3.82)	13.96	(4.89)	32.05	(5.47)
Diff		6.09		13.93		31.95	

(a) LT

Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-0.46	0.08	(0.07)	0.50	(0.35)	0.81	(0.32)
Middle	-0.01	2.82	(2.11)	6.72	(2.67)	17.74	(3.20)
High	0.48	7.19	(4.37)	14.68	(4.94)	33.31	(5.25)
Diff		7.08		14.18		32.50	

(b) HKR

Panel B - without forward looking							
Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-0.45	0.70	(0.68)	1.06	(0.70)	1.98	(0.78)
Middle	-0.01	1.15	(1.12)	2.13	(1.35)	4.90	(1.89)
High	0.59	6.12	(3.56)	13.75	(4.58)	32.65	(5.20)
Diff		5.42		12.96		30.67	

(a) LT

Rank	R_t	$R_{t,t+1}$	t -stat	$R_{t,t+2}$	t -stat	$R_{t,t+4}$	t -stat
Low	-0.40	-0.33	(-0.33)	-0.40	(-0.30)	0.35	(0.15)
Middle	-0.01	3.83	(2.54)	8.99	(3.13)	21.36	(3.33)
High	0.48	7.19	(4.37)	14.68	(4.94)	33.31	(5.25)
Diff		7.52		15.08		32.99	

(b) HKR

Table IA5: Summary statistics (Table I in LTW)

	Number	Mcap Mean	Mcap Median	Vol Mean	Vol Median
2014	109	239.8	3.9	1146.1	36.2
2015	77	134.5	2.8	1187.6	11.5
2016	155	160.6	3.4	1795.0	24.0
2017	795	439.4	9.0	18661.1	131.4
2018	1559	363.3	8.8	21184.2	124.9
2019	1085	300.5	5.4	59115.1	139.7
2020	665	440.2	5.4	125249.2	210.8
Full	1827	353.3	6.6	44991.0	121.9

(a) LTW

	Number	Mcap Mean	Mcap Median	Vol Mean	Vol Median
2014	130	250.7	4.0	1194.0	31.3
2015	74	144.8	3.1	1270.1	11.2
2016	151	184.3	3.7	2079.6	21.2
2017	556	618.0	12.0	26000.8	172.4
2018	732	618.2	16.0	30234.5	301.3
2019	852	364.3	7.0	63482.2	239.6
2020	882	367.4	6.6	107386.4	365.5
Full	1400	452.2	8.7	55345.1	225.5

(b) HKR

Table IA6: (Cross-sectional) momentum strategy returns (Table IV in LTW)

	1	2	3	4	5	5-1
r 1,0 Mean	-0.002	0.000	0.010	0.036**	0.023**	0.025**
r 1,0 t(Mean)	(-0.19)	(0.04)	(1.45)	(2.52)	(2.03)	(2.19)
r 2,0 Mean	0.000	0.005	0.009	0.017**	0.031***	0.031***
r 2,0 t(Mean)	(0.01)	(0.66)	(1.33)	(2.15)	(2.93)	(2.90)
r 3,0 Mean	0.005	0.002	0.016*	0.017**	0.036***	0.031***
r 3,0 t(Mean)	(0.60)	(0.28)	(1.94)	(2.30)	(3.21)	(2.65)
r 4,0 Mean	0.002	0.005	0.009	0.020**	0.025**	0.022**
r 4,0 t(Mean)	(0.30)	(0.66)	(1.28)	(2.45)	(2.32)	(2.26)
r 4,1 Mean	0.003	0.007	0.021**	0.011	0.020**	0.017*
r 4,1 t(Mean)	(0.35)	(0.94)	(2.34)	(1.51)	(2.02)	(1.82)

(a) LTW

	1	2	3	4	5	5-1
r 1,0 Mean	-0.003	0.003	0.013	0.033**	0.031**	0.034***
r 1,0 t(Mean)	(-0.38)	(0.34)	(1.62)	(2.32)	(2.31)	(2.65)
r 2,0 Mean	0.000	0.002	0.011	0.021*	0.042***	0.042***
r 2,0 t(Mean)	(0.00)	(0.21)	(1.49)	(1.94)	(3.40)	(3.54)
r 3,0 Mean	0.001	-0.004	0.013	0.029**	0.038***	0.037***
r 3,0 t(Mean)	(0.14)	(-0.53)	(1.66)	(2.44)	(3.21)	(3.06)
r 4,0 Mean	0.001	0.009	0.014	0.028***	0.024**	0.022**
r 4,0 t(Mean)	(0.16)	(1.11)	(1.39)	(2.99)	(2.20)	(2.22)
r 4,1 Mean	0.000	0.007	0.019	0.018**	0.010	0.010
r 4,1 t(Mean)	(0.06)	(0.83)	(1.48)	(2.16)	(1.03)	(1.17)

(b) HKR

Table IA7: Attention and cryptocurrency momentum (Table AI.XI (A) in LTW)

Panel (b) replicates the results in LTW using the same filtering method (market capitalization > 1 million USD) and sample period (January 2014 to July 2020). Panel (c) uses LTW's filtering method and our sample period, January 2014 to August 2023, and Panel (d) uses our filtering method (both market capitalization and trading volume > 1 million USD) and LTW's sample period.

	1	2	3	4	5	5 - 1
Low	0.028* (1.723)	0.010 (1.253)	0.020** (2.114)	0.020** (2.223)	0.023** (1.989)	-0.006 (-0.343)
High	-0.003 (-0.279)	0.008* (0.835)	0.010 (1.277)	0.013 (1.563)	0.032** (2.410)	0.034** (2.328)

(a) LTW

	1	2	3	4	5	5 - 1
Low	0.017 (1.642)	0.022** (2.285)	0.011 (1.260)	0.007 (0.769)	0.004 (0.361)	-0.012 (-1.022)
High	-0.002 (-0.252)	0.005 (0.617)	0.016* (1.838)	0.013 (1.581)	0.043** (3.066)	0.045** (3.290)

(b) HKR

	1	2	3	4	5	5 - 1
Low	0.038** (2.847)	0.016** (2.258)	0.009 (1.483)	0.009 (1.340)	0.004 (0.452)	-0.034** (-2.297)
High	-0.005 (-0.683)	0.004 (0.756)	0.014** (2.115)	0.014** (2.122)	0.032** (3.224)	0.037** (3.795)

(c) HKR (January 2014-August 2023)

	1	2	3	4	5	5 - 1
Low	0.001 (0.158)	0.010 (1.451)	0.014* (1.810)	0.030** (2.549)	0.027* (1.655)	0.026 (1.638)
High	0.012 (1.318)	0.005 (0.878)	0.020** (2.822)	0.021** (2.532)	0.019** (2.300)	0.007 (0.776)

(d) HKR (market cap and trading volume > 1 million USD)

C Time-series momentum

C.1 Further analysis

This section conducts various analyses to better understand time-series momentum in the cryptocurrency market and to check the robustness of earlier findings. All tests use the (28, 5) long-only strategy and take transaction costs into account unless otherwise noted.

C.1.1 Different weighting schemes

Table IA11 (a) compares the performance of the (28, 5) long-only strategy with the buy-and-hold strategy under different portfolio weighting schemes: capped-value-weight, volume-weight, capped-volume-weight, and equal-weight. The time-series momentum strategy outperforms the buy-and-hold strategy under all portfolio formation methods, yielding a higher Sharpe ratio and cumulative return, and a lower MDD. While the capped-value-weighted portfolio achieves the highest Sharpe ratio of 1.65, the performance variation across the weighting schemes is small. The lowest Sharpe ratio obtained from the capped-volume-weighted portfolio is still 1.40.

C.1.2 Different entry thresholds

This section examines the impact of the entry threshold, *i.e.*, when to buy the market, on the performance of the time-series momentum strategy. If higher look-back period returns are associated with higher holding period returns, a stricter threshold would yield higher returns during the holding period. However, a more stringent threshold also leads to fewer trading opportunities. Given these opposing effects, the impact of the threshold on portfolio performance is ambiguous. Table IA11 (b) reports the results, where the first column represents the entry threshold: *e.g.*, 10% means the strategy buys the market if its look-back period return falls within the top 10% of its historical returns.

The holding period return—the mean return annualized based on the days positions are held—

increases as the threshold tightens until it reaches 10% and then drops thereafter. Thresholds below 10% yield lower mean returns and higher standard deviations, suggesting that the market tends to decline following exceptional appreciation. While the 10% threshold yields the best performance during the holding period, the overall performance over the sample period is better when the threshold is higher, owing to more frequent trading. An optimal balance between return per trade and trading opportunity appears to emerge at thresholds between 30% and 50%.

C.1.3 Different rebalancing days

Table IA11 (c) examines the performance variation when the entire wealth is invested on a particular day of the week, where the rows represent the rebalancing day.¹⁷ The impact of the rebalancing day is nontrivial. The portfolio trading on Tuesdays yields the highest Sharpe ratio of 1.40, whereas the portfolio trading on Mondays yields the lowest Sharpe ratio of 1.09. The difference mainly results from the mean return. The cumulative returns in Figure IA2 show that the portfolio rebalanced on Mondays performs worse, especially when the market is bearish.

C.1.4 Individual coin time-series momentum

This section examines whether a long-short portfolio based on the time-series momentum of individual coins can generate profits. Two long-short strategies are considered. The first strategy buys (sells) the coins whose look-back period return is in the top (bottom) 20% of their historical returns. The second strategy sorts the coins on their time-series percentile ranks of the look-back period return and buys (sells) those in the top (bottom) 20%. Table IA13 reports the performance of these strategies.¹⁸

Both strategies perform poorly. They generate negative profits for the majority of the look-back and holding period pairs. Only one portfolio from the first strategy, (7, 5), and one from the second strategy, (7, 7), marginally outperform the market in terms of the Sharpe ratio, suggesting that a

¹⁷We use a holding period of seven days instead of five to examine the effect of the rebalancing day in a week.

¹⁸The pooled regression result of individual coins' time-series momentum is reported in Table IA12.

cross-sectional comparison of time-series momentum carries little predictive power.

C.2 Tables and Figures

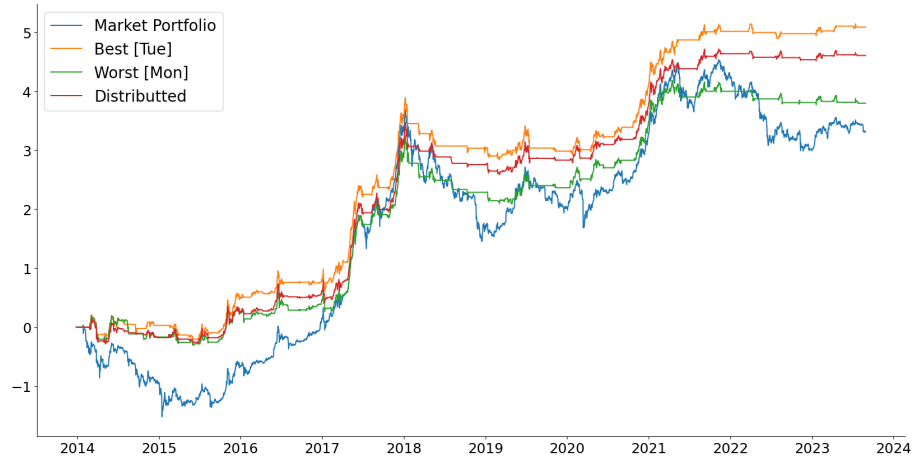


Figure IA2: The (28, 7) time-series momentum portfolio cumulative returns under different rebalancing days

Table IA8: Performance of time-series momentum portfolios (before transaction costs)

This table reports the performance of time-series momentum portfolios of various look-back and holding periods, (j, k) . ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 26, 2014 to August 28, 2023.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(7, 1)	58.86	43.70	1.35	11697.4	52.0	-12.23	47.85	-0.26	-89.8	92.8	48.75	64.73	0.75	1390.5	66.6
(7, 3)	59.08	48.22	1.23	9743.9	61.0	-5.80	50.97	-0.11	-83.5	91.8	60.76	68.38	0.89	3678.0	67.0
(7, 5)	55.63	52.76	1.05	5389.3	74.8	-13.47	54.49	-0.25	-93.5	96.4	35.95	70.19	0.51	187.8	92.3
(7, 7)	50.22	54.74	0.92	2817.0	79.4	-9.61	53.69	-0.18	-90.1	94.6	30.28	70.11	0.43	67.4	88.2
(7, 14)	69.81	60.79	1.15	13773.1	79.0	-27.62	61.05	-0.45	-98.8	99.5	58.24	72.76	0.80	2008.6	93.9
(14, 1)	57.50	45.46	1.26	9392.0	52.4	0.22	48.75	0.00	-67.2	79.6	58.69	66.39	0.88	3374.2	51.3
(14, 3)	61.41	47.96	1.28	12262.2	53.4	-1.88	49.33	-0.04	-74.0	83.5	61.36	67.90	0.90	3996.6	66.2
(14, 5)	64.81	49.34	1.31	16007.0	56.2	-0.61	50.05	-0.01	-72.0	82.4	66.76	68.28	0.98	6584.3	59.0
(14, 7)	67.36	50.58	1.33	19278.7	55.5	-7.30	52.27	-0.14	-86.8	93.3	73.33	69.01	1.06	12005.5	68.1
(14, 14)	53.33	56.20	0.95	3609.5	72.6	-3.61	54.20	-0.07	-82.6	92.3	32.08	71.29	0.45	85.5	95.6
(21, 1)	59.85	45.26	1.32	11923.4	58.5	2.60	48.10	0.05	-57.7	79.4	62.66	65.43	0.96	5286.5	58.9
(21, 3)	59.72	47.76	1.25	10521.6	59.0	5.44	48.72	0.11	-45.9	76.8	66.81	67.26	0.99	7078.9	63.6
(21, 5)	49.70	49.51	1.00	3571.2	67.0	10.14	49.48	0.20	-17.6	69.9	51.50	68.27	0.75	1410.7	77.8
(21, 7)	50.90	51.52	0.99	3642.7	70.2	12.99	49.06	0.26	10.0	70.1	52.25	69.10	0.76	1428.5	81.4
(21, 14)	58.86	54.70	1.08	6728.0	76.6	-4.52	53.57	-0.08	-83.9	94.4	63.87	71.04	0.90	4024.5	87.7
(28, 1)	69.16	44.99	1.54	29842.2	62.7	-4.42	47.16	-0.09	-77.7	87.5	65.12	64.69	1.01	7025.3	61.0
(28, 3)	68.38	47.14	1.45	25139.9	61.1	-0.56	47.65	-0.01	-68.3	84.6	67.45	66.13	1.02	8077.2	63.8
(28, 5)	75.37	48.34	1.56	46787.0	61.0	0.18	48.44	0.00	-67.4	86.5	77.15	67.23	1.15	19167.2	59.2
(28, 7)	73.96	48.26	1.53	40961.4	61.6	2.22	48.90	0.05	-60.9	79.1	77.74	67.49	1.15	20045.6	61.9
(28, 14)	66.30	54.09	1.23	14431.1	71.3	4.24	52.64	0.08	-60.0	86.7	68.10	70.09	0.97	6613.6	81.8

Table IA9: Performance of time-series momentum portfolio by year

The table reports the annual performance of the market portfolio and the (28, 5) time-series momentum portfolio. The columns ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 25, 2014 to August 28, 2023.

	Market					Time-series momentum				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
2014	-73.52	72.77	-1.01	-60.7	65.0	-35.26	50.12	-0.70	-37.9	44.6
2015	51.31	67.90	0.76	34.4	78.9	81.27	45.18	1.80	103.1	43.6
2016	90.26	46.02	1.96	120.1	57.7	53.66	35.79	1.50	60.5	24.7
2017	366.32	87.92	4.17	2420.6	43.4	304.64	69.32	4.39	1521.6	31.4
2018	-128.62	91.72	-1.40	-82.1	89.1	-42.07	46.54	-0.90	-41.2	59.3
2019	56.87	66.47	0.86	36.9	88.2	75.22	51.05	1.47	86.0	61.8
2020	158.63	74.22	2.14	261.9	86.3	141.49	46.46	3.05	269.1	21.8
2021	130.07	86.75	1.50	148.2	54.8	111.11	56.79	1.96	157.2	26.5
2022	-96.53	70.67	-1.37	-71.3	78.8	3.43	34.42	0.10	-2.5	30.1
2023	57.89	45.30	1.28	36.2	78.4	3.83	24.97	0.15	0.1	22.7
Total	62.37	73.63	0.85	2695.9	89.1	72.85	48.34	1.51	36785.8	61.8

Table IA10: Regression of time-series momentum in different market states

This table reports the results of the time-series momentum regression in good and bad market states. The market is assumed to be in a good (bad) state if the look-back period return is in the top (bottom) third of the historical returns. The figures are the estimates of β and Newey-West adjusted t -statistics.

Look-back (j)	Holding period (k)										
	1	3	5	7	14	21	28	35	42	49	56
(a) Good state											
1	0.016 (0.65)	0.087 (2.81)	0.099 (2.64)	0.112 (2.64)	0.130 (2.19)	0.130 (1.85)	0.152 (1.85)	0.144 (1.56)	0.132 (1.28)	0.133 (1.21)	0.137 (1.16)
3	0.077 (3.03)	0.142 (4.32)	0.152 (3.79)	0.151 (3.35)	0.178 (2.98)	0.169 (2.40)	0.190 (2.38)	0.194 (2.16)	0.185 (1.85)	0.181 (1.68)	0.171 (1.48)
5	0.085 (3.18)	0.150 (4.13)	0.157 (3.77)	0.159 (3.43)	0.198 (3.32)	0.181 (2.59)	0.204 (2.57)	0.204 (2.24)	0.197 (1.93)	0.190 (1.76)	0.179 (1.53)
7	0.111 (4.38)	0.160 (4.17)	0.170 (3.80)	0.173 (3.61)	0.225 (3.67)	0.204 (2.82)	0.226 (2.81)	0.219 (2.35)	0.209 (2.00)	0.192 (1.74)	0.179 (1.50)
14	0.117 (4.15)	0.181 (4.36)	0.198 (3.95)	0.209 (3.81)	0.223 (3.43)	0.225 (3.05)	0.231 (2.76)	0.224 (2.34)	0.205 (1.94)	0.192 (1.71)	0.186 (1.53)
21	0.115 (3.99)	0.164 (3.94)	0.177 (3.60)	0.188 (3.47)	0.217 (3.25)	0.209 (2.81)	0.214 (2.48)	0.199 (2.01)	0.185 (1.71)	0.174 (1.52)	0.184 (1.51)
28	0.126 (4.34)	0.170 (4.09)	0.183 (3.75)	0.202 (3.79)	0.234 (3.47)	0.223 (2.81)	0.221 (2.49)	0.205 (2.01)	0.199 (1.80)	0.202 (1.76)	0.216 (1.81)
35	0.118 (4.05)	0.173 (4.10)	0.188 (3.76)	0.205 (3.73)	0.229 (3.17)	0.216 (2.55)	0.219 (2.27)	0.211 (2.03)	0.206 (1.85)	0.216 (1.88)	0.232 (1.95)
42	0.124 (4.14)	0.177 (4.06)	0.176 (3.30)	0.183 (3.09)	0.200 (2.59)	0.194 (2.13)	0.204 (1.98)	0.209 (1.91)	0.214 (1.92)	0.223 (1.96)	0.237 (2.00)
49	0.109 (3.44)	0.140 (2.92)	0.141 (2.44)	0.149 (2.31)	0.171 (2.05)	0.167 (1.69)	0.192 (1.76)	0.216 (1.89)	0.222 (1.91)	0.228 (1.97)	0.241 (2.01)
56	0.107 (3.30)	0.139 (2.79)	0.140 (2.33)	0.145 (2.15)	0.155 (1.74)	0.166 (1.62)	0.201 (1.81)	0.229 (1.98)	0.240 (2.02)	0.247 (2.08)	0.255 (2.11)
(b) Bad state											
1	-0.082 (-3.01)	-0.049 (-1.56)	-0.073 (-2.05)	-0.051 (-1.25)	-0.047 (-0.84)	-0.054 (-0.81)	-0.050 (-0.64)	-0.047 (-0.52)	-0.058 (-0.59)	-0.044 (-0.43)	-0.049 (-0.43)
3	-0.032 (-1.21)	-0.032 (-0.90)	-0.018 (-0.43)	-0.019 (-0.42)	-0.013 (-0.22)	-0.027 (-0.39)	-0.012 (-0.15)	-0.013 (-0.15)	-0.038 (-0.40)	-0.025 (-0.25)	-0.028 (-0.24)
5	-0.039 (-1.51)	-0.008 (-0.22)	-0.011 (-0.26)	-0.015 (-0.32)	-0.009 (-0.14)	-0.029 (-0.40)	-0.012 (-0.15)	-0.020 (-0.22)	-0.040 (-0.40)	-0.025 (-0.23)	-0.032 (-0.27)
7	-0.021 (-0.83)	-0.011 (-0.29)	-0.011 (-0.24)	-0.020 (-0.41)	-0.001 (-0.02)	-0.021 (-0.28)	0.001 (0.01)	-0.015 (-0.15)	-0.029 (-0.28)	-0.007 (-0.06)	-0.010 (-0.09)
14	0.002 (0.07)	0.016 (0.40)	0.006 (0.11)	0.016 (0.29)	0.026 (0.36)	0.033 (0.44)	0.033 (0.39)	0.012 (0.12)	-0.000 (-0.00)	0.013 (0.11)	-0.002 (-0.02)
21	0.002 (0.10)	-0.001 (-0.02)	-0.009 (-0.18)	-0.003 (-0.05)	0.015 (0.20)	0.017 (0.22)	0.016 (0.18)	0.000 (0.00)	-0.011 (-0.09)	-0.006 (-0.05)	-0.013 (-0.11)
28	0.008 (0.36)	0.001 (0.03)	-0.012 (-0.26)	-0.000 (-0.01)	0.002 (0.03)	0.018 (0.21)	0.015 (0.15)	0.004 (0.04)	-0.013 (-0.11)	-0.010 (-0.08)	-0.011 (-0.09)
35	0.015 (0.62)	0.009 (0.21)	-0.012 (-0.22)	-0.013 (-0.21)	-0.001 (-0.01)	0.005 (0.05)	-0.001 (-0.01)	-0.019 (-0.17)	-0.024 (-0.20)	-0.017 (-0.14)	-0.020 (-0.15)
42	-0.006 (-0.23)	-0.015 (-0.32)	-0.040 (-0.70)	-0.026 (-0.39)	-0.018 (-0.20)	-0.022 (-0.21)	-0.036 (-0.31)	-0.041 (-0.33)	-0.036 (-0.29)	-0.024 (-0.19)	-0.018 (-0.14)
49	-0.006 (-0.20)	-0.007 (-0.14)	-0.021 (-0.35)	-0.007 (-0.10)	-0.013 (-0.14)	-0.031 (-0.30)	-0.031 (-0.28)	-0.023 (-0.19)	-0.017 (-0.14)	-0.007 (-0.05)	0.010 (0.08)
56	0.020 (0.84)	0.022 (0.49)	0.000 (0.00)	0.011 (0.18)	0.000 (0.01)	-0.004 (-0.04)	0.005 (0.05)	0.010 (0.08)	0.016 (0.14)	0.030 (0.25)	0.044 (0.36)

Table IA11: Further analysis of time-series momentum

This table reports the performance of the (28, 5) time-series momentum long-only portfolio under different conditions. Panel (a) reports the performance under different weighting schemes: value-weight ('Value'), capped-value-weight ('CapValue'), volume-weight ('Volume'), capped-volume-weight ('CapVolume'), and equal-weight ('Equal'). Panel (b) reports the performance under different entry thresholds. The value in the first column determines when to buy the market: *e.g.*, 10% means the strategy buys the market if the look-back period return is within the top 10% of the historical returns. The results under 'Sample period' are the performance over the entire sample period, and those under 'Holding period' are the performance during the holding period. Panel (c) reports the performance of the (28,7) for different rebalancing days. 'Distributed' means investing evenly throughout the week. The columns 'Mean', 'Std', 'Sharpe', 'Cum', and 'MDD' respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 26, 2014 to August 28, 2023. A transaction cost of 15 bps is assumed.

Scheme	Buy-and-hold					Time-series momentum				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
Value	61.48	73.55	0.84	2481.4	89.2	72.85	48.34	1.51	36783.0	61.8
CapValue	61.89	80.99	0.76	1506.3	94.3	86.02	52.28	1.65	107977.8	53.5
Volume	57.74	78.96	0.73	1197.3	92.4	74.01	50.65	1.46	36688.5	68.5
CapVolume	42.82	86.36	0.50	154.4	98.1	79.27	56.45	1.40	45158.7	61.3
Equal	74.54	88.26	0.84	2782.4	97.1	94.17	60.78	1.55	149367.6	75.6

(a) Different weighting schemes

	Sample period					Holding period			Entry	
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Count	Ratio
50%	74.52	54.51	1.37	31505.4	68.5	120.86	69.34	1.74	435	61.58%
40%	69.32	51.81	1.34	21959.9	59.6	129.54	70.68	1.83	378	53.51%
30%	62.38	45.73	1.36	15036.7	65.2	142.62	68.93	2.07	309	43.74%
20%	51.56	42.46	1.21	6072.9	51.4	150.53	72.28	2.08	242	34.26%
10%	33.59	35.63	0.94	1391.5	40.8	172.91	80.85	2.14	136	19.25%
5%	16.36	29.01	0.56	323.3	48.1	158.13	91.20	1.73	71	10.05%
1%	10.54	22.81	0.46	215.3	40.8	156.71	90.83	1.73	44	6.23%

(b) Different entry thresholds

	Mean	Std	Sharpe	Cum	MDD
Mon	49.91	45.87	1.09	4465.1	68.4
Tue	62.75	44.79	1.40	16223.0	65.2
Wed	52.48	44.35	1.18	6120.5	71.9
Thu	57.19	44.85	1.28	9418.5	50.3
Fri	57.49	45.43	1.27	9448.7	52.9
Sat	57.71	45.58	1.27	9617.9	49.5
Sun	61.19	45.93	1.33	13259.7	59.2
Distributed	56.96	42.91	1.33	10027.6	58.7

(c) Different rebalancing days

Table IA12: Pooled regression of individual coin time-series momentum

This table reports the results of the pooled time-series momentum regression of individual coins. The sample period is January 25, 2014 to August 28, 2023. The figures are the estimates of β and Newey-West adjusted t -statistics.

Look-back (j)	Holding period (k)										
	1	3	5	7	14	21	28	35	42	49	56
1	-0.058 (-45.81)	-0.018 (-14.73)	-0.014 (-11.5)	0.000 (-0.23)	0.006 (5.11)	0.009 (7.13)	0.012 (10.08)	0.019 (15.69)	0.008 (6.55)	0.006 (4.56)	0.002 (1.54)
3	-0.028 (-22.32)	0.004 (2.45)	0.013 (7.39)	0.014 (8.10)	0.025 (12.98)	0.022 (10.94)	0.032 (16.51)	0.036 (17.75)	0.020 (9.71)	0.013 (6.45)	0.005 (2.66)
5	-0.023 (-18.75)	0.014 (8.16)	0.015 (8.09)	0.017 (8.16)	0.031 (13.40)	0.028 (11.60)	0.043 (17.89)	0.040 (16.27)	0.021 (8.55)	0.015 (5.73)	0.006 (2.43)
7	-0.012 (-9.49)	0.014 (7.57)	0.015 (7.35)	0.018 (8.43)	0.036 (13.83)	0.034 (12.30)	0.052 (18.95)	0.043 (14.90)	0.023 (7.85)	0.015 (5.13)	0.004 (1.24)
14	-0.009 (-7.13)	0.017 (8.88)	0.023 (10.24)	0.028 (10.97)	0.043 (13.95)	0.059 (17.62)	0.065 (18.51)	0.045 (12.23)	0.026 (7.02)	0.012 (3.07)	-0.010 (-2.65)
21	-0.006 (-4.47)	0.017 (8.82)	0.023 (9.74)	0.029 (10.71)	0.060 (18.10)	0.069 (19.20)	0.062 (15.86)	0.043 (10.46)	0.022 (5.21)	-0.001 (-0.15)	-0.021 (-4.74)
28	0.000 (0.20)	0.029 (15.27)	0.042 (17.63)	0.053 (19.58)	0.072 (20.68)	0.070 (17.91)	0.063 (15.06)	0.042 (9.47)	0.013 (2.78)	-0.009 (-1.79)	-0.025 (-5.02)
35	0.008 (6.34)	0.035 (18.23)	0.042 (17.65)	0.047 (17.13)	0.055 (15.31)	0.055 (13.57)	0.047 (10.74)	0.021 (4.63)	-0.005 (-1.01)	-0.022 (-4.31)	-0.037 (-7.16)
42	-0.001 (-0.53)	0.021 (11.03)	0.025 (10.57)	0.030 (10.68)	0.040 (10.88)	0.040 (9.43)	0.025 (5.48)	0.004 (0.82)	-0.017 (-3.38)	-0.033 (-6.38)	-0.047 (-8.72)
49	-0.003 (-2.33)	0.017 (8.89)	0.021 (8.81)	0.026 (9.16)	0.033 (8.82)	0.023 (5.39)	0.011 (2.30)	-0.005 (-1.00)	-0.026 (-4.95)	-0.040 (-7.39)	-0.052 (-9.35)
56	-0.003 (-2.40)	0.014 (7.28)	0.019 (7.85)	0.021 (7.44)	0.020 (5.20)	0.013 (2.94)	0.004 (0.90)	-0.011 (-2.16)	-0.029 (-5.44)	-0.041 (-7.44)	-0.056 (-9.81)

Table IA13: Performance of portfolios based on individual coin time-series momentum

This table reports the performance of long-short portfolios formed on the time-series momentum of individual coins. In Panel (a), the long (short) leg consists of coins whose look-back period returns fall within the top (bottom) 20% of their historical returns. In Panel (b), coins are sorted on the time-series percentile rank of their look-back period return, and a long-short portfolio is formed by buying coins in the top 20% and shorting those in the bottom 20%. ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%).

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(a) Time-series momentum															
(7, 1)	42.61	129.40	0.33	-91.6	99.9	-31.86	110.99	-0.29	-99.9	99.9	-5.66	136.86	-0.04	-99.9	100.0
(7, 3)	55.77	110.83	0.50	-30.3	99.5	50.58	103.27	0.49	-23.5	92.2	97.27	125.02	0.78	280.9	92.4
(7, 5)	80.76	106.03	0.76	430.5	92.6	41.02	91.55	0.45	-7.1	92.9	112.23	100.27	1.12	6237.1	68.4
(7, 7)	63.08	108.84	0.58	53.8	97.1	45.40	95.56	0.48	-7.9	95.6	117.45	115.91	1.01	3781.7	82.6
(7, 14)	89.28	111.74	0.80	698.0	94.5	-54.64	146.17	-0.37	-100.0	100.0	27.29	103.18	0.26	-81.9	95.6
(14, 1)	8.35	118.08	0.07	-98.6	99.8	-26.31	118.08	-0.22	-100.0	100.0	-37.43	131.93	-0.28	-100.0	100.0
(14, 3)	61.89	114.30	0.54	-22.3	96.3	37.03	101.87	0.36	-73.1	92.7	48.68	118.50	0.41	-87.1	97.0
(14, 5)	71.71	107.34	0.67	159.0	96.0	50.05	104.01	0.48	-75.1	97.4	76.99	107.18	0.72	167.2	85.1
(14, 7)	54.05	103.39	0.52	4.6	97.0	70.38	90.68	0.78	582.7	89.3	96.29	96.90	0.99	2410.0	79.4
(14, 14)	52.29	103.43	0.51	-5.7	97.7	32.85	87.69	0.37	-34.5	95.0	34.35	84.98	0.40	-9.8	89.8
(21, 1)	53.87	114.02	0.47	-53.7	99.2	-13.91	104.54	-0.13	-99.2	99.7	4.21	113.08	0.04	-98.5	99.8
(21, 3)	60.21	106.88	0.56	20.3	98.3	34.35	97.52	0.35	-61.2	96.1	55.80	95.16	0.59	97.5	93.7
(21, 5)	53.03	104.33	0.51	-8.4	98.2	45.38	95.69	0.47	-9.8	95.0	60.78	94.17	0.65	180.4	89.6
(21, 7)	52.24	103.51	0.50	-8.2	98.2	48.42	88.95	0.54	74.2	86.7	67.07	89.88	0.75	530.3	60.8
(21, 14)	34.95	94.54	0.37	-49.8	98.6	23.67	109.15	0.22	-93.2	99.3	5.90	85.34	0.07	-89.0	98.2
(28, 1)	30.04	110.10	0.27	-87.8	99.5	-26.59	111.99	-0.24	-99.8	100.0	-30.56	113.01	-0.27	-99.8	99.9
(28, 3)	36.70	104.36	0.35	-69.2	98.7	8.19	98.14	0.08	-93.7	99.5	6.70	96.15	0.07	-92.6	98.0
(28, 5)	36.41	104.40	0.35	-68.0	98.6	23.74	94.03	0.25	-76.6	98.2	24.39	95.46	0.26	-71.8	96.0
(28, 7)	44.96	103.94	0.43	-41.5	98.3	29.17	93.99	0.31	-66.4	93.4	36.92	92.37	0.40	-19.7	94.7
(28, 14)	24.58	96.34	0.26	-77.4	99.3	20.77	97.40	0.21	-82.5	99.0	13.57	80.88	0.17	-71.5	97.3
(b) Cross-section of time-series momentum															
(7, 1)	132.62	120.87	1.10	7653.0	94.6	-60.25	91.34	-0.66	-99.9	99.9	72.04	107.49	0.67	296.3	91.0
(7, 3)	144.47	113.59	1.27	28529.9	93.7	-53.22	92.57	-0.57	-99.8	99.9	91.98	100.46	0.92	2196.1	84.8
(7, 5)	146.05	114.41	1.28	32441.7	96.2	-49.93	94.05	-0.53	-99.8	99.9	98.90	99.00	1.00	4282.6	81.6
(7, 7)	140.26	114.64	1.22	21698.4	95.7	-35.43	94.72	-0.37	-99.5	99.8	118.87	102.75	1.16	13005.8	76.0
(7, 14)	133.41	112.45	1.19	16437.2	95.1	-714.87	227.35	-3.14	-100.0	100.0	62.37	105.32	0.59	103.1	98.4
(14, 1)	104.32	111.98	0.93	1830.0	93.0	-82.67	93.32	-0.89	-100.0	100.0	21.31	102.78	0.21	-85.1	97.1
(14, 3)	112.67	108.85	1.04	4043.8	93.4	-55.12	92.87	-0.59	-99.9	100.0	44.78	97.33	0.46	-10.5	98.0
(14, 5)	118.40	107.76	1.10	6300.4	96.0	-55.02	95.88	-0.57	-99.9	100.0	59.40	96.85	0.61	149.3	98.0
(14, 7)	113.38	109.04	1.04	4476.2	95.3	-28.46	112.51	-0.25	-99.8	99.9	72.73	105.38	0.69	282.4	91.3
(14, 14)	103.44	103.74	1.00	3129.9	94.3	31.68	228.95	0.14	-99.9	100.0	32.26	112.90	0.29	-90.9	97.7
(21, 1)	81.16	112.64	0.72	288.2	98.6	-80.81	90.17	-0.90	-100.0	100.0	0.06	99.14	0.00	-95.8	98.3
(21, 3)	84.95	98.90	0.86	978.5	96.8	-54.77	88.87	-0.62	-99.8	99.9	26.50	80.73	0.33	-35.2	90.3
(21, 5)	83.46	98.09	0.85	927.4	96.7	-58.42	93.02	-0.63	-99.9	100.0	25.51	84.38	0.30	-69.9	96.2
(21, 7)	81.44	96.99	0.84	859.6	95.9	-60.63	97.87	-0.62	-100.0	100.0	31.26	87.93	0.36	-85.1	98.4
(21, 14)	71.25	94.26	0.76	486.6	96.7	-609.17	202.90	-3.00	-100.0	100.0	3.28	82.85	0.04	-90.2	99.4
(28, 1)	91.79	113.21	0.81	640.2	95.3	-97.67	91.22	-1.07	-100.0	100.0	-6.21	95.45	-0.07	-96.7	99.0
(28, 3)	93.53	104.91	0.89	1261.7	94.7	-60.43	90.89	-0.66	-99.9	99.9	32.65	89.13	0.37	-43.0	93.9
(28, 5)	85.52	101.11	0.85	908.2	95.6	-44.73	112.16	-0.40	-99.9	100.0	22.79	86.92	0.26	-79.8	97.4
(28, 7)	74.29	99.29	0.75	436.0	96.0	-67.30	101.19	-0.67	-100.0	100.0	9.14	87.24	0.10	-88.2	96.8
(28, 14)	58.90	96.69	0.61	119.1	97.0	-86.62	111.82	-0.77	-100.0	100.0	0.99	77.82	0.01	-85.7	95.2
Market											78.86	77.80	1.01	2333.3	89.1

D Cross-sectional momentum

D.1 Further analysis

This section conducts various analyses to better understand the cross-sectional momentum in the cryptocurrency market and to check the robustness of earlier findings. All the results presented in this section are net of transaction costs.

D.1.1 Different weighting schemes

Table IA22 reports the performance of the (1, 14) and (14, 7) portfolios constructed using different weighting schemes: value-, capped-value-, volume-, capped-volume-, and equal-weight. The value-weighted portfolios outperform the volume-weighted portfolios, yielding higher Sharpe ratios and lower MDDs. The inferior performance of the volume-weighted portfolios is consistent with the low mean returns in high-volume groups and provides further evidence against the attention-based overreaction explanation. The equal-weighted portfolios perform poorly due to greater weightings on small coins. Capping the weights reduces the MDD by mitigating concentration, but also lowers overall performance, perhaps because of reduced exposure to large coins.

D.1.2 Excluding the last day in the look-back period

The regression analysis suggests a strong reversal effect when the look-back period is one day. Therefore, we test momentum strategies excluding the last day from the look-back period, similarly to the momentum strategy of Jegadeesh and Titman (1993) that excludes the last month. Table IA23 reports the results.

Excluding the last day from the look-back period improves performance, and 12 of the 18 long-short portfolios yield higher Sharpe ratios. Nevertheless, the improvement is not significant, and some portfolios earn lower cumulative returns despite higher Sharpe ratios. For instance, the (14, 7) portfolio’s Sharpe ratio improves from 1.28 to 1.33, but its cumulative return decreases from

101,218% to 37,389%.

D.1.3 Different rebalancing days

This section examines the performance variation when the entire wealth is invested on a particular day of the week. Table IA24 reports the results for the (14, 7) strategy. Performance varies significantly depending on the rebalancing day. The portfolio rebalancing on Mondays yields the highest Sharpe ratio of 1.43, whereas the portfolios rebalancing on Wednesdays, Thursdays, and Fridays are all liquidated. The liquidated portfolios also perform inconsistently until they are liquidated. Figure IA4 shows that the performance gap between the best and worst portfolios (excluding those liquidated) is not driven by a few events. The worst portfolio consistently underperforms the best portfolio until the end of 2020. The significant performance variation across rebalancing days implies that the findings in previous studies that rebalance portfolios on a particular day of the week or month may not be credible.

D.1.4 Leverage and margin mode effects

This section investigates the case of no leverage (1x leverage) and investing 50% of the wealth in each leg of a long-short portfolio. By investing only 50% of the wealth, the portfolio sacrifices potential profits but also has a lower chance of liquidation. Figure IA5 compares the cumulative returns of the (14, 7) strategy under no leverage (50% investment) and 2x leverage (full investment). The leveraged portfolio earns a higher cumulative return of 101,218% despite a lower Sharpe ratio of 1.28, compared to 4,786% and 1.42 of the unleveraged portfolio. However, three of the seven accounts are liquidated under 2x leverage, while no account is liquidated without leverage. Considering the substantial liquidation risk and the investment constraints of a leveraged portfolio, an unleveraged strategy appears to be a better choice for institutional investors.

Many investors lean towards the cross-margin mode as it can avoid the liquidation of individual positions. Yet, there is pronounced tail risk in the cryptocurrency market, and a portfolio can suffer

a huge loss by a single dramatic event under the cross-margin mode. The isolated-margin mode prevents the propagation of losses at the cost of more frequent liquidation of individual positions. Figure IA6 compares the cumulative returns of the (14, 7) momentum strategy under the cross-margin and isolated-margin modes. The portfolio performs slightly better under the cross-margin mode, yielding a Sharpe ratio of 1.28, compared to 1.18 under the isolated-margin mode. Although one might be tempted to conclude that enduring temporary shocks is preferable to realizing losses, it is important to recognize that this result is based on the best-performing portfolio. As shown previously, many portfolios are liquidated under the cross-margin mode.

D.2 Tables and Figures

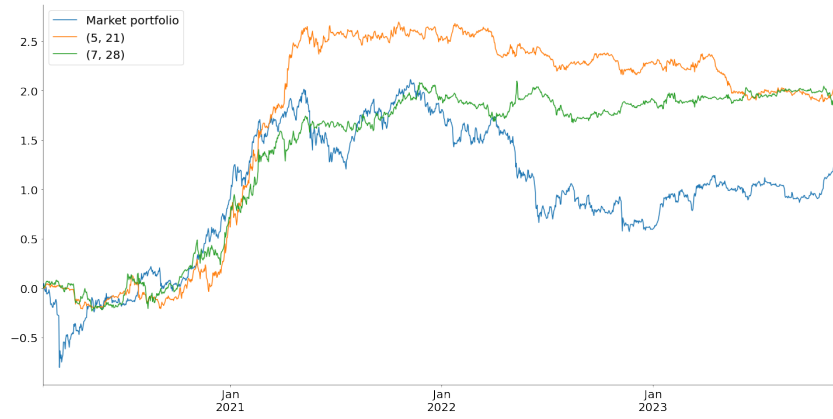


Figure IA3: Cross-sectional momentum long-short portfolio cumulative returns (Binance futures)

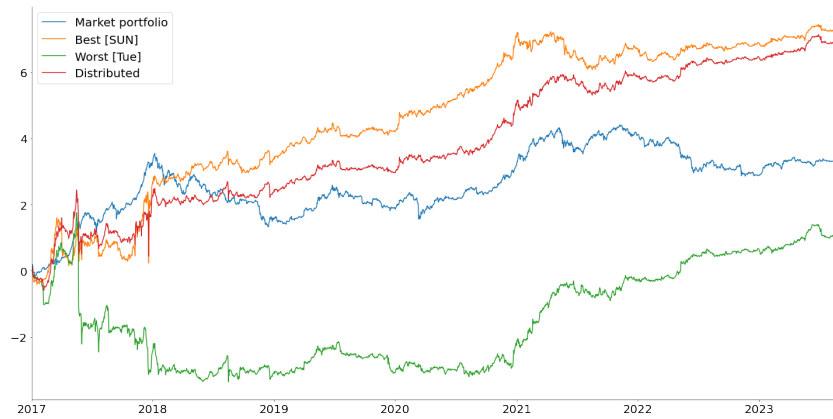


Figure IA4: The (14, 7) cross-sectional momentum portfolio cumulative returns under different rebalancing days

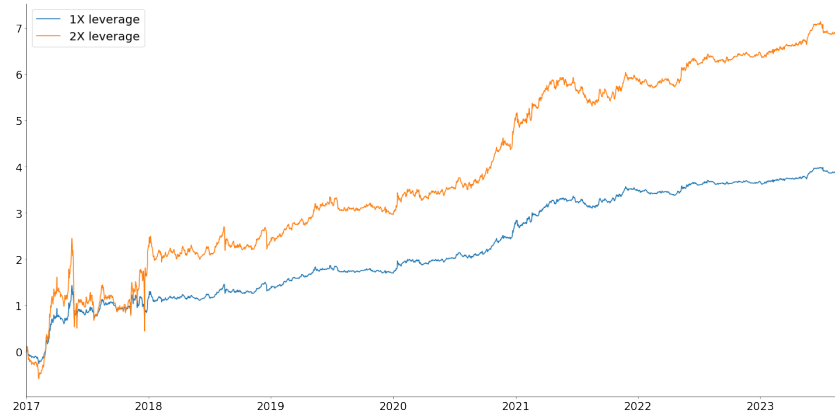


Figure IA5: The (14, 7) cross-sectional momentum portfolio cumulative returns under different leverage levels

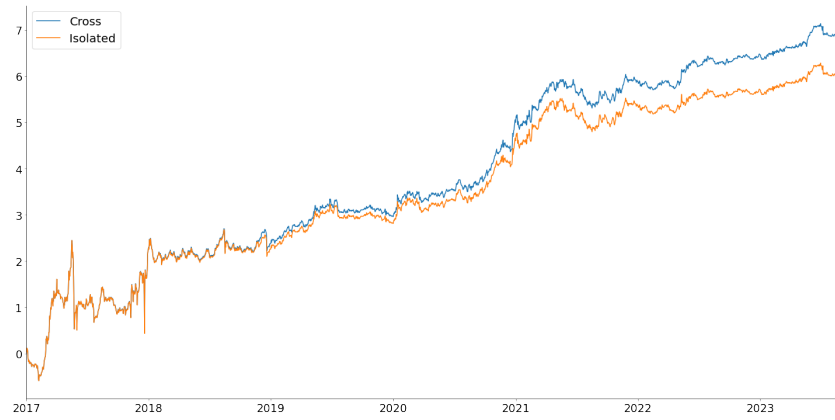


Figure IA6: The (14, 7) cross-sectional momentum portfolio cumulative returns under different margin modes

Table IA14: Performance of cross-sectional momentum portfolios (before transaction costs)

This table reports the performance of cross-sectional momentum portfolios of various look-back and holding periods, (j, k) . ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2017 to August 28, 2023.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(a) All coins															
(1, 14)	114.04	95.86	1.19	9110.0	95.1	-70.14	108.88	-0.64	-99.9	100.0	63.16	61.46	1.03	1903.6	61.7
(1, 21)	105.74	94.42	1.12	5563.1	94.7	-1352.78	263.44	-5.14	-100.0	100.0	34.25	104.54	0.33	-74.9	98.4
(1, 28)	108.99	102.04	1.07	4804.0	94.8	-564.43	226.54	-2.49	-100.0	100.0	53.99	257.59	0.21	-100.0	100.0
(3, 14)	112.88	95.03	1.19	8852.6	95.3	-726.89	198.62	-3.66	-100.0	100.0	53.13	76.03	0.70	379.1	97.1
(3, 21)	88.63	93.32	0.95	1801.5	95.2	-570.93	361.14	-1.58	-100.0	100.0	-338.95	292.18	-1.16	-100.0	100.0
(3, 28)	99.21	93.94	1.06	3682.1	94.0	-653.77	235.60	-2.77	-100.0	100.0	10.88	328.70	0.03	-100.0	100.0
(5, 14)	126.55	103.53	1.22	15143.4	95.3	-54.94	121.55	-0.45	-99.9	100.0	122.53	93.54	1.31	16121.8	91.4
(5, 21)	111.50	101.45	1.10	5937.7	93.8	46.26	899.12	0.05	-100.0	100.0	203.81	366.57	0.56	-100.0	100.0
(5, 28)	107.91	102.79	1.05	4363.2	94.2	-693.03	218.44	-3.17	-100.0	100.0	102.55	220.67	0.46	-100.0	100.0
(7, 14)	141.90	113.84	1.25	27066.2	95.5	-2.45	117.59	-0.02	-99.4	99.9	123.86	101.41	1.22	19467.5	87.5
(7, 21)	122.27	113.46	1.08	7178.5	94.3	-992.39	222.39	-4.46	-100.0	100.0	82.33	118.06	0.70	219.6	97.5
(7, 28)	120.89	113.91	1.06	6393.1	94.9	-631.09	252.31	-2.50	-100.0	100.0	135.40	293.56	0.46	-100.0	100.0
(14, 7)	159.33	105.74	1.51	102953.9	93.3	90.55	232.35	0.39	-98.1	99.8	195.92	136.89	1.43	415219.3	85.5
(14, 14)	131.79	103.36	1.28	19482.4	94.2	-526.27	237.90	-2.21	-100.0	100.0	160.07	162.79	0.98	11370.8	96.5
(14, 21)	115.31	106.90	1.08	5109.4	93.2	-761.66	251.57	-3.03	-100.0	100.0	162.31	215.54	0.75	1796.4	98.8
(21, 5)	139.43	106.14	1.31	26981.9	95.0	-34.90	114.60	-0.30	-99.9	100.0	126.56	110.34	1.15	9132.2	94.9
(21, 7)	124.35	104.41	1.19	10887.3	94.9	-569.53	257.71	-2.21	-100.0	100.0	98.18	111.94	0.88	787.3	98.3
(21, 14)	118.54	103.25	1.15	7793.0	94.7	-26.98	137.51	-0.20	-100.0	100.0	106.57	116.25	0.92	990.8	98.0
(28, 3)	133.10	108.23	1.23	15132.0	93.6	-46.30	107.78	-0.43	-99.9	100.0	86.45	98.01	0.88	1153.8	95.0
(28, 5)	122.26	105.60	1.16	8592.0	93.7	-39.60	115.71	-0.34	-99.9	100.0	83.36	97.76	0.85	967.7	95.4
(28, 7)	104.49	104.08	1.00	2794.7	94.0	-31.20	122.14	-0.26	-99.9	100.0	58.66	97.25	0.60	98.6	97.6
(b) Top 5%															
(1, 14)	105.22	85.71	1.23	9242.1	88.7	-98.67	94.58	-1.04	-99.9	100.0	23.95	59.49	0.40	44.5	91.3
(1, 21)	97.73	85.01	1.15	5789.9	90.6	-647.32	204.76	-3.16	-100.0	100.0	-27.26	181.66	-0.15	-100.0	100.0
(1, 28)	92.44	83.25	1.11	4429.0	89.7	-586.75	197.34	-2.97	-100.0	100.0	-88.38	198.61	-0.45	-100.0	100.0
(3, 14)	108.39	88.25	1.23	9747.5	89.6	-62.31	98.11	-0.64	-99.9	99.9	44.19	68.63	0.64	277.5	92.0
(3, 21)	102.35	83.99	1.22	8279.3	90.2	-26.24	191.67	-0.14	-99.9	100.0	73.99	109.41	0.68	470.6	80.2
(3, 28)	100.43	87.27	1.15	6046.0	91.3	-619.34	218.32	-2.84	-100.0	100.0	-61.64	235.24	-0.26	-100.0	100.0
(5, 14)	123.07	90.03	1.37	24098.2	90.3	-99.50	106.25	-0.94	-100.0	100.0	52.20	71.49	0.73	506.5	71.1
(5, 21)	100.63	85.13	1.18	6894.1	90.1	-804.57	196.61	-4.09	-100.0	100.0	8.56	94.52	0.09	-91.8	98.2
(5, 28)	105.05	88.75	1.18	7712.5	93.5	-543.29	206.49	-2.63	-100.0	100.0	112.74	216.06	0.52	-100.0	100.0
(7, 14)	114.31	89.20	1.28	14072.6	89.5	-701.64	225.46	-3.11	-100.0	100.0	-66.60	205.23	-0.32	-100.0	100.0
(7, 21)	97.30	87.09	1.12	4996.1	88.9	-747.46	190.19	-3.93	-100.0	100.0	20.68	87.39	0.24	-69.0	96.3
(7, 28)	86.74	87.11	1.00	2430.1	93.3	-586.11	210.53	-2.78	-100.0	100.0	-33.34	183.83	-0.18	-100.0	100.0
(14, 7)	138.21	94.60	1.46	52066.2	89.1	-37.04	86.52	-0.43	-99.3	99.7	103.51	74.66	1.39	15802.3	66.6
(14, 14)	117.46	92.92	1.26	14288.3	91.2	-65.54	93.94	-0.70	-99.9	99.9	72.54	74.62	0.97	1903.2	77.2
(14, 21)	90.48	89.98	1.01	2648.7	92.0	-716.43	190.23	-3.77	-100.0	100.0	0.91	93.63	0.01	-94.4	99.0
(21, 5)	122.14	95.03	1.29	17148.7	89.8	-42.88	86.11	-0.50	-99.5	99.8	80.24	72.83	1.10	3594.3	83.3
(21, 7)	119.92	93.90	1.28	15807.7	89.9	-54.22	85.90	-0.63	-99.7	99.9	72.09	72.56	0.99	2079.9	79.2
(21, 14)	95.86	90.52	1.06	3716.6	91.2	-86.69	97.75	-0.89	-99.9	99.9	30.75	69.87	0.44	54.5	92.1
(28, 3)	118.96	98.50	1.21	10928.1	89.8	-39.39	86.17	-0.46	-99.4	99.7	75.37	76.80	0.98	2057.3	89.7
(28, 5)	112.00	94.76	1.18	8565.3	90.7	-47.04	86.43	-0.54	-99.6	99.8	65.41	74.52	0.88	1112.6	92.0
(28, 7)	109.05	93.64	1.16	7554.6	91.5	-60.08	87.47	-0.69	-99.8	99.9	50.24	75.43	0.67	321.0	94.5
Market											78.86	77.80	1.01	2333.3	89.1

Table IA15: Liquidation of cross-sectional momentum portfolios

Panel (a) reports the number of liquidated accounts of the cross-sectional momentum portfolios and their liquidation dates. ‘Accounts’ and ‘Liquidated’ respectively denote the total number of accounts and the liquidated accounts. Panel (b) reports the liquidation ratio of each rebalancing frequency.

	Accounts	Liquidated	Liquidation dates
(1, 14)	7	5	2017-03-16, 2017-05-04, 2017-05-16, 2017-12-04, 2021-02-19
(1, 21)	7	6	2017-03-15, 2017-03-31, 2017-05-07, 2017-11-28, 2020-01-14
(1, 28)	7	7	2017-02-28, 2017-03-17, 2017-05-06, 2017-05-18, 2017-06-17, 2017-11-28, 2017-12-17
(3, 14)	7	5	2017-03-15, 2017-04-02, 2017-05-07, 2017-05-21, 2017-12-04
(3, 21)	7	7	2017-03-15, 2017-03-29, 2017-03-31, 2017-04-02, 2017-05-06, 2017-05-30, 2017-12-20
(3, 28)	7	7	2017-02-27, 2017-03-15, 2017-04-02, 2017-05-07, 2017-12-16, 2017-12-20, 2017-12-23
(5, 14)	7	5	2017-04-02, 2017-12-05, 2017-12-14
(5, 21)	7	7	2017-03-15, 2017-04-02, 2017-05-07, 2017-05-08, 2017-06-19, 2017-12-21
(5, 28)	7	7	2017-02-27, 2017-04-02, 2017-11-28, 2017-12-07, 2017-12-14, 2017-12-16
(7, 14)	7	4	2017-05-07, 2017-12-04, 2017-12-08, 2017-12-17
(7, 21)	7	5	2017-05-07, 2017-05-08, 2017-05-30, 2017-12-23
(7, 28)	7	7	2017-02-27, 2017-03-01, 2017-11-28, 2017-12-04, 2017-12-07, 2017-12-14, 2017-12-17
(14, 7)	7	3	2017-05-23, 2017-05-24, 2017-11-28
(14, 14)	7	5	2017-05-07, 2017-11-28, 2017-12-14, 2017-12-20
(14, 21)	7	5	2017-05-04, 2017-05-05, 2017-05-07, 2017-05-08, 2017-12-14
(21, 5)	5	2	2017-12-14
(21, 7)	7	5	2017-05-24, 2017-12-14
(21, 14)	7	5	2017-05-07, 2017-12-14, 2017-12-14, 2017-12-20, 2017-12-21
(28, 3)	3	0	
(28, 5)	5	0	
(28, 7)	7	1	2017-12-14

(a) Liquidation dates

Holding period	Accounts	Liquidated	Liquidation ratio
3	3	0	0.00%
5	10	2	20.00%
7	21	9	42.86%
14	42	29	69.05%
21	35	30	85.71%
28	28	28	100.00%

(b) Rebalancing frequency and liquidation

Table IA16: Impacts of mark-to-market and transaction costs

This table compares the performance of cross-sectional momentum portfolios i) without mark-to-market, ii) with mark-to-market, and iii) with mark-to-market and transaction costs of 15 bps. ‘LQ’ means the portfolio has been liquidated. The sample period is January 1, 2017 to August 28, 2023.

(j, k)	Without MTM						Daily MTM						Daily MTM + Cost					
	L		S		L-S		L		S		L-S		L		S		L-S	
	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR	Mean	SR
(1, 14)	114.1	1.19	-36.4	-0.39	77.7	1.70	114.0	1.19	LQ		63.2	1.03	107.2	1.12	LQ		49.2	0.80
(1, 21)	106.1	1.12	-55.2	-0.58	50.9	1.19	105.7	1.12	LQ		34.2	0.33	101.2	1.07	LQ		24.1	0.23
(1, 28)	108.5	1.16	-53.9	-0.59	54.1	1.35	108.9	1.07	LQ		LQ		105.6	1.03	LQ		LQ	
(3, 14)	112.8	1.19	-34.5	-0.36	78.3	1.33	112.9	1.19	LQ		53.1	0.70	106.1	1.12	LQ		39.4	0.52
(3, 21)	88.7	0.95	-64.6	-0.67	24.1	0.46	88.6	0.95	LQ		LQ		84.1	0.90	LQ		LQ	
(3, 28)	99.2	1.06	-52.2	-0.55	49.6	0.90	99.2	1.06	LQ		LQ		95.8	1.02	LQ		LQ	
(5, 14)	126.5	1.22	-20.7	-0.21	105.8	1.44	126.5	1.22	LQ		122.5	1.31	119.9	1.16	LQ		108.7	1.16
(5, 21)	112.0	1.10	-55.2	-0.56	56.7	0.83	111.5	1.10	LQ		LQ		107.0	1.05	LQ		LQ	
(5, 28)	107.9	1.05	-49.3	-0.50	61.1	0.88	107.9	1.05	LQ		LQ		104.5	1.02	LQ		LQ	
(7, 14)	141.9	1.25	-12.1	-0.12	129.8	1.38	141.9	1.25	-2.45	-0.02	123.9	1.22	135.4	1.19	-9.5	-0.08	110.2	1.09
(7, 21)	122.7	1.08	-31.4	-0.32	91.4	1.00	122.3	1.08	LQ		82.3	0.70	117.7	1.04	LQ		72.7	0.62
(7, 28)	120.9	1.06	-38.4	-0.39	84.9	0.93	120.9	1.06	LQ		LQ		117.5	1.03	LQ		LQ	
(14, 7)	159.4	1.51	-0.3	-0.00	159.0	1.78	159.3	1.51	90.5	0.39	195.9	1.43	150.0	1.42	77.7	0.34	174.7	1.28
(14, 14)	131.8	1.27	-27.0	-0.27	104.8	1.26	131.8	1.28	LQ		160.1	0.98	125.4	1.21	LQ		146.6	0.90
(14, 21)	115.3	1.08	-44.1	-0.44	60.3	0.79	115.3	1.08	LQ		162.3	0.75	110.8	1.04	LQ		153.2	0.71
(21, 5)	139.4	1.31	-31.9	-0.30	107.5	1.15	139.4	1.31	LQ		126.6	1.15	129.5	1.22	LQ		103.4	0.94
(21, 7)	124.3	1.19	-32.9	-0.32	91.4	1.01	124.4	1.19	LQ		98.2	0.88	116.2	1.11	LQ		79.1	0.71
(21, 14)	118.5	1.15	-40.8	-0.40	65.5	0.79	118.5	1.15	LQ		106.6	0.92	112.9	1.09	LQ		94.4	0.81
(28, 3)	133.0	1.23	-45.4	-0.43	87.6	0.92	133.1	1.23	LQ		86.4	0.88	121.6	1.12	LQ		58.0	0.59
(28, 5)	122.1	1.16	-43.9	-0.42	78.2	0.87	122.3	1.16	LQ		83.4	0.85	113.7	1.08	LQ		62.4	0.64
(28, 7)	104.4	1.00	-46.6	-0.45	57.9	0.66	104.5	1.00	LQ		58.7	0.60	97.4	0.94	LQ		41.3	0.42

Table IA17: Cross-sectional momentum portfolio factor regression (two-week look-back)

This table reports the factor regression results of the (14, 7) cross-sectional momentum long-short portfolio. The difference from the factor regression in the manuscript is that MOM and CS-CO factors are generated using the past two-week returns instead of three.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Const	0.0294 (4.28)	0.0302 (4.36)	0.0176 (2.86)	0.0217 (3.12)	0.0291 (4.44)	0.0177 (3.03)	0.0176 (3.11)
MKT	0.0926 (1.32)				0.0779 (1.18)	-0.0090 (-0.19)	-0.0101 (-0.22)
SIZE		0.1039 (0.77)			0.0887 (0.66)	0.0060 (0.04)	0.0055 (0.04)
MOM			0.7121 (4.93)			0.7127 (4.88)	0.6972 (2.94)
CS-CO				0.7961 (8.34)			0.0363 (0.15)
Adj R^2	0.004	0.004	0.291	0.144	0.007	0.290	0.290

Table IA18: Performance of cross-sectional momentum portfolios (Binance futures)

This table reports the performance of cross-sectional momentum portfolios formed of the coins listed on the Binance futures market. ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is February 12, 2020 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(1, 14)	60.93	86.05	0.71	132.0	84.1	-86.15	93.24	-0.92	-99.2	99.7	-14.16	40.11	-0.35	-58.8	72.4
(1, 21)	75.56	87.68	0.86	286.7	79.7	-57.96	98.91	-0.59	-98.2	99.4	12.70	43.21	0.29	12.8	66.8
(1, 28)	58.87	85.47	0.69	119.3	82.5	-64.04	95.61	-0.67	-98.4	99.6	0.55	33.67	0.02	-18.0	47.6
(3, 14)	77.05	86.80	0.89	317.2	82.1	-88.58	99.12	-0.89	-99.5	99.7	6.47	40.94	0.16	-7.0	53.1
(3, 21)	82.99	89.13	0.93	383.0	83.3	-54.31	95.47	-0.57	-97.6	98.8	43.85	44.10	0.99	262.5	36.3
(3, 28)	70.58	86.64	0.81	231.2	83.8	-52.07	95.99	-0.54	-97.5	99.2	27.82	40.72	0.68	108.9	42.6
(5, 14)	94.06	90.02	1.04	615.5	81.8	-63.15	97.25	-0.65	-98.4	99.3	43.22	53.38	0.81	200.3	44.6
(5, 21)	99.27	88.98	1.12	799.9	83.9	-33.43	92.37	-0.36	-94.2	97.0	69.00	52.45	1.32	708.8	55.4
(5, 28)	85.62	91.19	0.94	399.4	80.8	-34.21	92.70	-0.37	-94.4	97.6	53.48	49.52	1.08	375.4	39.0
(7, 14)	90.42	89.44	1.01	538.6	85.9	-68.15	101.66	-0.67	-98.9	99.4	35.12	56.11	0.63	110.0	49.6
(7, 21)	85.16	94.53	0.90	356.8	82.6	-44.32	101.39	-0.44	-97.4	98.7	32.21	52.26	0.62	102.2	56.5
(7, 28)	84.13	90.37	0.93	384.5	83.4	-21.86	95.39	-0.23	-91.9	97.8	61.30	50.76	1.21	522.4	34.4
(14, 7)	98.80	90.03	1.10	763.1	86.1	-39.81	96.47	-0.41	-96.1	98.3	52.61	63.68	0.83	240.4	56.4
(14, 14)	88.55	90.36	0.98	480.6	87.1	-39.54	98.51	-0.40	-96.4	98.0	45.30	61.54	0.74	171.7	62.4
(14, 21)	71.44	92.35	0.77	189.1	82.3	-56.94	119.88	-0.47	-99.4	99.6	1.23	60.86	0.02	-48.5	73.2
(21, 5)	99.40	90.45	1.10	778.3	87.3	-48.18	98.84	-0.49	-97.4	98.7	41.38	67.46	0.61	101.7	64.9
(21, 7)	95.96	90.40	1.06	672.9	86.1	-45.94	97.06	-0.47	-97.0	98.4	41.44	65.14	0.64	114.4	56.7
(21, 14)	82.61	91.37	0.90	346.7	87.9	-36.66	99.47	-0.37	-96.1	98.2	42.32	63.66	0.66	131.6	54.6
(28, 3)	95.15	93.82	1.01	562.4	85.9	-51.44	98.40	-0.52	-97.7	99.0	39.93	69.70	0.57	80.8	55.5
(28, 5)	89.80	93.57	0.96	445.7	86.1	-43.48	96.96	-0.45	-96.7	98.7	41.83	67.04	0.62	108.3	52.4
(28, 7)	80.94	92.89	0.87	299.2	86.2	-44.84	97.27	-0.46	-97.0	98.7	33.88	65.66	0.52	58.2	52.3
Market											60.66	73.85	0.82	238.1	78.5

Table IA19: Performance of cross-sectional reversal portfolios

This table reports the performance of cross-sectional reversal portfolios of various look-back and holding periods, (j, k) . ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2018 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(1,1)	-149.89	102.01	-1.47	-100.0	100.0	-145.47	105.33	-1.38	-100.0	100.0	-295.21	86.58	-3.41	-100.0	100.0
(1,3)	-87.90	96.47	-0.91	-100.0	100.0	-50.16	92.25	-0.54	-99.5	99.9	-149.31	55.00	-2.71	-100.0	100.0
(3,1)	-104.02	107.02	-0.97	-100.0	100.0	-124.76	97.20	-1.28	-100.0	100.0	-228.56	85.46	-2.67	-100.0	100.0
(3,3)	-87.75	100.42	-0.87	-100.0	100.0	-75.58	91.93	-0.82	-100.0	100.0	-161.70	71.98	-2.25	-100.0	100.0
(42,1)	-32.18	101.63	-0.32	-99.2	99.4	-45.05	92.98	-0.48	-99.3	99.9	-76.98	75.66	-1.02	-99.8	99.9
(42,3)	-19.58	98.64	-0.20	-98.0	98.4	-18.78	90.54	-0.21	-96.5	99.6	-43.36	71.60	-0.61	-98.0	99.1
(49,1)	-26.59	99.93	-0.27	-98.8	99.1	-46.88	94.33	-0.50	-99.5	99.9	-73.18	74.87	-0.98	-99.7	99.8
(49,3)	-23.81	97.78	-0.24	-98.4	98.7	-13.87	91.74	-0.15	-95.7	99.6	-44.47	71.63	-0.62	-98.1	99.2
(49,42)	25.86	93.98	0.28	-66.0	94.3	-12.45	97.03	-0.13	-96.7	99.8	37.59	65.43	0.64	153.4	81.6
(49,49)	1.90	91.07	0.02	-89.8	95.7	14.44	137.74	0.10	-98.5	99.9	8.66	73.48	0.12	-64.7	87.1
(56,42)	24.22	94.25	0.26	-69.1	91.8	-2.53	98.08	-0.03	-94.4	99.8	38.50	67.67	0.57	218.1	82.6
(56,49)	6.32	90.94	0.07	-86.0	94.8	-25.50	111.83	-0.23	-99.3	100.0	11.25	66.87	0.17	-46.1	84.7
Market											28.07	75.59	0.37	-6.7	89.1

Table IA20: Fama-MacBeth regression of the future return on the capital gains overhang

This table reports the Fama-MacBeth regression results of the form

$$r_{t,t+7}^i = \beta_0 + \beta_1 g_t^i + e_t^i,$$

where $r_{t,t+7}^i$ is coin i 's return over the period t to $t+7$ and g_t^i is the capital gains overhang of coin i , $g_t^i = \frac{P_t^i - R_t^i}{P_t^i}$, where P_t^i is the price of coin i and R_t^i is its reference price defined as

$$R_t^i = \sum_{n=1}^{84} \left(V_{t-n}^i \prod_{\tau=1}^{n-1} (1 - V_{t-n+\tau}^i) \right) P_{t-n}^i,$$

where V_t^i is the daily turnover. The weights on P_{t-n}^i are normalized so that their sum becomes 1. The regression is repeated three times using i) all samples, ii) samples with $g > 0$, and iii) samples with $g < 0$. The sample period is January 1, 2017 to August 28, 2023.

	All	$g > 0$	$g < 0$
Const	0.013 (2.038)	0.018 (2.432)	0.001 (0.112)
g	0.050 (3.629)	0.034 (0.800)	0.071 (2.119)

Table IA21: Fama-MacBeth regression of cross-sectional momentum on Google trends

This table reports the Fama-MacBeth regression results of the form

$$k_{t,t+7}^i = \beta_0 + \beta_1 k_{t-14,t}^i + \beta_2 (k_{t-14,t}^i \times G_t^i) + \beta_3 G_t^i + \beta_4 \ln(M_t^i) + e_t^i,$$

where $k_{t,t+k}^i$ is the cross-sectional percentile rank of coin i 's return over the period t to $t+k$, G_t^i is its Google trend, and M_t^i is its market capitalization.

	(1)	(2)
Const	0.537 (70.590)	0.495 (11.204)
$k_{t-j,t}^i$	-0.044 (-3.031)	-0.050 (-3.640)
$k_{t-j,t}^i \times G_t^i$	-0.020 (-1.357)	-0.018 (-1.266)
G_t^i	0.010 (1.069)	0.010 (1.108)
$\ln(M_t^i)$		0.002 (0.974)

Table IA22: Performance of cross-section momentum under different weighting schemes

This table reports the performance of the (1, 14) and (14, 7) cross-sectional momentum portfolios under different weighting schemes: value-weight ('Value'), capped-value-weight ('CapValue'), volume-weight ('Volume'), capped-volume-weight ('CapVolume'), and equal-weight ('Equal'). The columns 'Mean', 'Std', 'Sharpe', 'Cum', and 'MDD' respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2017 to August 28, 2023. A transaction cost of 15 bps is assumed.

Scheme	(1, 14)					(14, 7)				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
Value	49.18	61.55	0.80	687.6	62.8	174.73	136.93	1.28	101218.3	86.5
CapValue	43.22	53.53	0.81	600.3	58.6	113.42	95.95	1.18	9334.3	85.3
Volume	35.50	81.99	0.43	18.0	81.6	231.64	240.91	0.96	35454.4	92.1
CapVolume	30.85	66.00	0.47	87.4	71.4	49.63	96.06	0.52	30.1	89.4
Equal	-9.60	622.82	-0.02	-100.0	100.0	55.87	73.44	0.76	569.9	84.4
Market						78.86	77.80	1.01	2333.3	89.1

Table IA23: Performance of cross-sectional momentum portfolios (excluding the last day from the look-back period)

This table reports the performance of cross-sectional momentum portfolios when the last day is excluded from the look-back period. ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2017 to August 28, 2023. A transaction cost of 15 bps is assumed.

(j, k)	Long					Short					Long-Short				
	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD	Mean	Std	Sharpe	Cum	MDD
(3, 14)	99.48	95.72	1.04	3470.7	96.9	-47.04	142.70	-0.33	-100.0	100.0	82.45	96.15	0.86	1042.6	87.5
(3, 21)	88.17	94.19	0.94	1690.8	96.3	-381.79	294.10	-1.30	-100.0	100.0	-205.06	260.90	-0.79	-100.0	100.0
(3, 28)	84.59	94.65	0.89	1291.1	96.6	-1439.25	270.43	-5.32	-100.0	100.0	41.53	111.01	0.37	-62.8	97.4
(5, 14)	114.82	99.25	1.16	8354.8	96.0	-625.26	250.31	-2.50	-100.0	100.0	96.64	76.30	1.27	9280.8	61.9
(5, 21)	98.96	98.86	1.00	2838.2	94.4	-627.52	205.36	-3.06	-100.0	100.0	510.06	1073.48	0.48	75201.5	86.8
(5, 28)	97.96	100.16	0.98	2442.0	95.8	-565.63	253.64	-2.23	-100.0	100.0	-178.59	235.31	-0.76	-100.0	100.0
(7, 14)	130.43	113.06	1.15	13137.4	96.0	-680.85	217.89	-3.12	-100.0	100.0	115.58	94.00	1.23	17062.5	68.4
(7, 21)	113.21	113.23	1.00	3914.2	95.4	-642.02	217.49	-2.95	-100.0	100.0	95.14	115.78	0.82	1253.7	96.2
(7, 28)	105.19	114.16	0.92	2100.9	95.9	-1205.84	287.16	-4.20	-100.0	100.0	-143.30	300.87	-0.48	-100.0	100.0
(14, 7)	137.79	104.78	1.31	26030.1	93.6	20.65	141.97	0.15	-98.6	99.7	141.15	106.03	1.33	37389.9	88.5
(14, 14)	116.98	101.63	1.15	7949.0	94.3	-20.50	124.01	-0.17	-99.9	100.0	127.45	110.18	1.16	13757.2	94.9
(14, 21)	85.92	100.05	0.86	977.0	93.7	-891.75	224.66	-3.97	-100.0	100.0	40.28	99.17	0.41	-53.9	99.1
(21, 5)	113.26	104.58	1.08	5050.0	95.9	-27.56	123.71	-0.22	-100.0	100.0	144.03	205.80	0.70	1228.4	97.9
(21, 7)	106.91	103.51	1.03	3520.9	95.2	-767.15	219.49	-3.50	-100.0	100.0	66.63	110.05	0.61	-10.1	98.7
(21, 14)	89.22	102.51	0.87	1074.7	94.9	-34.83	150.72	-0.23	-99.9	100.0	96.40	129.14	0.75	580.5	95.7
(28, 3)	112.23	106.21	1.06	4269.8	93.2	-44.93	114.77	-0.39	-100.0	100.0	64.02	100.15	0.64	139.5	91.3
(28, 5)	99.20	104.36	0.95	1919.4	94.0	-44.47	119.91	-0.37	-100.0	100.0	49.34	96.76	0.51	16.9	95.5
(28, 7)	87.55	103.41	0.85	877.7	94.1	-70.35	115.86	-0.61	-100.0	100.0	15.73	97.05	0.16	-89.0	98.9
Market											78.86	77.80	1.01	2333.3	89.1

Table IA24: Performance of cross-sectional momentum under different rebalancing days

This table reports the performance of the (14, 7) cross-sectional momentum portfolio under different rebalancing days. ‘Distributed’ means investing evenly throughout the week. The columns ‘Mean’, ‘Std’, ‘Sharpe’, ‘Cum’, and ‘MDD’ respectively denote the annualized mean return (%), annualized standard deviation (%), annualized Sharpe ratio, cumulative return (%), and maximum drawdown (%). The sample period is January 1, 2017 to August 28, 2023. A transaction cost of 15 bps is assumed.

	Mean	Std	Sharpe	Cum	MDD
Mon	176.56	123.18	1.43	133392.1	74.1
Tue	92.04	114.09	0.81	197.2	99.4
Wed	238.84	272.98	0.87	-100.0	100.0
Thu	-130.54	296.64	-0.44	-100.0	100.0
Fri	612.34	300.22	2.04	-100.0	100.0
Sat	216.62	200.65	1.08	47418.4	97.2
Sun	217.97	199.17	1.09	146435.3	88.4
Distributed	174.73	136.93	1.28	101218.2	86.5